

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\140-C1 Final-23rd March.pvdb

Date of Analysis : Sep 9, 2026 9:34am

PV Elite 28, January 2026



PV Elite 28 Licensee: SL User
 FileName : 140-C1 Final-23rd March Page 2 of 216
 Vessel Design Summary: Step: 34 9:34am Sep 9,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1750.000 mm ID
 Vessel Design Length, Tangent to Tangent 8239.35 mm
 Specified Datum Line Distance 0.00 mm
 Shell Side Design Temperature 275 °C
 Channel Side Design Temperature 250 °C
 Shell Side Design Pressure 4950.000 KPa.
 Channel Side Design Pressure 1250.000 KPa.
 Shop Shell Side Test Pressure 7100.959 KPa.
 Field Shell Side Test Pressure 6435.000 KPa.
 Shop Channel Side Test Pressure 2413.724 KPa.
 Field Channel Side Test Pressure 1625.000 KPa.
 Wind Design Code ASCE/SEI 7-16
 Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-387 11	1	...	K11789	
Flange	SA-336 F11	3	...	K11572	
Nozzle	SA-336 F11	3	...	K11572	
Nozzle	SA-106 B	...	...	K03006	
Nozzle Flg	SA-182 F11	2	...	K11572	
Nozzle Flg	SA-336 F11	2	...	K11572	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-213 T11	...	...	K11597	
Tubesheet	SA-336 F11	2	...	K11572	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-307 A	...	6 <= t <= 100		
Saddles	SA-387 11	1	...	K11789	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
CH Cover	1270.682	103.42	1447.000	3.0000	N/A	No
LHS Ch Flg	1267.448	103.42	1450.000	3.0000	N/A	No
LHS Ch Shell	1267.183	103.42	1450.000	3.0000	N/A	No
Main Shell Flange	4950.000	0.00	5186.000	3.0000	N/A	No
Main Shell	4967.183	0.00	5088.000	3.0000	N/A	No

Liquid Level: 2110.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:

Element	"To" Elev	Element Length	Nominal Thickness	Finished Thickness	Reqd Thk Internal	Reqd Thk External	Long Eff	Circ Eff
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Vessel Design Summary:

Step: 34

9:34am

Sep 9, 2026

Type	mm	mm	mm	mm	mm	mm		
Body Flg	208.2	205.0	185.0	177.0	170.5	170.5	1.00	1.00
Body Flg	413.2	205.0	185.0	177.0	170.5	170.5	1.00	1.00
Cylinder	1613.2	1200.0	14.0	14.0	12.5	7.9	1.00	1.00
Body Flg	2239.3	298.0	240.0	232.0	226.3	226.3	1.00	1.00
Cylinder	8239.3	6000.0	42.0	42.0	40.9	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N3-Tube Inlet/	WRC-297	0.927	Shell	Passed
Noz N2 Fr30	WRC-297	0.924	Shell	Passed
3" Nozzle	WRC-297	1384.801	Shell	Failed
Noz N1 750	WRC-297	0.958	Shell	Passed
K7-Temperature	WRC-297	0.958	Shell	Passed
Noz N4 750	WRC-297	0.966	Shell	Passed
D4-Shell side	WRC-297	0.966	Shell	Passed

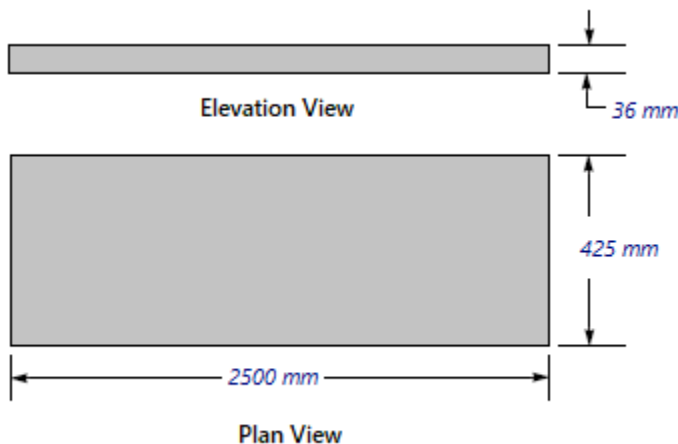
Foundation/Support Summary

Saddle Parameters:

Saddle Width	400.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	1535.000 mm
Wear Pad Width	780.000 mm
Wear Pad Thickness	14.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	1200.000 mm

Baseplate Length	2500.000 mm
Baseplate Thickness	36.000 mm
Baseplate Width	425.000 mm
Number of Ribs (including outside ribs)	4
Rib Thickness	14.000 mm
Web Thickness	14.000 mm
Height of Center Web	600.000 mm
Number of Bolts in Baseplate	6

Baseplate Sketch





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Vessel Design Summary:

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Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	62191.95	Kgf
Maximum Transverse Saddle Shear Load	2385.37	Kgf
Maximum Longitudinal Saddle Shear Load	1610.80	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	58318.97	Kgf
Maximum Transverse Saddle Shear Load	193.66	Kgf
Maximum Longitudinal Saddle Shear Load	201.04	Kgf

Weights:

Fabricated - Bare w/o removable internals	41654.3	kg.
Shop Test - Fabricated + water (full)	60558.6	kg.
Shipping - Fab. + rem. intls.+ shipping app.	43343.7	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	43343.7	kg.
Empty - Fab. + intls. + details + wghts.	43343.7	kg.
Operating - Empty + operating liquid (no ca)	61508.3	kg.
Field Test - Empty weight + test medium (full)	61150.0	kg.

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Nozzle Summary:

Step: 32 9:34am Sep 9,2026

Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N3-Tube Inlet/L	1450	OK	...	OK	11.33	OK	Passed
Noz N2 Fr30	1450	OK	...	OK	11.33	OK	Passed
3" Nozzle	1450	...	...	OK	8.73	OK	No Calc[*]
Noz N1 750	5059	...	...	OK	15.56	OK	Passed
K7-Temperature	5088	...	...	OK	6.42	OK	No Calc[*]
M1-Manway	5088	...	...	OK	12.61	OK	Passed
Noz N4 750	5042	...	...	OK	15.61	OK	Passed
D4-Shell side d	5088	...	...	OK	6.42	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 5042 Nozzle : Noz N4 750 [Shellside]
 Minimum MAWP Nozzles : 1450 Nozzle : N3-Tube Inlet/L [Tubeside]

[*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N3-Tube Inlet/L	1013.175	180.000	589.901
30	Noz N2 Fr30	1013.175	0.000	589.901
30	3" Nozzle	628.440	90.000	164.400
50	Noz N1 750	3439.349	0.000	1471.992
50	K7-Temperature	4439.349	0.000	138.100
50	M1-Manway	7039.349	90.000	1117.732
50	Noz N4 750	5739.350	180.000	1471.992
50	D4-Shell side d	7970.489	180.000	138.100

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
K7-Temperature	1.500 in	Actual	RTJWn	2.366	11.000	...	...	242.52	2500
D4-Shell side d	1.500 in	Actual	RTJWn	2.366	11.000	...	...	242.52	2500
3" Nozzle	3.000 in	Actual	LWN	4.620	20.574	...	...	221.99	300
N3-Tube Inlet/L	12.750 in	Actual	WNF	12.750	17.450	...	...	335.11	300
Noz N2 Fr30	12.750 in	Actual	WNF	12.750	17.450	...	...	335.11	300
M1-Manway	22.890 in	Actual	WNF	22.890	14.270	...	...	541.70	600
Noz N1 750	28.740 in	Actual	None	30.000	16.000	...	...	617.00	...
Noz N4 750	28.740 in	Actual	None	30.000	16.000	...	...	617.00	...

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K7-Temperat	SA-106 B	44.000	10.000	...	...	...
D4-Shell si	SA-106 B	44.000	10.000	...	...	...
3" Nozzle	SA-336 F11	14.000	10.000	...	...	...
N3-Tube Inl	SA-336 F11	14.000	10.000	...	...	...
Noz N2 Fr30	SA-336 F11	14.000	10.000	...	...	...
M1-Manway	SA-336 F11	44.000	10.000	...	...	...
Noz N1 750	SA-336 F11	44.000	10.000	...	...	...
Noz N4 750	SA-336 F11	44.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
K7-Temperature	4439.349	0.0	200.00	0.00	Main Shell
D4-Shell side d	7970.488	180.0	200.00	0.00	Main Shell
3" Nozzle	628.440	90.0	200.02	0.00	LHS Ch Shell
N3-Tube Inlet/L	1013.175	180.0	300.00	0.00	LHS Ch Shell
Noz N2 Fr30	1013.175	0.0	300.00	0.00	LHS Ch Shell
M1-Manway	7039.349	90.0	450.00	0.00	Main Shell
Noz N1 750	3439.349	0.0	400.00	0.00	Main Shell
Noz N4 750	5739.350	180.0	400.00	0.00	Main Shell

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Bill of Material:

Step: 30 9:34am

Sep 9, 2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	WELD NECK FLANGE: 185.0mm THK X 1990.0mm OD	SA-336 F11
1	WELD NECK FLANGE: 240.0mm THK X 2240.0mm OD	SA-336 F11
1	INSULATION: 136mm X 100mm THK	
1	INSULATION: 212mm X 100mm THK	
1	INSULATION: 1200mm X 100mm THK	
1	TUBESHEET: 2240.0mm X 325.0mm THK	SA-336 F11
638	TUBES: 2500.0mm X 25.4mm DIA X 2.1mm THK	SA-213 T11
1	INSULATION: 325mm X 100mm THK	
2	SADDLE: 400.0mm X 120 DEG	SA-387 11
1	INSULATION: 6000mm X 100mm THK	
1	CLASS 600 GR 1.1, 22.9" BLIND FLANGE(S)	SA-105
1	GASKET: 1820.0mm OD X 1782.0mm ID	...
56	BODY FLANGE BOLTS: 34.9mm DIA	SA-193 B7
112	NUTS FOR BODY FLANGE BOLTS: 34.9mm DIA	...
1	GASKET: 1872.0mm OD X 1782.0mm ID	...
28	BODY FLANGE BOLTS: 76.2mm DIA	SA-193 B7
28	NUTS FOR BODY FLANGE BOLTS: 76.2mm DIA	...
1	CYLINDRICAL SEGMENT 14.0mm THK X 3054.3mm X 1200.0mm	SA-387 11
1	CYLINDRICAL SEGMENT 14.0mm THK X 2443.4mm X 1200.0mm	SA-387 11
1	CYLINDRICAL SEGMENT 42.0mm THK X 3054.3mm X 6000.0mm	SA-387 11
1	CYLINDRICAL SEGMENT 42.0mm THK X 2443.4mm X 6000.0mm	SA-387 11
1	NAMEPLATE	...



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MDMT Summary:

Step: 33 9:34am Sep 9,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
Main Shell Fl[11]	B	14	9		0.921	44.000	1.00	No
Bolting [21]		-40						
Main Shell [8]	A	32	32		0.987	42.000	1.00	Yes
Left Saddle [24]	A	2	1		0.987	14.000	1.00	Yes
New Saddle [24]	A	2	1		0.987	14.000	1.00	Yes
Noz N1 750 [1]	A	32	31		0.969	42.000	1.00	No
K7-Temperature[1]	A	-5	-104	-29	0.119	11.000	1.00	Yes
Nozzle Flg [5]	A	-18	-104					
M1-Manway [1]	A	32	31		0.971	42.000	1.00	Yes
Nozzle Flg [5]	A	-18	-48					
Noz N4 750 [1]	A	32	31		0.972	42.000	1.00	No
D4-Shell side [1]	A	-5	-104	-29	0.119	11.000	1.00	Yes
Nozzle Flg [5]	A	-18	-104					
Tubesheet: SS[13]	B	27	21		0.885	81.250	1.00	No
Warmest MDMT:		32	32					
CH Cover [11]	B	-18	-31		0.774	14.000	1.00	No
LHS Ch Flg [11]	B	-18	-31		0.774	14.000	1.00	No
LHS Ch Shell [8]	A	2	1		0.986	14.000	1.00	Yes
N3-Tube Inlet/[1]	A	12	-14		0.558	20.000	1.00	Yes
Nozzle Flg [5]	B	-18	-104					
Noz N2 Fr30 [1]	A	12	-14		0.551	20.000	1.00	Yes
Nozzle Flg [5]	B	-18	-104					
3" Nozzle [1]	A	12	-14		0.554	20.000	1.00	No
Nozzle Flg [5]	B	-18	-104					
Tubesheet: CS[14]	A	45	39		0.885	81.250	1.00	No
Bolting [21]		-48						
Warmest MDMT:		45	39					
Exchanger Side	Computed MDMT		Required MDMT		Pass/Fail			
Shell	32.0		-12.0		Fail			
Channel/Tube	39.0		-12.0		Fail			

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material



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MDMT Summary:

Step: 33 9:34am Sep 9, 2026

[23] - Stiffening Ring to Shell Weld

[24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Warnings and Errors:

Step: 0 9:34am Sep 9, 2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Input Echo:

Step: 1 9:34am Sep 9, 2026

Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	4950	KPa.
Channel Side Design Pressure	1250	KPa.
Shell Side Design Temperature	275.0	°C
Channel Side Design Temperature	250.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	None	
Service Type, Channel Side	None	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes



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Input Echo:

Step: 1 9:34am Sep 9,2026

Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	25500 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.100
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



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Use ASME VIII-1 Appendix 1-9 N

Perform Blast Load Analysis No

Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes
 Do not use Nozzle MDMT Interpretation VIII-1 01-37 No
 Use Table G instead of exact equation for "A" Yes
 Shell Head Joints are Tapered Yes
 Compute "K" in corroded condition Yes
 Use Code Case 2286 No
 Use the MAWP to compute the MDMT Yes
 For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes
 For PWHT & P1 Materials the MDMT can be < -55F (-48C) No

Using Metric Material Databases, ASME II D Yes
 Calculate B31.3 type stress for Nozzles with Loads Yes
 Reduce the MDMT due to lower membrane stress Yes
 Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/5
Element From Node	10
Element To Node	20
Element Type	Flange
Description	CH Cover
Distance "FROM" to "TO"	205 mm
Flange Inside Diameter	1750 mm
Element Thickness	177 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	185 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1250 KPa.
Design Temperature Internal Pressure	250 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-336 F11
Allowable Stress, Ambient	148 N./mm ²
Allowable Stress, Operating	148 N./mm ²
Allowable Stress, Hydrotest	278.99 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	16.002 mm
Yield Stress, Operating	253 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-3
UNS Number	K11572
Class / Thickness / Grade	3 ::
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2110 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	10
Detail Type	Insulation
Detail ID	Ins: 50



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Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	136	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element#	2/5
Element From Node	20
Element To Node	30
Element Type	Flange
Description	LHS Ch Flg
Distance "FROM" to "TO"	205 mm
Flange Inside Diameter	1750 mm
Element Thickness	177 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	185 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1250 KPa.
Design Temperature Internal Pressure	250 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-336 F11
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1780 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	20
Detail Type	Insulation
Detail ID	Ins: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	212 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element#	3/5
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	LHS Ch Shell
Distance "FROM" to "TO"	1200 mm
Inside Diameter	1750 mm
Element Thickness	14 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1250 KPa.
Design Temperature Internal Pressure	250 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11
Allowable Stress, Ambient	118 N./mm ²
Allowable Stress, Operating	118 N./mm ²
Allowable Stress, Hydrotest	216 N./mm ²
Material Density	0.00775 kg./cm ³



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P Number Thickness	16.002	mm
Yield Stress, Operating	197	N./mm ²
UCS-66 Chart Curve Designation	A	
External Pressure Chart Name	CS-2	
UNS Number	K11789	
Class / Thickness / Grade	1 ::	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1750	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	30	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1200	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N3-Tube Inlet/L	
Dist. from "FROM" Node / Offset dist	600	mm
Nozzle Diameter	12.75	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	157.94	Kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-336 F11	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	Noz N2 Fr30	
Dist. from "FROM" Node / Offset dist	600	mm
Nozzle Diameter	12.75	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	157.94	Kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-336 F11	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	3" Nozzle	
Dist. from "FROM" Node / Offset dist	215.26	mm
Nozzle Diameter	3	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	5.5949	Kgf
Grade of Attached Flange	GR 1.9	
Nozzle Matl	SA-336 F11	



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Element From Node	30
Detail Type	Weight
Detail ID	Partition plate
Dist. from "FROM" Node / Offset dist	600 mm
Miscellaneous Weight	200 Kg
Offset from Element Centerline	0 mm

Element#	4/5
Element From Node	40
Element To Node	50
Element Type	Flange
Description	Main Shell Flange
Distance "FROM" to "TO"	298 mm
Flange Inside Diameter	1750 mm
Element Thickness	232 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	240 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4950 KPa.
Design Temperature Internal Pressure	275 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	275 °C
Effective Diameter Multiplier	1.2
Material Name	SA-336 F11
Allowable Stress, Ambient	148 N./mm ²
Allowable Stress, Operating	148 N./mm ²
Allowable Stress, Hydrotest	278.99 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	16.002 mm
Yield Stress, Operating	249 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-3
UNS Number	K11572
Class / Thickness / Grade	3 ::
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	40
Detail Type	Insulation
Detail ID	Ins: 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	326 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element#	5/5
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	Main Shell
Distance "FROM" to "TO"	6000 mm
Inside Diameter	1750 mm
Element Thickness	42 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	42 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4950 KPa.
Design Temperature Internal Pressure	275 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	275 °C



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Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	
Allowable Stress, Ambient	118	N./mm ²
Allowable Stress, Operating	118	N./mm ²
Allowable Stress, Hydrotest	216	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	16.002	mm
Yield Stress, Operating	241	N./mm ²
UCS-66 Chart Curve Designation	A	
External Pressure Chart Name	CS-2	
UNS Number	K11789	
Class / Thickness / Grade	1 ::	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	1200	mm
Width of Saddle	400	mm
Height of Saddle at Bottom	1535	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	780	mm
Thickness of Wear Plate	14	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.30000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	50	
Detail Type	Saddle	
Detail ID	New Saddle	
Dist. from "FROM" Node / Offset dist	4800	mm
Width of Saddle	400	mm
Height of Saddle at Bottom	1535	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	780	mm
Thickness of Wear Plate	14	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1750	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	50	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	6000	mm



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Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element From Node	50	
Detail Type	Nozzle	
Detail ID	Noz N1 750	
Dist. from "FROM" Node / Offset dist	1200	mm
Nozzle Diameter	28.74	in.
Nozzle Schedule	30	
Nozzle Class	None	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	682.74	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F11	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	K7-Temperature	
Dist. from "FROM" Node / Offset dist	2200	mm
Nozzle Diameter	1.5	in.
Nozzle Schedule	XXS	
Nozzle Class	2500	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	14.547	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	M1-Manway	
Dist. from "FROM" Node / Offset dist	4800	mm
Nozzle Diameter	22.89	in.
Nozzle Schedule	30	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	1493.6	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-336 F11	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	Noz N4 750	
Dist. from "FROM" Node / Offset dist	3500	mm
Nozzle Diameter	28.74	in.
Nozzle Schedule	30	
Nozzle Class	None	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	682.74	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F11	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	D4-Shell side d	
Dist. from "FROM" Node / Offset dist	5731.1	mm
Nozzle Diameter	1.5	in.
Nozzle Schedule	XXS	
Nozzle Class	2500	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	14.547	Kgf



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Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B
Element From Node	50
Detail Type	Weight
Detail ID	Distributor wei
Dist. from "FROM" Node / Offset dist	4000 mm
Miscellaneous Weight	850 Kg
Offset from Element Centerline	0 mm
Element From Node	50
Detail Type	Weight
Detail ID	Distributor wt
Dist. from "FROM" Node / Offset dist	2500 mm
Miscellaneous Weight	200 Kg
Offset from Element Centerline	0 mm
Element From Node	50
Detail Type	Weight
Detail ID	Wght:[3 of 3]
Dist. from "FROM" Node / Offset dist	1200 mm
Miscellaneous Weight	200 Kg
Offset from Element Centerline	1500 mm
Element From Node	50
Detail Type	Weight
Detail ID	Direprofing wei
Dist. from "FROM" Node / Offset dist	4800 mm
Miscellaneous Weight	200 Kg
Offset from Element Centerline	1500 mm

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XY Coordinate Calculations:

Step: 2 9:34am

Sep 9, 2026

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
CH Cover		208.175	...	205	...
LHS Ch Flg		413.175	...	205	...
LHS Ch Shell		1613.18	...	1200	...
Main Shell Flan		2239.35	...	298	...
Main Shell		8239.35	...	6000	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flg Calc [Int P]: New Flange

Flng: 7 9:34am

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Flange Input Data Values

Description: New Flange :

CH Cover

Flange Type	Integral Weld Neck		
Design Pressure	P	1270.68	KPa.
Design Temperature		250	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1750.000	mm
Flange Outside Diameter	A	1990.000	mm
Flange Thickness	t	177.0000	mm
Thickness of Hub at Small End	go	14.0000	mm
Thickness of Hub at Large End	gl	21.0000	mm
Length of Hub	h	28.0000	mm
Flange Material	SA-336 F11		
Flange Material UNS number	K11572		
Flange Allowable Stress At Temperature	Sfo	148.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	148.00	N./mm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	1910.000	mm
Nominal Bolt Diameter	a	34.9250	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	56		
Flange Face Outside Diameter	Fod	1831.000	mm
Flange Face Inside Diameter	Fid	1750.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	1820.000	mm
Gasket Inside Diameter	Gi	1782.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	1781.9999	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



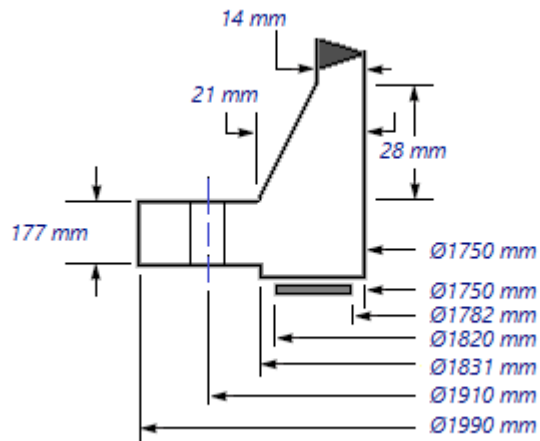
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Flg Calc [Int P]: New Flange

Flng: 7 9:34am Sep 9, 2026



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1271 \cdot (1750/2 + 3)) / (118 \cdot 1 - 0.6 \cdot 1271) + C_a$$

$$= 12.5170 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118 \cdot 1 \cdot 11) / (878 + 0.6 \cdot 11)$$

$$= 1467.218 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118 \cdot 1 \cdot 14) / (875 + 0.6 \cdot 14)$$

$$= 1869.905 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	174.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1756.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	18.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	11.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	59.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	19.000 mm
Basic Gasket Width, $b_o = N/2$	9.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	7.767 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1804.466 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1804^2 \cdot 1271$$

$$= 331375.875 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 7.77 \cdot \pi \cdot 1804 \cdot 3 \cdot 1271$$

$$+ 2 \cdot 1782 \cdot 5 \cdot 3 \cdot 1271$$

$$= 41159.266 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 1756^2 \cdot 1271$$

$$= 313814.094 \text{ Kgf}$$



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 Flg Calc [Int P]: New Flange Flng: 7 9:34am Sep 9,2026

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 331376 - 313814$$

$$= 17561.805 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(331376 + 41159 + 0, 0)$$

$$= 372535.156 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 7.77 \cdot \pi \cdot 1804 + 68.9 \cdot 5 \cdot 1782$$

$$= 372210.281 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((372535 + 0 + 4 \cdot 0 / 1804) / 172, 372210 / 172)$$

$$= 212.410 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 34.9 + 6 \cdot 174 / (3 + 0.5)$$

$$= 368.136 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1910 \cdot \sin(\pi / 56)$$

$$= 107.095 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 107 / (2 \cdot 34.9 + 174) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	212.410	417.289	
Radial Distance between Hub and Bolts:	47.625	59.000	
Radial Distance between Bolts and Edge:	34.925	40.000	
Circ. Spacing between the Bolts:	77.800	107.095	368.136

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 417 \cdot 172 / (68.9 \cdot \pi (1820 + 1782))$$

$$= 9.199 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172 \cdot 417$$

$$= 731862.31 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 372535 - 331376$$

$$= 41159.27 \text{ Kgf}$$



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 Flg Calc [Int P]: New Flange Flng: 7 9:34am Sep 9, 2026

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
 $= (C - G) / 2$
 $= (1910 - 1804) / 2$
 $= 52.7669 \text{ mm}$

Distance to Face Pressure Reaction [ht]:
 $= 0.5 ((C - B) / 2 + hg)$
 $= 0.5 ((1910 - 1756) / 2 + 52.8)$
 $= 64.8835 \text{ mm}$

Distance to End Pressure Reaction [hd]:
 $= (C - B - g1) / 2$
 $= (1910 - 1756 - 18) / 2$
 $= 68.0000 \text{ mm}$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	313814.	68.0000	1.0000	21340.
Face Pressure, Mt	17562.	64.8835	1.0000	1139.
Gasket Load, Mg	41159.	52.7669	1.0000	2172.
Gasket Seating, Matm	731862.	52.7669	1.0000	38619.

Total Moment for Operation, Mo 24651. Kg-m.
 Total Moment for Gasket seating, Mg 38619. Kg-m.

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_{oCor}}$ 138.982 mm
 Hub Ratio, $h/h_o = H_L / H_o$ 0.201
 Thickness Ratio, $g1/g_o = (g1_{cor}/g_{oCor})$ 1.636

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.898
 Factor V per 4.16.5 0.394
 Factor f per 4.16.5 1.774

Stress factor from Table 4.16.4 [K]:

$= A / B$
 $= 1990/1756$
 $= 1.133$

Stress factor from Table 4.16.4 [Y]:

$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$
 $= (0.66845 + (5.7169 (1.13^2 \cdot \log_{10}(1.13))) / (1.13^2 - 1)) / (1.13 - 1)$
 $= 15.546$

Stress factor from Table 4.16.4 [T]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 $= (1.13^2 (1 + 8.55246 \cdot \log_{10}(1.13)) - 1) / ((1.0472 + 1.9448 \cdot 1.13^2) \cdot (1.13 - 1))$
 $= 1.865$

Stress factor from Table 4.16.4 [U]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$
 $= (1.13^2 (1 + 8.55246 \cdot \log_{10}(1.13)) - 1) / (1.36136 (1.13^2 - 1) \cdot (1.13 - 1))$
 $= 17.084$

Stress factor from Table 4.16.4 [Z]:

$= (K^2 + 1) / (K^2 - 1)$
 $= (1.13^2 + 1) / (1.13^2 - 1)$
 $= 8.036$

Stress factor from Table 4.16.4 for integral flanges [d]:

$= U \cdot g_o^2 \cdot h_o / V$



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Flg Calc [Int P]: New Flange

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$$= 17.1 \cdot 11^2 \cdot 139 / 0.39$$

$$= 728601.000 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.9 / 139$$

$$= 0.006 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3 / d$$

$$= (174 \cdot 0.0065 + 1) / 1.87 + 174^3 / 728601$$

$$= 8.369$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g1^2)$$

$$= (1.77 \cdot 24651 / 1756) / (8.37 \cdot 18^2)$$

$$= 90.07 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g1^2)$$

$$= (1.77 \cdot 38619 / 1756) / (8.37 \cdot 18^2)$$

$$= 141.10 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.5 \cdot 24651 / 1756) / (8.37 \cdot 174^2)$$

$$= 1.36 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.5 \cdot 38619 / 1756) / (8.37 \cdot 174^2)$$

$$= 2.13 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (15.5 \cdot 24651 / (174^2 \cdot 1756)) - 8.04 \cdot 1.36$$

$$= 59.78 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (15.5 \cdot 38619 / (174^2 \cdot 1756)) - 8.04 \cdot 2.13$$

$$= 93.65 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (90.1 + \max(1.36, 59.8)) / 2$$

$$= 74.92 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (141 + \max(2.13, 93.7)) / 2$$

$$= 117.38 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 372535 / 417$$

$$= 87.55 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (372210 / 417)$$

$$= 87.47 \text{ N./mm}^2$$



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Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	90.07	222.00	141.10	222.00
Radial Flange	1.36	148.00	2.13	148.00
Tangential Flange	59.78	148.00	93.65	148.00
Maximum Average	74.92	148.00	117.38	148.00
Bolting	87.55	172.00	87.47	172.00

Minimum Required Flange Thickness [Rigidity] 170.536 mm

Estimated M.A.W.P. (Operating) 1467 KPa.
 Estimated Finished Weight of Flange at given Thk. 988.2 kg.
 Estimated Unfinished Weight of Forging at given Thk 1120.1 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 38619 / 1 \cdot 9807 \cdot 0.39 / (7.57 \cdot 204092 \cdot 11^2 \cdot 139 \cdot 0.3)$
 $= 0.999$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 24651 / 1 \cdot 9807 \cdot 0.39 / (7.57 \cdot 189669 \cdot 11^2 \cdot 139 \cdot 0.3)$
 $= 0.686$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 38619 / 1 \cdot 9807 \cdot 0.39 / (8.37 \cdot 204092 \cdot 11^2 \cdot 139 \cdot 0.3)$
 $= 0.903$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 24651 / 1 \cdot 9807 \cdot 0.39 / (8.37 \cdot 189669 \cdot 11^2 \cdot 139 \cdot 0.3)$
 $= 0.620$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.77, Temperature Reduction per Fig. UCS 66.1 = 13 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -31 °C

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	88.47	222.00	135.89	222.00
Radial Flange	1.87	148.00	2.87	148.00
Tangential Flange	53.48	148.00	82.15	148.00
Maximum Average	70.97	148.00	109.02	148.00
Bolting	87.55	172.00	87.47	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 160.376 mm



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Flange Input Data Values

Description: LHS Ch Flg :

LHS Ch Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	1267.45	KPa.
Design Temperature		250	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1750.000	mm
Flange Outside Diameter	A	1990.000	mm
Flange Thickness	t	177.0000	mm
Thickness of Hub at Small End	go	14.0000	mm
Thickness of Hub at Large End	gl	21.0000	mm
Length of Hub	h	28.0000	mm
Flange Material	SA-336 F11		
Flange Material UNS number	K11572		
Flange Allowable Stress At Temperature	Sfo	148.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	148.00	N./mm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	1910.000	mm
Nominal Bolt Diameter	a	34.9250	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	56		
Flange Face Outside Diameter	Fod	1831.000	mm
Flange Face Inside Diameter	Fid	1750.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	1820.000	mm
Gasket Inside Diameter	Gi	1782.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	1781.9999	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



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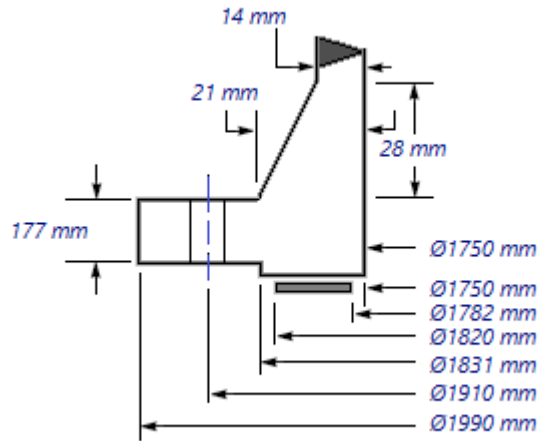
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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1267 \cdot (1750/2 + 3)) / (118 \cdot 1 - 0.6 \cdot 1267) + C_a$$

$$= 12.4926 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118 \cdot 1 \cdot 11) / (878 + 0.6 \cdot 11)$$

$$= 1467.218 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118 \cdot 1 \cdot 14) / (875 + 0.6 \cdot 14)$$

$$= 1869.905 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	174.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1756.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	18.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	11.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	59.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	19.000 mm
Basic Gasket Width, $b_o = N/2$	9.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	7.767 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1804.466 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1804^2 \cdot 1267$$

$$= 330532.344 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{part} \cdot m_{part} \cdot P$$

$$= 2 \cdot 7.77 \cdot \pi \cdot 1804 \cdot 3 \cdot 1267$$

$$+ 2 \cdot 1782 \cdot 5 \cdot 3 \cdot 1267$$

$$= 41054.492 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 1756^2 \cdot 1267$$

$$= 313015.250 \text{ Kgf}$$



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Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 330532 - 313015$$

$$= 17517.096 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(330532 + 41054 + 0, 0)$$

$$= 371586.812 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 7.77 \cdot \pi \cdot 1804 + 68.9 \cdot 5 \cdot 1782$$

$$= 372210.281 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((371587 + 0 + 4 \cdot 0 / 1804) / 172, 372210 / 172)$$

$$= 212.225 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 34.9 + 6 \cdot 174 / (3 + 0.5)$$

$$= 368.136 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1910 \cdot \sin(\pi / 56)$$

$$= 107.095 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 107 / (2 \cdot 34.9 + 174) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	212.225	417.289	
Radial Distance between Hub and Bolts:	47.625	59.000	
Radial Distance between Bolts and Edge:	34.925	40.000	
Circ. Spacing between the Bolts:	77.800	107.095	368.136

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 417 \cdot 172 / (68.9 \cdot \pi (1820 + 1782))$$

$$= 9.199 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172 \cdot 417$$

$$= 731862.31 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 371587 - 330532$$

$$= 41054.48 \text{ Kgf}$$



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Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
 $= (C - G) / 2$
 $= (1910 - 1804) / 2$
 $= 52.7669 \text{ mm}$

Distance to Face Pressure Reaction [ht]:
 $= 0.5 ((C - B) / 2 + hg)$
 $= 0.5 ((1910 - 1756) / 2 + 52.8)$
 $= 64.8835 \text{ mm}$

Distance to End Pressure Reaction [hd]:
 $= (C - B - g1) / 2$
 $= (1910 - 1756 - 18) / 2$
 $= 68.0000 \text{ mm}$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	313015.	68.0000	1.0000	21285.
Face Pressure, Mt	17517.	64.8835	1.0000	1137.
Gasket Load, Mg	41054.	52.7669	1.0000	2166.
Gasket Seating, Matm	731862.	52.7669	1.0000	38619.

Total Moment for Operation, Mo 24588. Kg-m.
 Total Moment for Gasket seating, Mg 38619. Kg-m.

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_{oCor}}$ 138.982 mm
 Hub Ratio, $h/h_o = H_L / H_o$ 0.201
 Thickness Ratio, $g1/g_o = (g1_{cor}/g_{oCor})$ 1.636

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.898
 Factor V per 4.16.5 0.394
 Factor f per 4.16.5 1.774

Stress factor from Table 4.16.4 [K]:

$= A / B$
 $= 1990/1756$
 $= 1.133$

Stress factor from Table 4.16.4 [Y]:

$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$
 $= (0.66845 + (5.7169 (1.13^2 \cdot \log_{10}(1.13))) / (1.13^2 - 1)) / (1.13 - 1)$
 $= 15.546$

Stress factor from Table 4.16.4 [T]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 $= (1.13^2 (1 + 8.55246 \cdot \log_{10}(1.13)) - 1) / ((1.0472 + 1.9448 \cdot 1.13^2) \cdot (1.13 - 1))$
 $= 1.865$

Stress factor from Table 4.16.4 [U]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$
 $= (1.13^2 (1 + 8.55246 \cdot \log_{10}(1.13)) - 1) / (1.36136 (1.13^2 - 1) \cdot (1.13 - 1))$
 $= 17.084$

Stress factor from Table 4.16.4 [Z]:

$= (K^2 + 1) / (K^2 - 1)$
 $= (1.13^2 + 1) / (1.13^2 - 1)$
 $= 8.036$

Stress factor from Table 4.16.4 for integral flanges [d]:

$= U \cdot g_o^2 \cdot h_o / V$



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$$= 17.1 \cdot 11^2 \cdot 139 / 0.39$$

$$= 728601.000 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.9 / 139$$

$$= 0.006 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3 / d$$

$$= (174 \cdot 0.0065 + 1) / 1.87 + 174^3 / 728601$$

$$= 8.369$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g1^2)$$

$$= (1.77 \cdot 24588 / 1756) / (8.37 \cdot 18^2)$$

$$= 89.84 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g1^2)$$

$$= (1.77 \cdot 38619 / 1756) / (8.37 \cdot 18^2)$$

$$= 141.10 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.5 \cdot 24588 / 1756) / (8.37 \cdot 174^2)$$

$$= 1.35 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.5 \cdot 38619 / 1756) / (8.37 \cdot 174^2)$$

$$= 2.13 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (15.5 \cdot 24588 / (174^2 \cdot 1756)) - 8.04 \cdot 1.35$$

$$= 59.63 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (15.5 \cdot 38619 / (174^2 \cdot 1756)) - 8.04 \cdot 2.13$$

$$= 93.65 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (89.8 + \max(1.35, 59.6)) / 2$$

$$= 74.73 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (141 + \max(2.13, 93.7)) / 2$$

$$= 117.38 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 371587 / 417$$

$$= 87.33 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (372210 / 417)$$

$$= 87.47 \text{ N./mm}^2$$



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Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	89.84	222.00	141.10	222.00
Radial Flange	1.35	148.00	2.13	148.00
Tangential Flange	59.63	148.00	93.65	148.00
Maximum Average	74.73	148.00	117.38	148.00
Bolting	87.33	172.00	87.47	172.00

Minimum Required Flange Thickness [Rigidity] 170.536 mm

Estimated M.A.W.P. (Operating) 1467 KPa.
 Estimated Finished Weight of Flange at given Thk. 988.2 kg.
 Estimated Unfinished Weight of Forging at given Thk 1120.1 kg.

Flange Stress results per WRC-538 at computed Sfmax (N./mm²):

	Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub Sh	578.29	578.29	1.000	578.29	619.99	0.933
Tangential Flange St	385.34	433.71	0.888	385.34	464.99	0.829
Tangential Flange Sto	108.53	289.14	0.375	108.53	309.99	0.350
Combined Sh/f + St	712.60	867.43	0.822	712.60	929.98	0.766
Combined Sr + St	394.09	867.43	0.454	394.09	929.98	0.424
Combined Sh + Sto	686.82	867.43	0.792	686.82	929.98	0.739
Radial Flange Sr	8.75	578.29	0.015	8.75	619.99	0.014
Maximum Stress Ratio			1.000			0.933



Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)
 = max(1; 1.89/1.07)
 = 1.7671

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if h/ho < 0.35, else max (0.25; fn3)
 = max(0.25; 0.31/0.93) if h/ho < 0.35, else max (0.25; 0.31/0.79)
 = 0.3316

In this case, the hub ratio h/ho, 0.2015 was < 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	707.7 N./mm ²
Target assembly gasket stress	SgT	208.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700



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Root Area of a Single Bolt	Abs	7.4516 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	310.0 N./mm ²
Flange Material Operating Yield Stress	Syo	253.0 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.611 deg

Gasket Contact Area [Ag]:
 $= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$
 $= \pi/4 (1820^2 - 1782^2) + (0.5 \cdot 1782 \cdot 5)$
 $= 1119.6 \text{ cm}^2$

Target Bolt Stress [Sbsel]: (Eq. O-1)
 $= SgT \cdot Ag / (nb \cdot Ab)$
 $= 208 \cdot 1120 / (56 \cdot 7.45)$
 $= 558.1 \text{ N./mm}^2$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)
 $= \min(Sbsel; Sbmax)$
 $= \min(558; 508)$
 $= 507.5 \text{ N./mm}^2$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)
 $= \max(Sbsel; Sbmin)$
 $= \max(508; 290)$
 $= 507.5 \text{ N./mm}^2$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)
 $= \min(Sbsel; Sfmax)$
 $= \min(508; 708)$
 $= 507.5 \text{ N./mm}^2$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)
 $:: Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$
 $:: 508 \geq 140 (1120 / (56 \cdot 7.45))$
 $:: 507.5 \geq 375.6 \text{ N./mm}^2$ Check Passes

Check if Gasket Operating Stress is maintained. (Eq. O-8)
 $:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$
 $:: 508 \geq (97 \cdot 1120 + \pi/4 \cdot 1267 \cdot 1782^2) / (0.7 \cdot 7.45 \cdot 56)$
 $:: 507.5 \geq 480.0 \text{ N./mm}^2$ Check Passes

Check if Gasket Maximum Stress is achieved. (Eq. O-9)
 $:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$
 $:: 508 \leq 276 (1120 / (56 \cdot 7.45))$
 $:: 507.5 \leq 740.5 \text{ N./mm}^2$ Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)
 $:: Sbsel \leq Sfmax (thetagmax / thetafmax)$
 $:: 508 \leq 708 (1/0.61)$
 $:: 507.5 \leq 1157.6 \text{ N./mm}^2$ Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)
 $= Sbsel \cdot K \cdot Abs \cdot \phi b$
 $= 508 \cdot 0.2 \cdot 7.45 \cdot 34.9$
 $= 269.4 \text{ Kg-m.}$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 89 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since $Sb \geq SbB8$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 38619 / 1 \cdot 9807 \cdot 0.39 / (7.57 \cdot 204092 \cdot 11^2 \cdot 139 \cdot 0.3)$
 $= 0.999$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 24588 / 1 \cdot 9807 \cdot 0.39 / (7.57 \cdot 189669 \cdot 11^2 \cdot 139 \cdot 0.3)$
 $= 0.684$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 38619 / 1 \cdot 9807 \cdot 0.39 / (8.37 \cdot 204092 \cdot 11^2 \cdot 139 \cdot 0.3)$
 $= 0.903$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 24588 / 1 \cdot 9807 \cdot 0.39 / (8.37 \cdot 189669 \cdot 11^2 \cdot 139 \cdot 0.3)$
 $= 0.619$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.77, Temperature Reduction per Fig. UCS 66.1 = 13 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -31 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket	Seating
			Actual	Allowed
Longitudinal Hub	88.24	222.00	135.89	222.00
Radial Flange	1.86	148.00	2.87	148.00
Tangential Flange	53.34	148.00	82.15	148.00
Maximum Average	70.79	148.00	109.02	148.00
Bolting	87.33	172.00	87.47	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 160.376 mm

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Flange Input Data Values

Description: New Flange :

Main Shell Flange

Flange Type	Integral Weld Neck		
Design Pressure	P	4950.00	KPa.
Design Temperature		275	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1750.000	mm
Flange Outside Diameter	A	2240.000	mm
Flange Thickness	t	232.0000	mm
Thickness of Hub at Small End	go	44.0000	mm
Thickness of Hub at Large End	gl	55.0000	mm
Length of Hub	h	66.0000	mm
Flange Material	SA-336 F11		
Flange Material UNS number	K11572		
Flange Allowable Stress At Temperature	Sfo	148.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	148.00	N./mm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	159.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00	N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Diameter of Bolt Circle	C	2070.000	mm
Nominal Bolt Diameter	a	76.2000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		28	
Flange Face Outside Diameter	Fod	1883.000	mm
Flange Face Inside Diameter	Fid	1750.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	1872.000	mm
Gasket Inside Diameter	Gi	1782.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	1800.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



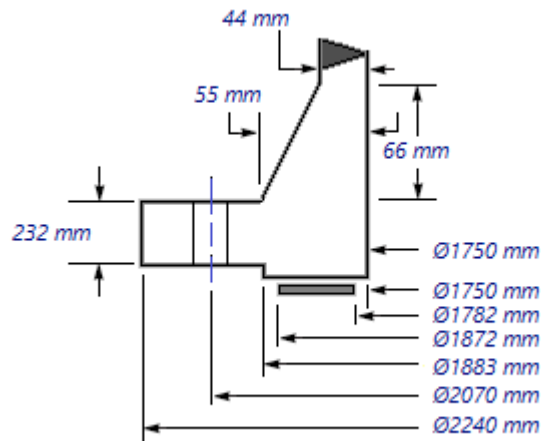
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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (4950 \cdot (1750/2 + 3)) / (118 \cdot 1 - 0.6 \cdot 4950) + C_a$$

$$= 40.7853 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118 \cdot 1 \cdot 41) / (878 + 0.6 \cdot 41)$$

$$= 5359.663 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118 \cdot 1 \cdot 44) / (875 + 0.6 \cdot 44)$$

$$= 5759.491 \text{ KPa.}$$

The hub small end thickness is now taken as the attached shell thickness.

Corroded Flange Thickness, $t_c = T - c_i$	229.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1756.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	52.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	39.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	105.000 mm
Gasket Contact Width, $N = (G_o - G_i)/2$	45.000 mm
Basic Gasket Width, $b_o = N/2$	22.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	11.953 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1848.094 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1848^2 \cdot 4950$$

$$= 1354065.750 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{part} \cdot m_{part} \cdot P$$

$$= 2 \cdot 12 \cdot \pi \cdot 1848 \cdot 3 \cdot 4950$$

$$+ 2 \cdot 1800 \cdot 5 \cdot 3 \cdot 4950$$

$$= 237444.859 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$



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$$= \pi/4 \cdot 1756^2 \cdot 4950$$

$$= 1222476.750 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 1354065 - 1222476$$

$$= 131588.906 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(1354065 + 237445 + 0, 0)$$

$$= 1591510.500 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 12 \cdot \pi \cdot 1848 + 68.9 \cdot 5 \cdot 1800$$

$$= 551207.812 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((1591510 + 0 + 4 \cdot 0 / 1848) / 159, 551208 / 159)$$

$$= 981.632 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 76.2 + 6 \cdot 229 / (3 + 0.5)$$

$$= 544.971 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2070 \cdot \sin(\pi / 28)$$

$$= 231.766 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 232 / (2 \cdot 76.2 + 229) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	981.632	1142.398	
Radial Distance between Hub and Bolts:	92.075	105.000	
Radial Distance between Bolts and Edge:	73.025	85.000	
Circ. Spacing between the Bolts:	158.750	231.766	544.971

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 1142 \cdot 159 / (68.9 \cdot \pi (1872 + 1782))$$

$$= 22.948 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 159 \cdot 1142$$

$$= 1852157.50 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 1591510 - 1354065$$



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= 237444.86 Kgf

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
 $= (C - G) / 2$
 $= (2070 - 1848) / 2$
 $= 110.9530 \text{ mm}$

Distance to Face Pressure Reaction [ht]:
 $= 0.5 ((C - B) / 2 + hg)$
 $= 0.5 ((2070 - 1756) / 2 + 111)$
 $= 133.9765 \text{ mm}$

Distance to End Pressure Reaction [hd]:
 $= (C - B - g1) / 2$
 $= (2070 - 1756 - 52) / 2$
 $= 131.0000 \text{ mm}$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	1222477.	131.0000	1.0000	160148.
Face Pressure, Mt	131589.	133.9765	1.0000	17630.
Gasket Load, Mg	237445.	110.9530	1.0000	26346.
Gasket Seating, Matm	1852158.	110.9530	1.0000	205507.

Total Moment for Operation, Mo 204124. Kg-m.
 Total Moment for Gasket seating, Mg 205507. Kg-m.

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_{oCor}}$ 261.694 mm
 Hub Ratio, $h/h_o = H_L / H_o$ 0.252
 Thickness Ratio, $g1/g_o = (g1_{cor}/g_{oCor})$ 1.333

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.896
 Factor V per 4.16.5 0.423
 Factor f per 4.16.5 1.051

Stress factor from Table 4.16.4 [K]:

$= A / B$
 $= 2240 / 1756$
 $= 1.276$

Stress factor from Table 4.16.4 [Y]:

$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$
 $= (0.66845 + (5.7169 (1.28^2 \cdot \log_{10}(1.28))) / (1.28^2 - 1)) / (1.28 - 1)$
 $= 8.114$

Stress factor from Table 4.16.4 [T]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 $= (1.28^2 (1 + 8.55246 \cdot \log_{10}(1.28)) - 1) / ((1.0472 + 1.9448 \cdot 1.28^2) \cdot (1.28 - 1))$
 $= 1.808$

Stress factor from Table 4.16.4 [U]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$
 $= (1.28^2 (1 + 8.55246 \cdot \log_{10}(1.28)) - 1) / (1.36136 (1.28^2 - 1) \cdot (1.28 - 1))$
 $= 8.917$

Stress factor from Table 4.16.4 [Z]:

$= (K^2 + 1) / (K^2 - 1)$
 $= (1.28^2 + 1) / (1.28^2 - 1)$
 $= 4.189$



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Stress factor from Table 4.16.4 for integral flanges [d]:

$$\begin{aligned}
 &= U \cdot g_0^2 \cdot h_0 / V \\
 &= 8.92 \cdot 39^2 \cdot 262 / 0.42 \\
 &= 8390490.000 \text{ mm}^3
 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned}
 &= F / h_0 \\
 &= 0.9 / 262 \\
 &= 0.003 \text{ 1/mm}
 \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned}
 &= (t \cdot e + 1) / T + t^3 / d \\
 &= (229 \cdot 0.0034 + 1) / 1.81 + 229^3 / 8390490 \\
 &= 2.418
 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned}
 &= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\
 &= (1.05 \cdot 204124 / 1756) / (2.42 \cdot 52^2) \\
 &= 183.19 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned}
 &= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\
 &= (1.05 \cdot 205507 / 1756) / (2.42 \cdot 52^2) \\
 &= 184.43 \text{ N./mm}^2
 \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned}
 &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\
 &= (2.05 \cdot 204124 / 1756) / (2.42 \cdot 229^2) \\
 &= 18.39 \text{ N./mm}^2
 \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned}
 &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\
 &= (2.05 \cdot 205507 / 1756) / (2.42 \cdot 229^2) \\
 &= 18.51 \text{ N./mm}^2
 \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned}
 &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\
 &= (8.11 \cdot 204124 / (229^2 \cdot 1756)) - 4.19 \cdot 18.4 \\
 &= 99.36 \text{ N./mm}^2
 \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$\begin{aligned}
 &= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa \\
 &= (8.11 \cdot 205507 / (229^2 \cdot 1756)) - 4.19 \cdot 18.5 \\
 &= 100.03 \text{ N./mm}^2
 \end{aligned}$$

Average Flange Stress, Operating [SAo]:

$$\begin{aligned}
 &= (SHo + \max(SRo, STo)) / 2 \\
 &= (183 + \max(18.4, 99.4)) / 2 \\
 &= 141.27 \text{ N./mm}^2
 \end{aligned}$$

Average Flange Stress, Seating [SAa]:

$$\begin{aligned}
 &= (SHa + \max(SRa, STa)) / 2 \\
 &= (184 + \max(18.5, 100)) / 2 \\
 &= 142.23 \text{ N./mm}^2
 \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned}
 &= Wm1 / Ab \\
 &= 1591510 / 1142 \\
 &= 136.62 \text{ N./mm}^2
 \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned}
 &= (Wm2 / Ab) \\
 &= (551208 / 1142)
 \end{aligned}$$



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= 47.32 N./mm²

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	183.19	222.00	184.43	222.00
Radial Flange	18.39	148.00	18.51	148.00
Tangential Flange	99.36	148.00	100.03	148.00
Maximum Average	141.27	148.00	142.23	148.00
Bolting	136.62	159.00	47.32	159.00

Minimum Required Flange Thickness 226.263 mm

Estimated M.A.W.P. (Operating) 5186 KPa.
 Estimated Finished Weight of Flange at given Thk. 2901.2 kg.
 Estimated Unfinished Weight of Forging at given Thk 3546.5 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	557.72	569.14	0.980	557.72	619.99	0.900
Tangential Flange	St	309.93	426.86	0.726	309.93	464.99	0.667
Tangential Flange	Sto	138.48	284.57	0.487	138.48	309.99	0.447
Combined	Sh/f + St	853.71	853.71	1.000	853.71	929.98	0.918
Combined	Sr + St	367.30	853.71	0.430	367.30	929.98	0.395
Combined	Sh + Sto	696.20	853.71	0.816	696.20	929.98	0.749
Radial Flange	Sr	57.37	569.14	0.101	57.37	619.99	0.093
Maximum Stress Ratio				1.000			0.918



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Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.21/1.18)

= 1.0256

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.28/1.11) if $h/h0 < 0.35$, else max (0.25; 0.22/0.79)

= 0.2547

In this case, the hub ratio $h/h0$, 0.2522 was < 0.35 .

PV Elite iterates on the assembly bolt load stress, Sf_{max} , until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sb_{max}	507.5 N./mm ²
Minimum permissible bolt stress	Sb_{min}	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sf_{max}	492.6 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sg_{max}	276.0 N./mm ²
Minimum gasket operating stress	Sg_{min-O}	97.0 N./mm ²
Minimum gasket seating stress	Sg_{min-S}	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	$Phig$	0.700



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 Flg Calc [Int P]: New Flange Flng: 9 9:34am Sep 9,2026

Root Area of a Single Bolt	Abs	40.7999 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	310.0 N./mm ²
Flange Material Operating Yield Stress	Syo	249.0 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.519 deg

Gasket Contact Area [Ag]:
 $= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$
 $= \pi/4 (1872^2 - 1782^2) + (0.5 \cdot 1800 \cdot 5)$
 $= 2627.9 \text{ cm}^2$

Target Bolt Stress [Sbsel]: (Eq. O-1)
 $= SgT \cdot Ag / (nb \cdot Ab)$
 $= 207 \cdot 2628 / (28 \cdot 40.8)$
 $= 476.2 \text{ N./mm}^2$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)
 $= \min(Sbsel; Sbmax)$
 $= \min(476; 508)$
 $= 476.2 \text{ N./mm}^2$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)
 $= \max(Sbsel; Sbmin)$
 $= \max(476; 290)$
 $= 476.2 \text{ N./mm}^2$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)
 $= \min(Sbsel; Sfmax)$
 $= \min(476; 493)$
 $= 476.2 \text{ N./mm}^2$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)
 $:: Sbsel \geq Sgmin \cdot S(Ag / (nb \cdot Abs))$
 $:: 476 \geq 140 (2628 / (28 \cdot 40.8))$
 $:: 476.2 \geq 322.0 \text{ N./mm}^2$ Check Passes

Check if Gasket Operating Stress is maintained. (Eq. O-8)
 $:: Sbsel \geq (Sgmin \cdot O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$
 $:: 476 \geq (97 \cdot 2628 + \pi/4 \cdot 4950 \cdot 1782^2) / (0.7 \cdot 40.8 \cdot 28)$
 $:: 476.2 \geq 473.1 \text{ N./mm}^2$ Check Passes

Check if Gasket Maximum Stress is achieved. (Eq. O-9)
 $:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$
 $:: 476 \leq 276 (2628 / (28 \cdot 40.8))$
 $:: 476.2 \leq 634.9 \text{ N./mm}^2$ Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)
 $:: Sbsel \leq Sfmax(thetagmax / thetafmax)$
 $:: 476 \leq 493 (1/0.52)$
 $:: 476.2 \leq 948.5 \text{ N./mm}^2$ Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)
 $= Sbsel \cdot K \cdot Abs \cdot \phi b$
 $= 476 \cdot 0.2 \cdot 40.8 \cdot 76.2$
 $= 3019.1 \text{ Kg-m.}$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 87 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 158 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 205507 / 1 \cdot 9807 \cdot 0.42 / (2.3 \cdot 204092 \cdot 39^2 \cdot 262 \cdot 0.3)$
 $= 0.792$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 204124 / 1 \cdot 9807 \cdot 0.42 / (2.3 \cdot 187994 \cdot 39^2 \cdot 262 \cdot 0.3)$
 $= 0.854$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 205507 / 1 \cdot 9807 \cdot 0.42 / (2.42 \cdot 204092 \cdot 39^2 \cdot 262 \cdot 0.3)$
 $= 0.754$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 204124 / 1 \cdot 9807 \cdot 0.42 / (2.42 \cdot 187994 \cdot 39^2 \cdot 262 \cdot 0.3)$
 $= 0.813$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.92, Temperature Reduction per Fig. UCS 66.1 = 4 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 9 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	175.23	222.00	174.66 222.00
Radial Flange	19.23	148.00	19.17 148.00
Tangential Flange	92.36	148.00	92.05 148.00
Maximum Average	133.79	148.00	133.36 148.00
Bolting	136.62	159.00	47.32 159.00

Minimum Req'd Thickness, ambient, no Corrosion: 216.586 mm

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 Internal Pressure Calculations: Step: 6 9:34am Sep 9,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
CH Cover		1270.7	185	3	1750	148	1
LHS Ch Flg		1267.4	185	3	1750	148	2
LHS Ch Shell		1267.2	14	3	1750	118	3
Main Shell Flang		4950	240	3	1750	148	4
Main Shell		4967.2	42	3	1750	118	5

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
CH Cover		1250	1447	1870	177	170.536	1
LHS Ch Flg		1250	1450	1870	177	170.536	2
LHS Ch Shell		1250	1450	1870	14	12.4906	3
Main Shell Flang		4950	5186	5475	232	226.263	4
Main Shell		4950	5088	5505	42	40.9198	5

Warning: Channel Side MAWP governed by a standard Flange!

Warning: Tube side flange MAWP was less than the design pressure!

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:
 For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Note:
 The MAWPs and MAPs for the Exchanger elements all appear to be zero or some could not be calculated. Please fill these values in on the main tubesheet tab.

Elements Suitable for Design Internal Pressure.

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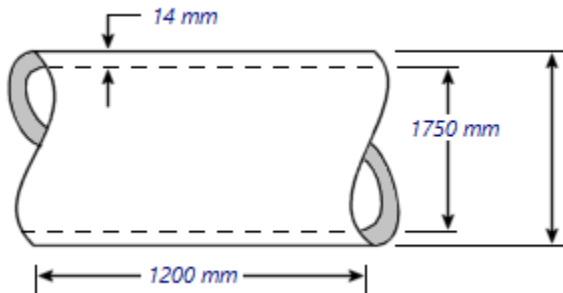
Element: LHS Ch Shell

Step: 7 9:34am

Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: LHS Ch Shell

Cylindrical Shell Design Information:			
Design Pressure P	1267.183 KPa.	Int. Design Temperature	250 °C
Material	SA-387 11	Ext. Design Temperature	200 °C
Length	1200.000 mm	External Pressure Chart	CS-2
		UNS Number	K11789
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	118.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	118.0 N./mm ²
Inside Diameter Di	1750.000 mm	Actual Stress	101.9 N./mm ²
Spec. Min. Thk t,e, or sa	14.0000 mm	Spec. Nominal Thk.	14.000 mm
Surface Area	67029.0 cm ²	SG of Contents	1.000
Element Volume	2886855.5 cm ³	Weight of Contents	2885.1 Kg
Empty Weight	721.6 Kg	Operating Weight	3606.7 Kg
		Comp. All. Ope. Stress	92.7 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1267 \cdot 878) / (118 \cdot 1 - 0.6 \cdot 1267) \\
 &= 9.4906 + 3.0000 = 12.4906 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 17.183 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (118 \cdot 1 \cdot 11) / (878 + 0.6 \cdot 11) \\
 &= 1467 - 17.2 = 1450 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (118 \cdot 1 \cdot 14) / (875 + 0.6 \cdot 14) \\
 &= 1870 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1267 \cdot (878 + 0.6 \cdot 11)) / (1 \cdot 11) \\
 &= 101.910 \text{ N./mm}^2
 \end{aligned}$$



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 Element: LHS Ch Shell Step: 7 9:34am Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.794 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 14$, $t_r = 10.8$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.99$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 2 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 1 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	1446.536 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	14.000 mm	Basic MDMT	2 °C
Stress/Thk. Ratio	0.986	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	1 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: LHS Ch Shell

Cylindrical Shell From 30 to 40 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193602528 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	1778.00	1200.00	161.64	0.6749	0.0009974	77.60

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 77.6) / (3 \cdot 162) = 640 \text{ KPa.}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.889	1778.00	1200.00	363.65	0.6749	0.0002914	28.21

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 28.2) / (3 \cdot 364) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	1778.00	8262.13	161.64	4.6469	0.0001296	12.54

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 12.5) / (3 \cdot 162) = 103 \text{ KPa.}$$



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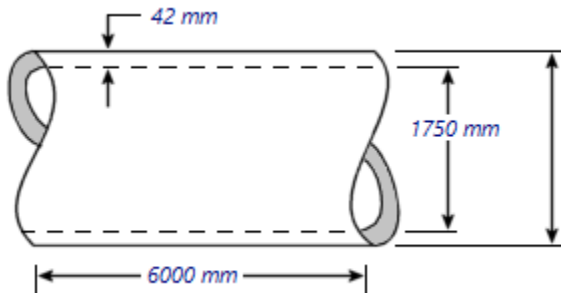
Element: Main Shell

Step: 8 9:34am

Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:			
Design Pressure P	4967.183 KPa.	Int. Design Temperature	275 °C
Material	SA-387 11	Ext. Design Temperature	275 °C
Length	6000.000 mm	External Pressure Chart	CS-2
		UNS Number	K11789
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	118.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	118.0 N./mm ²
Inside Diameter Di	1750.000 mm	Actual Stress	114.8 N./mm ²
Spec. Min. Thk t,e, or sa	42.0000 mm	Spec. Nominal Thk.	42.000 mm
Surface Area	345700.8 cm ²	SG of Contents	1.000
Element Volume	13395702.0 cm ³	Weight of Contents	13387.5 Kg
Empty Weight	10995.6 Kg	Operating Weight	24383.1 Kg
		Comp. All. Ope. Stress	109.3 N./mm ²
Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (4967 \cdot 878) / (118 \cdot 1 - 0.6 \cdot 4967)$$

$$= 37.9198 + 3.0000 = 40.9198 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 17.183 KPa.

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118 \cdot 1 \cdot 39) / (878 + 0.6 \cdot 39)$$

$$= 5105 - 17.2 = 5088 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118 \cdot 1 \cdot 42) / (875 + 0.6 \cdot 42)$$

$$= 5505 \text{ KPa.}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (4967 \cdot (878 + 0.6 \cdot 39)) / (1 \cdot 39)$$

$$= 114.812 \text{ N./mm}^2$$



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Element: Main Shell

Step: 8

9:34am

Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f (1 - R_f / R_o)$) 2.344 %

Minimum Design Metal Temperature Results:

Govern. thk, $t_g = 42$, $t_r = 38.5$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.99$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 32 °C

Min Metal Temp. at Required thickness (UCS 66.1) 32 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	5042.231 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	42.000 mm	Basic MDMT	32 °C
Stress/Thk. Ratio	0.987	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	32 °C
		Joint Efficiency E^*	100 %

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 Shop Pressure Test Results: Step: 9 9:34am Sep 9,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6554.900 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	6435.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	7100.959 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	5546.454 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	7078.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	7078.500 KPa.
App 27-4	= M.A.W.P.	5042.231 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	***** KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1625.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2413.724 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	***** KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1787.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1787.500 KPa.
App 27-4	= M.A.W.P.	0.000 KPa.

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Ch Shell	195.5	216.0	0.905	2430.88
Main Shell	164.5	216.0	0.762	7118.11

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3-Tube Inlet/L	Nozzle	148.00	148.00	1.000
Noz N2 Fr30	Nozzle	148.00	148.00	1.000
3" Nozzle	Nozzle	148.00	148.00	1.000
Noz N1 750	Nozzle	148.00	148.00	1.000
K7-Temperature	Nozzle	118.00	118.00	1.000
M1-Manway	Nozzle	148.00	148.00	1.000
Noz N4 750	Nozzle	148.00	148.00	1.000
D4-Shell side d	Nozzle	118.00	118.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
CH Cover	148.00	148.00	1.000
LHS Ch Flg	148.00	148.00	1.000
LHS Ch Shell	118.00	118.00	1.000
Main Shell Flange	148.00	148.00	1.000
Main Shell	118.00	118.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):



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Shop Pressure Test Results:

Step: 9 9:34am

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Description	Ambient	Operating	Ratio
Tube Material	118.00	110.50	1.068
Tubesheet Material	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N3-Tube Inlet/L	26.27	278.99	0.094
Noz N2 Fr30	26.27	278.99	0.094
3" Nozzle	7.14	278.99	0.026
Noz N1 750	205.78	278.99	0.738
K7-Temperature	23.89	216.00	0.111
M1-Manway	180.77	278.99	0.648
Noz N4 750	205.78	278.99	0.738
D4-Shell side d	23.89	216.00	0.111

Elements Suitable for Test Pressure.

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 Field Pressure Test Results: Step: 10 9:34am Sep 9,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6554.900 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	6435.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	7100.959 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	5546.454 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	7078.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	7078.500 KPa.
App 27-4	= M.A.W.P.	5042.231 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	***** KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1625.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2413.724 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	***** KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1787.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1787.500 KPa.
App 27-4	= M.A.W.P.	0.000 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 4950 • 1
 = 6435.000 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 1250 • 1
 = 1625.000 KPa.

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Ch Shell	132.1	216.0	0.611	1642.15
Main Shell	149.1	216.0	0.690	6452.15

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3-Tube Inlet/L	Nozzle	148.00	148.00	1.000
Noz N2 Fr30	Nozzle	148.00	148.00	1.000
3" Nozzle	Nozzle	148.00	148.00	1.000
Noz N1 750	Nozzle	148.00	148.00	1.000
K7-Temperature	Nozzle	118.00	118.00	1.000
M1-Manway	Nozzle	148.00	148.00	1.000
Noz N4 750	Nozzle	148.00	148.00	1.000
D4-Shell side d	Nozzle	118.00	118.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
-------------	---------	-----------	-------



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Field Pressure Test Results:

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CH Cover	148.00	148.00	1.000
LHS Ch Flg	148.00	148.00	1.000
LHS Ch Shell	118.00	118.00	1.000
Main Shell Flange	148.00	148.00	1.000
Main Shell	118.00	118.00	1.000

Minimum

1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	118.00	110.50	1.068
Tubesheet Material	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N3-Tube Inlet/L	17.75	278.99	0.064
Noz N2 Fr30	17.75	278.99	0.064
3" Nozzle	4.83	278.99	0.017
Noz N1 750	186.53	278.99	0.669
K7-Temperature	21.66	216.00	0.100
M1-Manway	163.86	278.99	0.587
Noz N4 750	186.53	278.99	0.669
D4-Shell side d	21.66	216.00	0.100

Elements Suitable for Test Pressure.

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ASME TS Calc: Pressure Test

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Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Main Shell

Shell Test Pressure		6435.00 KPa.
Shell Thickness	ts	42.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1750.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K11789
Shell Allowable Stress at Temperature	Ss	118.00 N./mm ²
Shell Allowable Stress at Ambient		118.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Test Pressure		1625.00 KPa.
Channel Thickness	tc	14.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1750.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K11789
Channel Allowable Stress at Temperature	Sc	118.00 N./mm ²
Channel Allowable Stress at Ambient		118.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	638
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	2500.00 mm
Straight Tube Length (bet. inner tubsht faces) L		2175.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-213 T11
Tube Material UNS Number		K11597
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		118.00 N./mm ²
Tube Allowable Stress At Ambient		118.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	205.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	849.916 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	5810.7 cm ²
Perimeter of Tube Layout	Cp	4024.3 mm



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ASME TS Calc: Pressure Test

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Tubesheet Data:

Tubesheet TYPE: U-tube, Integral with Channel, Conf. e

Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material		SA-336 F11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K11572
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	325.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	2240.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		Yes
Is the Tubesheet Flat on Shellside?		No
Step 1 Diameter on the Shellside	ds1	0.00 mm
Step 1 Depth on the Shellside	hs1	0.00 mm
Step 2 Diameter on the Shellside	ds2	1747.00 mm
Step 2 Depth on the Shellside	hs2	7.00 mm
Calculated Tubesheet Diameter per Div 2, 4.18.4		2240.00 mm

Note: Tubesheet diameter is now: 2240.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes	AL	0.0 cm ²
Junction Stress Reduction option		Perform Plastic Calculation
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Shell
Flange Outside Diameter	A	2240.000 mm
Flange Inside Diameter	B	1750.000 mm
Flange Face Outside Diameter	Fod	1883.000 mm
Flange Face Inside Diameter	Fid	1750.000 mm
Gasket Outside Diameter	Go	1872.000 mm
Gasket Inside Diameter	Gi	1782.000 mm
Small end Hub thk.	g0	44.0000 mm
Large end Hub thk.	g1	55.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	3.1750 mm
Full face Gasket Flange Option		Program Selects
Length of Partition Gasket	lp	1800.000 mm
Width of Partition Gasket	wp	10.0000 mm
Partition Gasket Factor,	mPart	3.0000
Partition Gasket Design Seating Stress	yPart	68.95 N./mm ²
Partition Gasket Facing Sketch		Code Sketch 1a
Partition Gasket Column for Gasket Seating		Code Column II

Bolting Information:

Diameter of Bolt Circle	C	2070.000 mm
Nominal Bolt Diameter	dB	76.2000 mm
Thread Series	TEMA Thread Series	
Number of Bolts	n	28
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	159.00 N./mm ²



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ASME TS Calc: Pressure Test

Case: 1

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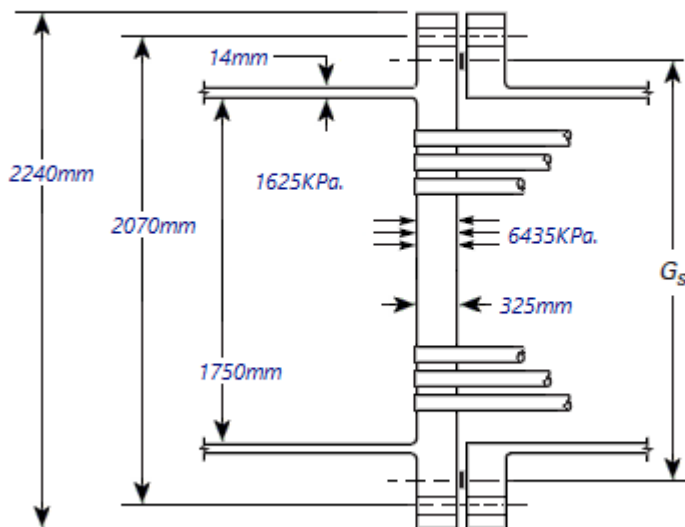
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Bolt Allowable Stress At Ambient	Sa	159.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Note: The full bolt load $Sa \cdot Ab$, was used for all load cases.

Tubesheet Integral with	Channel
Tubesheet Extended as Flange	Yes
Thickness of Extended Portion of Tubesheet	Tf 309.0000 mm
Is Bolt Load Transferred to the Tubesheet	Yes

Tubesheet Gasketed w/Shell and Integral w/Channel, Extended as a Flange



Configuration e:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Material Group C

Elastic Mod. at Design Temperature	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Channel - TM-1 Material Group C

Elastic Mod. at Design Temperature	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Tubes - TM-1 Material Group C

Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

TubeSheet - TM-1 Material Group C

Elastic Mod. at Design Temperature	21.1 °C	0.20408E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):



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Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1625 \cdot (25.4/2 - 0)) / (118 \cdot 1 + 0.4 \cdot 1625)$$

$$= 0.1739 + 0.0000 = 0.1739 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-2 at 21.11 °C
Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLen	D/T	L/D	Factor A	B
2.1100	25.40	3024.92	12.04	50.0000	0.0075908	122.73
EMAWP = (4*B)/(3*(D/T)) = (4 *123)/(3 *12) = 13593 KPa.						

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLen	D/T	L/D	Factor A	B
1.1519	25.40	3024.92	22.05	50.0000	0.0022623	106.44
EMAWP = (4*B)/(3*(D/T)) = (4 *106)/(3 *22.1) = 6436 KPa.						

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.1519 mm
Allowable Internal Pressure at Corroded thickness	20998.74 KPa.
Required Internal Design Pressure	1625.00 KPa.
Allowable External Pressure at Corroded thickness	13593.04 KPa.
Required External Design Pressure	6435.00 KPa.
Required Thickness due to Shell Side pressure	1.1519 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	N = (Goc-Gic) / 2	45.000 mm
Basic Gasket Width,	b0 = N / 2	22.500 mm
Effective Gasket Width,	b = sqrt(b0) * 2.5	11.949 mm
Gasket Reaction Diameter,	G = Go-2.0*b	1848.103 mm

Warning: Available bolt area, 1142 cm² for the flange is < required 1276 cm²

Note that for the pressure test case, ASME VIII-2 paragraph 4.18.13 (c)(9) does not require that the bolt area requirement be met.

Results for ASME U-tube Tubesheet Calculations for Configuration e:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 850 + 25.4$$

$$= 1725.232 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:



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Ratio [ps]:

$$= G_s / D_o \text{ (Configurations d, e, f)}$$

$$= 1848/1725$$

$$= 1.0712$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 1750/1725$$

$$= 1.0144$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 0/325$$

$$= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(d_t - 2t_t (E_t T / E) \cdot (S_t T / S) \cdot (\rho); d_t - 2t_t)$$

$$= \max(25.4 - 2 \cdot 2.11 (204080160/204080160) \cdot (118/138) \cdot (0); 25.4 - 2 \cdot 2.11)$$

$$= 25.4000 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 31.8 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 1725 \cdot 31.8) / (\pi \cdot 1725^2)) }$$

$$= 31.7500 \text{ mm}$$

Effective ligament efficiency for bending [μ*]:

$$= (p^* - d^*) / p^*$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.2000$$

h/p = 10.236220 ; μ* = 0.200000 ; E* = 31703676 KPa.

E*/E = 0.155349 ; ν* = 0.458572

Note: As h/p (10.236) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:

βs, ks, λs, δs and ωs are all = 0

Channel coefficient [βc]:

$$= ((12 (1 - \nu_c^2))^{1/4}) / \sqrt{ (D_c + t_c) t_c }$$

$$= ((12 (1 - 0.3^2))^{1/4}) / \sqrt{ (1750 + 14) 14 }$$

$$= 0.0116 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6 (1 - \nu_s^2))$$

$$= 0.012 \cdot 204080160 \cdot 14^3 / (6 (1 - 0.3^2))$$

$$= 1186409600.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$= 6 (D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6 (1750 + 14 / \cos(0)) \cdot 1186409600 / 325^3 \cdot (1 + 325 \cdot 0.012 \cdot (1 + 1) / 2 + (325 \cdot 0.012)^2 / 2 \cdot 1)$$

$$= 4325945.5 \text{ KPa}$$

Channel coefficient [δc]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2 (D_c + t_c)))$$



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$$= 1750^2 / (4 \cdot 204080160 \cdot 14) \cdot (1 - 1750 \cdot 0.3 / (2(1750 + 14)))$$

$$= 0.22808E+00 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

Parameter [wc]:

$$= p_c \cdot k_c [\delta C (C_m \Theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \Theta + h \cdot C_q \Theta \cdot h \cdot \beta C / 2)]$$

$$= 1.01 \cdot 1186409600 [0.22808082 (1 + 0.012 \cdot 1 \cdot 325) - 0.000 (1/325 + 1 \cdot 0.012/2)]$$

$$= 15112.6064 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2240/1725$$

$$= 1.2984$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda c + E \cdot \ln(K))$$

$$= (1 - 0.46) / 31703676 \cdot (4325945 + 204080160 \cdot \ln(1.3))$$

$$= 0.9839$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_t - 1) \cdot (p_t^2 + 1) \cdot P_t]$$

$$= 1725^2 / 16 \cdot [(1.07 - 1) \cdot (1.07^2 + 1) \cdot 6435 - (1.01 - 1) \cdot (1.01^2 + 1) \cdot 1625]$$

$$= 174282928.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= M_{ts} + (w_c \cdot P_t) + W^* \cdot (C - G_s) / (2 \cdot \pi \cdot D_o)$$

$$= 174282928 + (24557984) + 1852157 \cdot (2070 - 1848) / (2 \cdot \pi \cdot 1725)$$

$$= 570638016.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M^*) - D_o^2 / 32 \cdot F (P_s - P_t) / (1 + F)$$

$$= (570638016) - 1725^2 / 32 \cdot 0.98 (6435 - 1625) / (1 + 0.98)$$

$$= 65747032.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2 / 64 \cdot (3 + v^*) \cdot (P_s - P_t)$$

$$= 65747032 + 1725^2 / 64 \cdot (3 + 0.46) \cdot (6435 - 1625)$$

$$= 839419072.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|M_p|; |M_o|)$$

$$= \max(|65747032|; |839419072|)$$

$$= 839419072.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 839419072 / (0.2 (325 - 8)^2)$$

$$= 250.6148 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot \beta T \cdot S_y$$

$$= 2 \cdot 0.95 \cdot 275$$



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ASME TS Calc: Pressure Test

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$$= 522.49 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness $[\sigma]$:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 822059328 / (0.2 \cdot (225 - 8)^2)$$

$$= 522.4900 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress $[H_{reqB}]$:

$$= h + C_{ats} + C_{atc}$$

$$= 225 + 0 + 0$$

$$= 225.2630 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) $[H_{req}]$:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(225; 16.6)$$

$$= 225.2630 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential $[|P_s - P_t|]$:

$$= |P_s - P_t|$$

$$= |6435 - 1625|$$

$$= 4810.000 \text{ KPa.}$$

Shear Stress check $[\tau_{limit}]$:

$$= 4 \cdot \mu \cdot h / D_o \cdot 0.8 \cdot \beta_T \cdot S_y$$

$$= 4 \cdot 0.2 \cdot 325 / 1725 \cdot 0.8 \cdot 0.95 \cdot 275$$

$$= 31.497 \text{ N./mm}^2$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}) . Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear $[H_{reqS}]$:

$$= 1 / (4 \cdot \mu \cdot (0.8 \cdot \beta_T \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.2 \cdot 209) \cdot (4 \cdot 5811 / 4024) \cdot |6435 - 1625| + 0$$

$$= 16.6165 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Channel thickness of: 14.000 mm

Min. Channel len. adjacent to the tubesheet is: 281.745mm

Tube Metal Area $[A_t]$:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 2.11) 2.11$$

$$= 1.544 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength $[F_t]$:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.11 (25.4 - 2.11) 195$$

$$= 3065.793 \text{ Kg}$$

Groove Weld Strength $[F_g]$:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 195$$

$$= 3000.1914 \text{ Kg}$$

Weld Strength Factor $[f_w]$:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 195 / 195$$

$$= 1.000$$



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Design Strength Ratio [fd]:

= 1 for full strength welds

Fillet Weld Strength [Ff]:

= $0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw$ (but not > Ft)

= $0.55 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 195$

= 1941.3005 Kgf

Minimum Length of Weld [ar]:

= $2 \left(\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) fw \cdot fd)} - 0.75 \cdot do \right)$

= $2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.11 \cdot (25.4 - 2.11) 1 \cdot 1)} - 0.75 \cdot 25.4 \right)$

= 2.667 mm

Minimum Required Fillet Weld Leg [afr]: 1.3334 mm

Minimum Required Groove Weld Leg [agr]: 1.3334 mm

Tube-Tubesheet Jt allowable, 3066 is >= tube strength 3066 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	250.6 <=	522.5 N./mm ²	Ok
Tubesheet shear stress	10.7 <=	209.0 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	225.263	325.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.333	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.333	2.110	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/ Fail
D1uc	107.172 94.120	53	522 4 209	Ps+Pt D1	Ok
D2uc	248.463 ...	304	522 14 209	Ps+Pt D2	Ok
D3uc	225.263 ...	251	522 11 209	Ps+Pt D3	Ok
Max:	248.4630 94.120 mm	0.582	0.068 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)



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Shell and Channel Stress Summary:

-- Integral Cyl. Req'd. Thk.	Shell	Stress	Channel	Stress	Pass
Case#	Shell	Channel	Stress	Allwd	Fail
D1uc	...	...	...	...	342 Ok
D2uc	...	...	...	...	342 Ok
D3uc	...	...	...	...	342 Ok
Max	...	...	...	...	

Note:
 Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	248.463	325.000	Ok
Tubesheet thickness flanged extension :	94.120	309.000	Ok
Tube thickness :	1.152	2.110	Ok
Tube-tubesheet fillet weld leg :	1.333	2.110	Ok
Tube-tubesheet groove weld leg :	1.333	2.110	Ok

Given Shell Thickness: 42.0000 mm
 Given Channel Thickness: 14.0000 mm

Min Channel length of thk, (14.000) adj. to tubesheet: 281.745 mm

Note: This is a full strength Tube to Tubesheet Joint.

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 FileName : 140-C1 Final-23rd March Page 63 of 216
 External Pressure Calculations: Step: 11 9:34am Sep 9,2026

External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	...	174	No Calc	No Calc
20	30	No Calc	...	174	No Calc	No Calc
30	40	1200	1778	11	0.00099743	77.6017
40	50	No Calc	...	229	No Calc	No Calc
50	60	6000	1834	39	0.0011621	69.1045

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	177	170.536	103.419	No Calc
20	30	177	170.536	103.419	No Calc
30	40	14	7.88935	103.419	640
40	50	232	226.263	...	No Calc
50	60	42	No Calc	...	1959

Minimum

640

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	40	1200	8262.13	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	6000	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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 Element and Detail Weights: Step: 12 9:34am Sep 9,2026

Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	1031.91	515474	1004.64	515937	206.382
20	30	1031.91	515474	1004.64	515937	206.382
30	40	721.586	2886855	567.926	2906685	144.317
40	50	2999.42	740114	2960.24	741205	599.885
50	60	10995.6	13395702	10227.3	13494850	2199.11
Total		16780	18053620.00	15764	18174612.00	3356



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset, Dtl. Cent. mm	Description
10	Liqd	515.159	102.5	-180	...	Liquid: 10
10	Insl	36.3832	68	...	...	Ins: 50
20	Liqd	515.159	102.5	-15	...	Liquid: 20
20	Insl	36.3832	106	...	...	Ins: 50
30	Liqd	2885.09	600	...	...	Liquid: 30
30	Insl	249.504	600	...	...	Ins: 50
30	Nozl	189.527	600	-1036.92	...	N3-Tube Inlet/L
30	Nozl	189.527	600	1036.92	...	Noz N2 Fr30
30	Nozl	6.71389	215.265	...	913.1	3" Nozzle
30	Wght	200	600	...	...	Partition plate
40	Insl	82.3592	163	...	...	Ins: 40
50	Sadl	941.092	1200	1177	...	Left Saddle
50	Sadl	941.092	4800	1177	...	New Saddle
50	Liqd	13387.5	3000	...	...	Liquid: 40
50	Insl	1284.72	3000	...	...	Ins: 50
50	Nozl	819.284	1200	1240	...	Noz N1 750
50	Nozl	17.4562	2200	894.05	...	K7-Temperature
50	Nozl	1792.28	4800	...	1165.7	M1-Manway
50	Nozl	819.284	3500	-1240	...	Noz N4 750
50	Nozl	17.4562	5731.14	-894.05	...	D4-Shell side d
50	Wght	850	4000	...	...	Distributor wei
50	Wght	200	2500	...	...	Distributor wt
50	Wght	200	1200	-1500	...	Wght:[3 of 3]
50	Wght	200	4800	-1500	...	Direprofiging wei
30	FTsh	10934.7	1362.5	...	...	Tubesheet
30	Tube	3199.43	2450	...	...	Tubes

Total Weight of Each Detail Type:

Saddles	1882.2
Liquid	17302.9
Insulation	1689.4
Nozzles	3851.5
Weights	1650.0
Exchanger Components	14134.1
Liquid in Tubes	861.7
Sum of the Detail Weights	41371.8 kg.



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 Element and Detail Weights: Step: 12 9:34am Sep 9,2026

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	20136.5	20136.5	20136.5	20136.5	20136.5	20136.5
Saddles	1882.2	1882.2	1882.2	1882.2	1882.2	1882.2
Nozzles	3851.5	3851.5	3851.5	3851.5	3851.5	3851.5
Wld Weights	1650.0	1650.0	1650.0	1650.0	1650.0	1650.0
Exchanger	14134.1	14134.1	14134.1	14134.1	14134.1	14134.1
Insulation	...	...	1689.4	1689.4	1689.4	1689.4
Ope. Liquid	...	...	...	...	...	17302.9
Tube Ope Lqd	...	...	...	...	...	861.7
Test Liquid	...	18042.6	...	...	...	...
Tube Tst Lqd	...	861.7	...	...	...	...
Totals	41654.3	60558.6	43343.7	43343.7	43343.7	61508.3

Field Installation Options:

Miscellaneous Weight Percent: 20.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	41654.3 kg.
Shop Test Wt.	- Fabricated weight + water (full)	60558.6 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	43343.7 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	43343.7 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	43343.7 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	61508.3 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	60289.5 kg.
Field Test Wt.	- Empty weight + test medium (full)	61150.0 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	862205.4 cm ³
Weight of Ope Liq in tubes :	861.7 kg.
Weight of Water in tubes :	861.7 kg.

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	20191.8	11710.3
20	30	20191.8	11710.3
30	40	67029	65973.5
40	50	36101.2	16823.2
50	60	345701	329867

Total	489214.7	436084.5 cm ²
Total	526.6	469.4 ft ²



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 Nozzle Flange MAWP: Step: 13 9:34am Sep 9,2026

Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating Ope.	Flange Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	UG-44	Max Pressure 50%	DNV
N3-Tube Inlet/	4630.0	5170.0	250	300	GR 1.9	1730.4	4630.0	4630.0	4630.0
Noz N2 Fr30	4630.0	5170.0	250	300	GR 1.9	1730.4	4630.0	4630.0	4630.0
3" Nozzle	4630.0	5170.0	250	300	GR 1.9	*****	*****	0.0*	*****
K7-Temperature	34115.3	42510.0	275	2500	GR 1.1	...	...	...	...
M1-Manway	8185.1	10200.0	275	600	GR 1.1	...	...	...	...
D4-Shell side	34115.3	42510.0	275	2500	GR 1.1	...	...	...	...

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 8185.085 KPa.
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 10200.003 KPa.

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: ***** KPa.
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 5170.005 KPa.

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kg	MAWP PD KPa.	Maximum Pressure Pmax KPa.	Pass Fail
N3-Tube Inlet/L	389.001	0.50	1816	2294	0.0	4630.0	Pass
Noz N2 Fr30	389.001	0.50	1816	2295	0.0	4630.0	Pass
3" Nozzle	111.125	0.50	1267135	6000	0.0	*****	Fail

Evaluating Flange on Nozzle [N3-Tube Inlet/L]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(1816) + 4 \cdot \text{abs}(2294) \cdot 389.001 \leq \pi \cdot 389.001^3 [(4630.037 - 0.000) + .500 \cdot 4630.037]$$

$$= 0.32634E+05 \leq 0.13097E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N3-Tube Inlet/L [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4630.037, 0.500 \cdot 4630.037 + 4630.037 - 32633.6 / (\pi \cdot 389.001^3))$$

$$= \min(4630.037, 5214.607)$$

$$= 4630.037 \text{ KPa.}$$

Evaluating Flange on Nozzle [Noz N2 Fr30]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(1816) + 4 \cdot \text{abs}(2295) \cdot 389.001 \leq \pi \cdot 389.001^3 [(4630.037 - 0.000) + .500 \cdot 4630.037]$$

$$= 0.32634E+05 \leq 0.13097E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle Noz N2 Fr30 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4630.037, 0.500 \cdot 4630.037 + 4630.037 - 32633.7 / (\pi \cdot 389.001^3))$$

$$= \min(4630.037, 5214.607)$$

$$= 4630.037 \text{ KPa.}$$



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Nozzle Flange MAWP:

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Evaluating Flange on Nozzle [3" Nozzle]:

$$\begin{aligned}
 &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\
 &= 16(1267135) + 4 \cdot \text{abs}(6000) 111.125 \leq \pi \cdot 111.125^3 [(4630.037 - 0.000) + .500 \cdot 4630.037] \\
 &= 0.20277E+08 \leq 0.30533E+04 \text{ Check Fails}
 \end{aligned}$$

Maximum Pressure for Flange on Nozzle 3" Nozzle [Pmax]:

$$\begin{aligned}
 &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\
 &= \min(4630.037, 0.500 \cdot 4630.037 + 4630.037 - 20276822.0 / (\pi \cdot 111.125^3)) \\
 &= \min(4630.037, -46115212.000) \\
 &= -46115212.000 \text{ KPa.}
 \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	25500 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.19 \cdot 0.91) / (1 + 1.7 \cdot 3.4 \cdot 0.19)))$
 $= \min(0.85, 0.88)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.554
Structure Height to Diameter ratio	4.258

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Zg = 274320 \text{ mm}$

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1535 + 25500 = 27035 \text{ mm}$
 $= 88.7 \text{ ft. Imperial Units}$

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (88.7 ft.) $> 15 \text{ ft.}$
 $= 2.01(z / Zg)^{2/\alpha}$
 $= 2.01(88.7/900)^{2/9.5}$
 $= 1.23$

Type of Hill: No Hill



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Wind Load Calculation:

Step: 14

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Wind Directionality Factor [Kd]:

= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

= (1 + K1 • K2 • K3)²

= (1 + 0 • 0 • 0)²

= 1

Velocity Pressure evaluated at height z, Imperial Units [qz]:

= max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)

= max(16, 0.00256 • 1.23 • 1 • 1 • 1 • 60.4²)

= 16 psf [78.1] Kgs/m²

Force on the first element [F]:

= qz • G • Cf • WindArea

= 16 • 0.85 • 0.55 • 4.99

= 37.6 lbs. [17] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
CH Cover	27035.0	0.000	0.000	0.000	1.234	1.000	78.120
LHS Ch Flg	27035.0	0.000	0.000	0.000	1.234	1.000	78.120
LHS Ch Shell	27035.0	0.000	0.000	0.000	1.234	1.000	78.120
Main Shell Flan	27035.0	0.000	0.000	0.000	1.234	1.000	78.120
Main Shell	27035.0	0.000	0.000	0.000	1.234	1.000	78.120

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
Partition pla	0.00	27035.00	78.12	0.00
Distributor w	0.00	27035.00	78.12	0.00
Distributor w	0.00	27035.00	78.12	0.00
Wght:[3 of 3]	0.00	25500.00	78.12	0.00
Direprofing w	0.00	25500.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	27035	2259.22	4631.4	78.12	18.7512
20	30	27035	2340	4797	78.12	19.4216
30	40	27035	2373.6	28483.2	78.12	115.32
40	50	27035	2340	6973.2	78.12	28.2324
50	60	27035	2440.8	146448	78.12	592.925

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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 Earthquake Load Calculation: Step: 15 9:34am Page 70 of 216
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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.100
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\ SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot h_{nx}$ where $Ct = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 $= 0.020(7.910.75)$
 $= 0.094$ seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/33)$
 $= 0.030$

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.094, 0.03) \text{ per 12.8.2}$
 $= 0.030$
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 $= 0.3 \cdot Sds \cdot W \cdot I$
 $= 0.3 \cdot 0.16 \cdot 58407 \cdot 1.25$
 $= 3434.347 \text{ Kg}$

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.16 \cdot 58407$
 $= 1831.7 \text{ Kg}$

Final Base Shear, $V = 3777.78 \text{ Kg}$
 Final Vertical Load, $E_v = 2014.82 \text{ Kg}$

Earthquake Load Calculation:

From	To	Earthquake Height	Earthquake Weight	Element Ope Load
------	----	-------------------	-------------------	------------------



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Earthquake Load Calculation:

Step: 15

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		mm	Kgf	Kgf
10	20	875	8343.9	539.683
20	30	875	8343.9	539.683
30	40	875	8343.9	539.683
40	50	875	8343.9	539.683
50	Sad1	875	8343.9	539.683
Sad1	60	875	8343.9	539.683
50	60	875	8343.9	539.683

The Earthquake Weight does not include the weight of the saddles.

The Earthquake Loads calculated and printed in the Earthquake Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

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Center of Gravity Calculation: Step: 16 9:34am Sep 9,2026

Shop/Field Installation Options :

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	5239.349 mm
Center of Gravity of the Liquid	4234.996 mm
Center of Gravity of the Insulation	4244.892 mm
Center of Gravity of the Nozzles	5385.226 mm
Center of Gravity of the Added Weights (Operating)	5181.632 mm
Center of Gravity of the Added Weights (Empty)	5181.632 mm
Center of Gravity of the Tubesheet(s)	1775.675 mm
Center of Gravity of the Tubes	2863.175 mm
Center of Gravity of Bare Shell New and Cold	3875.396 mm
Center of Gravity of Bare Shell Corroded	3853.954 mm
Vessel CG in the Operating Condition	3708.000 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	3528.482 mm
Vessel CG in the Test Condition	3698.561 mm

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 Saddle Calcs: Bundle Pull New: Step: 17 9:34am Sep 9,2026

ASME Horizontal Vessel Analysis: Bundle pullout, Left Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	896.00	mm
Shell Thickness used in this Case	t	42.000	mm
Stiffened Vessel Length per 4.15.6	L	6298.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1200.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	780.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	118.00	N./mm ²
Shell Allowable Stress used in Calculation		118.00	N./mm ²
Head Allowable Stress used in Calculation		0.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1535.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		15500.00	Kgf
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	

Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force		481.18	Kgf
Pressure used in Analysis	P	0.000	KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		-0.02	-216.00
Long. Stress at Top of Midspan		0.02	118.00
Long. Stress at Bottom of Midspan		0.02	118.00
Long. Stress at Top of Saddles		0.07	118.00
Long. Stress at Bottom of Saddles		-0.04	-216.00
Long. Stress at Bottom of Saddles		0.04	118.00
Tangential Shear in Shell		0.09	94.40
Circ. Stress at Horn of Saddle		0.17	177.00
Circ. Compressive Stress in Shell		0.01	118.00

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:
 = Bundle Pull Out Force • Impact Factor • B / Ls
 = 15500 • 1.5 • 1535/3600
 = 9913.5 Kgf

Load Combination Results for Tube Bundle Pull Out [Q]:
 = Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = -9432 + max(9914, 0, 0, 0)



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= 481.2 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	1422.27	Kgf
Transverse Shear Load	0.00	Kgf
Longitudinal Shear Load	23250.00	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

= $\min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$
 = $\min(2(1750/2 + 42 + 14)\sin(120/2), 2 \cdot 896)$
 = 1612.539 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

= $-Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 = $-481 \cdot 1200 [1 - (1 - 1200/6298 + (896^2 - 0^2)/(2 \cdot 1200 \cdot 6298))/(1 + (4 \cdot 0)/(3 \cdot 6298))]$
 = -79.4 Kg-m.

Moment per Equation 4.15.2 [M2]:

= $Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$
 = $481 \cdot 6298/4 ((1 + 2(896^2 - 0^2)/(6298^2))/(1 + (4 \cdot 0)/(3 \cdot 6298)) - 4 \cdot 1200/6298)$
 = 210.9 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= $P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$
 = $0 \cdot 896/(2 \cdot 42) - 211/(\pi \cdot 896^2 \cdot 42)$
 = -0.02 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= $P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$
 = $0 \cdot 896/(2 \cdot 42) + 211/(\pi \cdot 896^2 \cdot 42)$
 = 0.02 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

= $P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$
 = $0 \cdot 896/(2 \cdot 42) - 79.4/(0.11 \cdot \pi \cdot 896^2 \cdot 42)$
 = 0.07 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

= $P \cdot R_m/(2t) + M1/(K1 \cdot \pi \cdot R_m^2 \cdot t)$
 = $0 \cdot 896/(2 \cdot 42) + 79.4/(0.19 \cdot \pi \cdot 896^2 \cdot 42)$
 = -0.04 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= $Q(L - 2a)/(L + (4 \cdot h_m)/3)$
 = $481(6298 - 2 \cdot 1200)/(6298 + (4 \cdot 0)/3)$
 = 297.8 Kgf

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

= $K2 \cdot T / (R_m \cdot t)$
 = $1.17 \cdot 298/(896 \cdot 42)$
 = 0.09 N./mm²



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Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(896 \cdot 42)}$$

$$= 151.312 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(400 + 1.56 \cdot \sqrt{(896 \cdot 42)}, 2 \cdot 1200)$$

$$= 702.62 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

= 1.0000 Materials are the same, bundle pull case

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ6,r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 481 \cdot 0.1 / (703 (42 + 1 \cdot 14))$$

$$= -0.01 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, L<8Rm (4.15.29) [σ7,r*]:

$$= -Q / (4(t + \eta \cdot tr) b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2)$$

$$= -481 / (4(42 + 1 \cdot 14) 703) -$$

$$12 \cdot 0.053 \cdot 481 \cdot 896 / (6298(42 + 1 \cdot 14)^2)$$

$$= -0.17 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2500.0000	mm
Baseplate Thickness	Bpthk	36.0000	mm
Baseplate Width	Bpwid	425.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	240.0	N./mm ²
Height of Web at Center	Hw,c	600.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(875 \cdot 42)} + 780) 42^3) / 12$$

$$= 666.209 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 666 + 453 (176 - 21)^2$$

$$= 109945.133 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	1079.1	42.0	21.0	453.2	951727.8	0.110E+06
Wearplate	780.0	14.0	49.0	109.2	535080.0	0.177E+05
Web	14.0	568.0	340.0	79.5	2703680.5	0.427E+05
BasePlate	425.0	36.0	642.0	153.0	9822602.0	0.332E+06
Totals	...	...	...	794.9	14013090.0	0.502E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 14013090 / 795$$

$$= 176.282 \text{ mm}$$

Angle [β]:



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$$\begin{aligned}
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 120/2 \\
 &= 120.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\
 &= 0.2035
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.2 \cdot 481 \\
 &= 97.9308 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh / As) = 0.0281 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 143.9974 \text{ N./mm}^2
 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned}
 &= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians}) \\
 &= 1535 - 875 \cdot \sin(120/2) / 1.05 \\
 &= 811.381 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 79.4607 \text{ Kg-m.}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 0.0273 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 159.9971 \text{ N./mm}^2
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\
 &= \sqrt{(3(481 + 941) \cdot 425) / (4 \cdot 2500 \cdot 160)} \\
 &= 3.334 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned}
 &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120/2) (875 + 42 + 14) \\
 &= 1612.539 \text{ mm}
 \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned}
 &= \text{Web Length} / (N_{\text{ribs}} - 1) \\
 &= 1613 / (4 - 1) \\
 &= 537.513 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned}
 &= e \cdot \text{Bpwid} / 2 \\
 &= 538 \cdot 425 / 2 \\
 &= 1142.215 \text{ cm}^2
 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned}
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 1422 / (2500 \cdot 425) \\
 &= 0.134 \text{ Kgf/cm}^2
 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned}
 &= Ap \cdot Bp \\
 &= 1142 \cdot 0.13 \\
 &= 152.898 \text{ Kgf}
 \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 54 + 37.6 \\
 &= 91.666 \text{ cm}^2
 \end{aligned}$$

Compressive Stress [Sc]:



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= P/Ar
 = 153/91.7
 = 0.164 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	400.0	...	56.0	...	0.747E+04

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 400 - 14
 = 386.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0/91.7
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 400/2
 = 200.000 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{7467/91.7}$
 = 90.253 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1535 - cos(120/2)(875 + 42 + 14) - 36
 = 1033.500 mm

Intermediate Term [Cc]:
 $= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 $= \sqrt{(2 \cdot \pi^2 \cdot 199943008/240)}$
 = 128.242

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 1034/90.3
 = 11.451

Bending Moment [Rm]:
 = FI / (2 • Bplen) • e • L / 2
 = 23250 / (2 • 2500) • 538 • 1034/2
 = 1291.609 Kg-m.

Compressive Allowable, KL/r < Cc (11.5 < 128) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (11.5)^2 / (2 \cdot 128^2)) 240 / (5/3 + 3 \cdot (11.5) / (8 \cdot 128) - (11.5)^3 / (8 \cdot 128^3))$
 = 140.6 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba
 = 0.16/141 + (1292 • 200/74666672) / 160
 = 0.213

Check of Inside Ribs:



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Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	386.0	0.0	54.0	0.0	0.747E+04
Web	537.5	14.0	0.0	75.3	0.0	12.3
Totals	...	...	...	129.3	...	0.748E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/129$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400/2$$

$$= 200.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1535 - \sqrt{(931 + 14)^2 - (538/2)^2} - 36$$

$$= 607.635 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(7479/129)}$$

$$= 76.055 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 608/76.1$$

$$= 7.989$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 23250 / (2 \cdot 2500)$$

$$= 4.650 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 4.65 \cdot 538 \cdot 608$$

$$= 1518.776 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1519/373932$$

$$= 39.831 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($7.99 < 128$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.99)^2 / (2 \cdot 128^2)) 240 / (5/3 + 3 \cdot (7.99) / (8 \cdot 128) - (7.99)^3 / (8 \cdot 128^3))$$

$$= 141.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / Sba$$

$$= 0.23/142 + (1519 \cdot 200/7479) / 160$$

$$= 0.251$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	6
Total Number of Bolts in Tension/Baseplate	Nbt	3
Bolt Material Specification	SA-307 A	
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm



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Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	118.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8674 cm ²
Saddle Load QO (Weight)	QO	1422.3 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	23250.0 Kgf
Maximum Transverse Force	Ft	0.0 Kgf
Maximum Longitudinal Force	F1	23250.0 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

= $F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= $23250 / (4.87 \cdot 6)$

= **78.1 N./mm². Must be less than 29.0 N./mm².**

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

= $Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= $0 / (4.87 \cdot 6)$

= 0.0 N./mm². Must be less than 29.0 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= $(QL - QO) / (Stba \cdot Rnbolts)$

= $(23250 - 1422) / (48.3 \cdot 6)$

= **7.3866 cm²**

Bolt Area due to Shear Load [Bltarears]:

= $F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$

= $23250 / (29 \cdot 6)$

= **13.1132 cm²**

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot Ft + \text{Sum of X Moments}$

= $1535 \cdot 0 + 0$

= 0.00 Kg-m.

Eccentricity (e):

= $Rmom / QO$

= $0 / 1422$

= 0.00 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[Bltarearl, Bltarears, Bltareart]$

= $\max[7.39, 13.1, 0]$

= **13.1132 cm²[Failed]**

Review notes about nozzle loadings and supports in the Warnings report.

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ASME Horizontal Vessel Analysis: Bundle pullout, Left Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	897.50	mm
Shell Thickness used in this Case	t	42.000	mm
Stiffened Vessel Length per 4.15.6	L	6298.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1200.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	780.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	118.00	N./mm ²
Shell Allowable Stress used in Calculation		118.00	N./mm ²
Head Allowable Stress used in Calculation		0.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1535.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		15500.00	Kgf
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	

Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force		128.24	Kgf
Pressure used in Analysis	P	0.000	KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		-0.01	-216.00
Long. Stress at Top of Midspan		0.01	118.00
Long. Stress at Bottom of Midspan		0.01	118.00
Long. Stress at Top of Saddles		0.02	118.00
Long. Stress at Bottom of Saddles		-0.01	-216.00
Long. Stress at Bottom of Saddles		0.01	118.00
Tangential Shear in Shell		0.03	94.40
Circ. Stress at Horn of Saddle		0.05	177.00
Circ. Compressive Stress in Shell		0.00	118.00

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:
 = Bundle Pull Out Force • Impact Factor • B / Ls
 = 15500 • 1.5 • 1535/3600
 = 9913.5 Kgf

Load Combination Results for Tube Bundle Pull Out [Q]:
 = Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = -9785 + max(9914, 0, 0, 0)



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= 128.2 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	1069.33	Kgf
Transverse Shear Load	0.00	Kgf
Longitudinal Shear Load	23250.00	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

= $\min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$

= $\min(2(1750/2 + 42 + 14)\sin(120/2), 2 \cdot 897)$

= 1612.539 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

= $-Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$

= $-128 \cdot 1200 [1 - (1 - 1200/6298 + (897^2 - 0^2)/(2 \cdot 1200 \cdot 6298))/(1 + (4 \cdot 0)/(3 \cdot 6298))]$

= -21.1 Kg-m.

Moment per Equation 4.15.2 [M2]:

= $Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$

= $128 \cdot 6298/4 ((1 + 2(897^2 - 0^2)/(6298^2))/(1 + (4 \cdot 0)/(3 \cdot 6298)) - 4 \cdot 1200/6298)$

= 56.2 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= $P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$

= $0 \cdot 897/(2 \cdot 39) - 56.2/(\pi \cdot 897^2 \cdot 39)$

= -0.01 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= $P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$

= $0 \cdot 897/(2 \cdot 39) + 56.2/(\pi \cdot 897^2 \cdot 39)$

= 0.01 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

= $P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$

= $0 \cdot 897/(2 \cdot 39) - 21.1/(0.11 \cdot \pi \cdot 897^2 \cdot 39)$

= 0.02 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

= $P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$

= $0 \cdot 897/(2 \cdot 39) + 21.1/(0.19 \cdot \pi \cdot 897^2 \cdot 39)$

= -0.01 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= $Q(L - 2a)/(L + (4 \cdot h_m)/3)$

= $128(6298 - 2 \cdot 1200)/(6298 + (4 \cdot 0/3))$

= 79.4 Kgf

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

= $K2 \cdot T / (R_m \cdot t)$

= $1.17 \cdot 79.4 / (897 \cdot 39)$

= 0.03 N./mm²



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Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(897 \cdot 39)}$$

$$= 145.930 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(400 + 1.56 \cdot \sqrt{(897 \cdot 39)}, 2 \cdot 1200)$$

$$= 691.86 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

= 1.0000 Materials are the same, bundle pull case

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 128 \cdot 0.1 / (692(39 + 1 \cdot 14))$$

$$= -0.00 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, L<8Rm (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -128 / (4(39 + 1 \cdot 14)692) -$$

$$12 \cdot 0.053 \cdot 128 \cdot 897 / (6298(39 + 1 \cdot 14)^2)$$

$$= -0.05 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2500.0000	mm
Baseplate Thickness	Bpthk	36.0000	mm
Baseplate Width	Bpwid	425.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	240.0	N./mm ²
Height of Web at Center	Hw,c	600.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(878 \cdot 39)} + 780)39^3) / 12$$

$$= 528.271 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 528 + 417(182 - 19.5)^2$$

$$= 109993.156 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	1068.7	39.0	19.5	416.8	812724.8	0.110E+06
Wearplate	780.0	14.0	46.0	109.2	502320.0	0.201E+05
Web	14.0	568.0	337.0	79.5	2679823.5	0.406E+05
BasePlate	425.0	36.0	639.0	153.0	9776700.0	0.320E+06
Totals	...	...	...	758.5	13771569.0	0.491E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 13771569 / 759$$

$$= 181.563 \text{ mm}$$

Angle [β]:



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$$\begin{aligned}
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 120/2 \\
 &= 120.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\
 &= 0.2035
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.2 \cdot 128 \\
 &= 26.0999 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh/As) = 0.0075 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 143.9974 \text{ N./mm}^2
 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned}
 &= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians}) \\
 &= 1535 - 878 \cdot \sin(120/2) / 1.05 \\
 &= 808.900 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 21.1126 \text{ Kg-m.}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 0.0077 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 159.9971 \text{ N./mm}^2
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\
 &= \sqrt{(3(128 + 941) \cdot 425) / (4 \cdot 2500 \cdot 160)} \\
 &= 2.891 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned}
 &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120/2) (875 + 42 + 14) \\
 &= 1612.539 \text{ mm}
 \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned}
 &= \text{Web Length} / (\text{Nr ribs} - 1) \\
 &= 1613 / (4 - 1) \\
 &= 537.513 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned}
 &= e \cdot \text{Bpwid} / 2 \\
 &= 538 \cdot 425 / 2 \\
 &= 1142.215 \text{ cm}^2
 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned}
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 1069 / (2500 \cdot 425) \\
 &= 0.101 \text{ Kgf/cm}^2
 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned}
 &= Ap \cdot Bp \\
 &= 1142 \cdot 0.1 \\
 &= 114.956 \text{ Kgf}
 \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 54 + 37.6 \\
 &= 91.666 \text{ cm}^2
 \end{aligned}$$

Compressive Stress [Sc]:



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= P/Ar
 = 115/91.7
 = 0.123 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	400.0	...	56.0	...	0.747E+04

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 400 - 14
 = 386.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0/91.7
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 400/2
 = 200.000 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{7467/91.7}$
 = 90.253 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1535 - cos(120/2)(875 + 42 + 14) - 36
 = 1033.500 mm

Intermediate Term [Cc]:
 $= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 $= \sqrt{(2 \cdot \pi^2 \cdot 199943008/240)}$
 = 128.242

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 1034/90.3
 = 11.451

Bending Moment [Rm]:
 = FI / (2 • Bplen) • e • L / 2
 = 23250 / (2 • 2500) • 538 • 1034/2
 = 1291.609 Kg-m.

Compressive Allowable, KL/r < Cc (11.5 < 128) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (11.5)^2 / (2 \cdot 128^2)) 240 / (5/3 + 3 \cdot (11.5) / (8 \cdot 128) - (11.5)^3 / (8 \cdot 128^3))$
 = 140.6 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba
 = 0.12/141 + (1292 • 200/74666672) / 160
 = 0.213

Check of Inside Ribs:



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Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	386.0	0.0	54.0	0.0	0.747E+04
Web	537.5	14.0	0.0	75.3	0.0	12.3
Totals	...	...	...	129.3	...	0.748E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/129$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400/2$$

$$= 200.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1535 - \sqrt{(931 + 14)^2 - (538/2)^2} - 36$$

$$= 607.635 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(7479/129)}$$

$$= 76.055 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 608/76.1$$

$$= 7.989$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 23250 / (2 \cdot 2500)$$

$$= 4.650 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 4.65 \cdot 538 \cdot 608$$

$$= 1518.776 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1519/373932$$

$$= 39.831 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($7.99 < 128$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.99)^2 / (2 \cdot 128^2)) 240 / (5/3 + 3 \cdot (7.99) / (8 \cdot 128) - (7.99^3) / (8 \cdot 128^3))$$

$$= 141.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / Sba$$

$$= 0.17/142 + (1519 \cdot 200/7479) / 160$$

$$= 0.250$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	6
Total Number of Bolts in Tension/Baseplate	Nbt	3
Bolt Material Specification	SA-307 A	
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm



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Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	118.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8674 cm ²
Saddle Load QO (Weight)	QO	1069.3 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	23250.0 Kgf
Maximum Transverse Force	Ft	0.0 Kgf
Maximum Longitudinal Force	F1	23250.0 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

= $F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= $23250 / (4.87 \cdot 6)$

= **78.1 N./mm². Must be less than 29.0 N./mm².**

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

= $Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= $0 / (4.87 \cdot 6)$

= 0.0 N./mm². Must be less than 29.0 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= $(QL - QO) / (Stba \cdot Rnbolts)$

= $(23250 - 1069) / (48.3 \cdot 6)$

= **7.5061 cm²**

Bolt Area due to Shear Load [Bltarears]:

= $F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$

= $23250 / (29 \cdot 6)$

= **13.1132 cm²**

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot Ft + \text{Sum of X Moments}$

= $1535 \cdot 0 + 0$

= 0.00 Kg-m.

Eccentricity (e):

= $Rmom / QO$

= $0 / 1069$

= 0.00 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[Bltarearl, Bltarears, Bltareart]$

= $\max[7.51, 13.1, 0]$

= **13.1132 cm²[Failed]**

Review notes about nozzle loadings and supports in the Warnings report.

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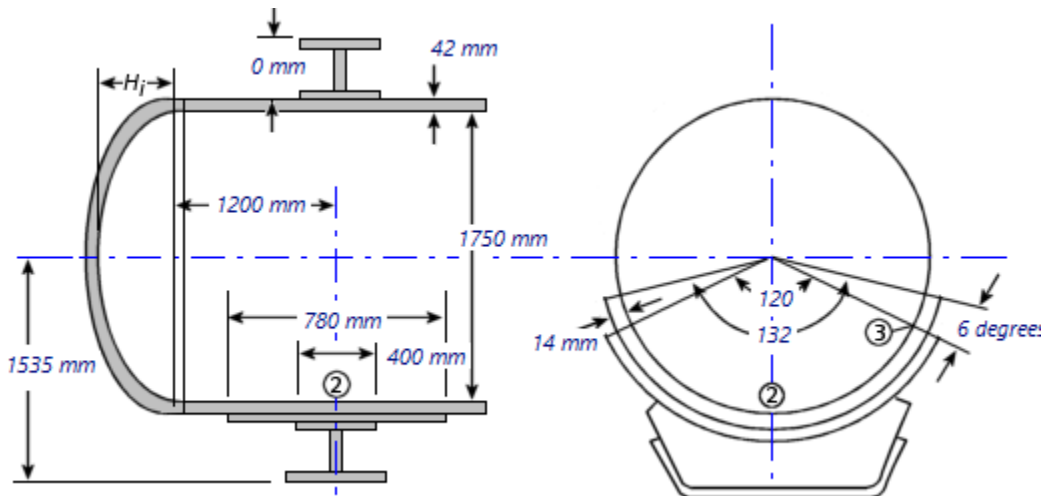
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Road Transport

Input and Calculated Values:

Vessel Mean Radius	Rm	896.00	mm
Shell Thickness used in this Case	t	42.000	mm
Stiffened Vessel Length per 4.15.6	L	6298.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1200.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	780.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	118.00	N./mm ²
Shell Allowable Stress used in Calculation		118.00	N./mm ²
Head Allowable Stress used in Calculation		148.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1535.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		44660.12	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr



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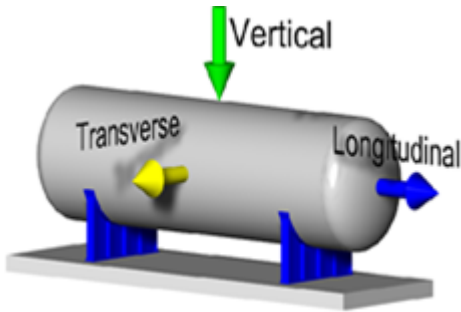
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Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-1.81	-216.00
Long. Stress at Top of Midspan	1.81	118.00
Long. Stress at Bottom of Midspan	1.81	118.00
Long. Stress at Top of Saddles	6.40	118.00
Long. Stress at Bottom of Saddles	-3.54	-216.00
Long. Stress at Bottom of Saddles	3.54	118.00
Tangential Shear in Shell	8.43	94.40
Circ. Stress at Horn of Saddle	15.38	177.00
Circ. Compressive Stress in Shell	0.85	118.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [F_{tw}]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 206 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \\
 &= 1749.4 \text{ lbf} \\
 &= 793.520 \text{ Kg}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [F_{lw}]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 6.67^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \cdot 1.2 \\
 &= 356.5 \text{ lbf} \\
 &= 161.7 \text{ Kg}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [F_t]:

$$\begin{aligned}
 &= g_z \cdot \text{Shipping Weight} \cdot g_w + F_{tw} \\
 &= 0.1 \cdot 39772 \cdot 1 + 794 \\
 &= 4770.7 \text{ Kg}
 \end{aligned}$$

Longitudinal Force due to Axial Acceleration [F_l]:

$$\begin{aligned}
 &= g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw} \\
 &= 0.1 \cdot 39772 \cdot 1 + 162 \\
 &= 4138.9 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Transverse Load [F_{wt}]:

$$\begin{aligned}
 &= F_t / \text{Num of Saddles} \cdot B / E \\
 &= 4771/2 \cdot 1535/1613 \\
 &= 2270.7 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Longitudinal Load [F_{wl}]:

$$\begin{aligned}
 &= F_l \cdot B / L_s \\
 &= 4139 \cdot 1535/3600 \\
 &= 1764.8 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [F_{rv}]:

$$\begin{aligned}
 &= \text{Saddle Load} \cdot g_y \\
 &= 40371 \cdot 0.05
 \end{aligned}$$



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= 2018.545 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)
 = 40371 + 2019 + max(1765, 2271)
 = 44660.1 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	45601.21	Kgf
Transverse Shear Load	2385.37	Kgf
Longitudinal Shear Load	4138.93	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2), 2•Rm)
 = min(2(1750/2 + 42 + 14)sin(120/2), 2•896)
 = 1612.539 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]
 = -44660•1200[1-(1-1200/6298+(896²-0²)/
 (2•1200•6298))/(1+(4•0)/(3•6298))]
 = -7365.0 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)
 = 44660•6298/4((1+2(896²-0²)/(6298²))/(1+(4•0)/
 (3•6298))-4•1200/6298)
 = 19572.1 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P • Rm/(2t) - M2/(π•Rm²t)
 = 0 • 896/(2•42) - 19572/(π•896²•42)
 = -1.81 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

= P • Rm/(2t) + M2/(π • Rm² • t)
 = 0 • 896/(2 • 42) + 19572/(π • 896² • 42)
 = 1.81 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*3]:

= P • Rm/(2t) - M1/(K1•π•Rm²t)
 = 0•896/(2•42)--7365/(0.11•π•896²•42)
 = 6.40 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)
 = 0•896/(2•42)+-7365/(0.19•π•896²•42)
 = -3.54 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))
 = 44660 (6298 - 2 • 1200)/(6298 + (4 • 0/3))
 = 27641.3 Kgf



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Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 27641 / (896 \cdot 42)$$

$$= 8.43 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(896 \cdot 42)}$$

$$= 151.312 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(400 + 1.56 \cdot \sqrt{(896 \cdot 42)}, 2 \cdot 1200)$$

$$= 702.62 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 44660 \cdot 0.1 / (703(42 + 1 \cdot 14))$$

$$= -0.85 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -44660 / (4(42 + 1 \cdot 14)703) -$$

$$12 \cdot 0.053 \cdot 44660 \cdot 896 / (6298(42 + 1 \cdot 14)^2)$$

$$= -15.38 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2500.0000	mm
Baseplate Thickness	Bpthk	36.0000	mm
Baseplate Width	Bpwid	425.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	240.0	N./mm ²
Height of Web at Center	Hw, c	600.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(875 \cdot 42)} + 780)42^3) / 12$$

$$= 666.209 \text{ cm}^4$$

Resolved Inertia for the Shell [I, shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 666 + 453(176 - 21)^2$$

$$= 109945.133 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	1079.1	42.0	21.0	453.2	951727.8	0.110E+06
Wearplate	780.0	14.0	49.0	109.2	535080.0	0.177E+05
Web	14.0	568.0	340.0	79.5	2703680.5	0.427E+05
BasePlate	425.0	36.0	642.0	153.0	9822602.0	0.332E+06
Totals	...	...	...	794.9	14013090.0	0.502E+06

Distance to Centroid [C1]:

$$= AY / A$$



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$$= 14013090/795$$

$$= 176.282 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 44660$$

$$= 9089.3027 \text{ Kgf}$$

Tension Stress, $St = (Fh/As) = 2.6085 \text{ N./mm}^2$

Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str} = 143.9974 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1535 - 875 \cdot \sin(120/2) / 1.05$$

$$= 811.381 \text{ mm}$$

Bending Moment, $M = Fh \cdot d = 7375.0312 \text{ Kg-m.}$

Bending Stress, $Sb = (M \cdot C1 / I) = 2.5379 \text{ N./mm}^2$

Allowed Stress, $Sa = 2/3 \cdot \text{Yield Str} = 159.9971 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(44660 + 941)425) / (4 \cdot 2500 \cdot 160)}$$

$$= 18.878 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (875 + 42 + 14)$$

$$= 1612.539 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1613 / (4 - 1)$$

$$= 537.513 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 538 \cdot 425/2$$

$$= 1142.215 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 45601 / (2500 \cdot 425)$$

$$= 4.292 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1142 \cdot 4.29$$

$$= 4902.250 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$



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$$= 54 + 37.6$$

$$= 91.666 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 4902/91.7$$

$$= 5.245 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	400.0	...	56.0	...	0.747E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 400 - 14$$

$$= 386.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/91.7$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400/2$$

$$= 200.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(7467/91.7)}$$

$$= 90.253 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1535 - \cos(120/2) (875 + 42 + 14) - 36$$

$$= 1033.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008/240)}$$

$$= 128.242$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1034/90.3$$

$$= 11.451$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 4139 / (2 \cdot 2500) \cdot 538 \cdot 1034/2$$

$$= 229.930 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($11.5 < 128$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.5)^2 / (2 \cdot 128^2)) 240 /$$

$$(5/3 + 3 \cdot (11.5) / (8 \cdot 128) - (11.5^3) / (8 \cdot 128^3))$$

$$= 140.6 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 5.24/141 + (230 \cdot 200/74666672) / 160$$



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= 0.075

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	386.0	0.0	54.0	0.0	0.747E+04
Web	537.5	14.0	0.0	75.3	0.0	12.3
Totals	...	...	...	129.3	...	0.748E+04

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/129

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 400/2

= 200.000 mm

Length of Inner Rib [L]:

= Saddle Height - $\sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2}$ - Bpthk

= 1535 - $\sqrt{(931 + 14)^2 - (538/2)^2}$ - 36

= 607.635 mm

Radius of Gyration [r]:

= $\sqrt{(Total\ Inertia / Total\ Area)}$

= $\sqrt{(7479/129)}$

= 76.055 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 608/76.1

= 7.989

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 4139 / (2 • 2500)

= 0.828 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 0.83 • 538 • 608

= 270.370 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σB,base,c]:

= Bending Moment / Section Modulus

= 270/373932

= 7.091 N./mm²

Compressive Allowable, KL/r < Cc (7.99 < 128) per AISC E2-1 [Sca]:

= $(1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$

= $(1 - (7.99)^2 / (2 \cdot 128^2)) 240 /$

$(5/3 + 3 \cdot (7.99) / (8 \cdot 128) - (7.99^3) / (8 \cdot 128^3))$

= 141.7 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 7.44/142 + (270 • 200/7479)/160

= 0.097

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:



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Vessel Mean Radius	Rm	896.00	mm
Shell Thickness used in this Case	t	42.000	mm
Stiffened Vessel Length per 4.15.6	L	6298.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1200.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	780.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	118.00	N./mm ²
Shell Allowable Stress used in Calculation		118.00	N./mm ²
Head Allowable Stress used in Calculation		0.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1535.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		2405.32	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-0.10	-216.00	
Long. Stress at Top of Midspan	0.10	118.00	
Long. Stress at Bottom of Midspan	0.10	118.00	
Long. Stress at Top of Saddles	0.34	118.00	
Long. Stress at Bottom of Saddles	-0.19	-216.00	
Long. Stress at Bottom of Saddles	0.19	118.00	
Tangential Shear in Shell	0.45	94.40	
Circ. Stress at Horn of Saddle	0.83	177.00	
Circ. Compressive Stress in Shell	0.05	118.00	

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 206 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \\
 &= 1749.4 \text{ lbf} \\
 &= 793.520 \text{ Kg}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [Flw]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 6.67^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \cdot 1.2 \\
 &= 356.5 \text{ lbf} \\
 &= 161.7 \text{ Kg}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [Ft]:

$$\begin{aligned}
 &= gz \cdot \text{Shipping Weight} \cdot gw + Ftw \\
 &= 0.1 \cdot 39772 \cdot 1 + 794
 \end{aligned}$$



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= 4770.7 Kgf

Longitudinal Force due to Axial Acceleration [FI]:

= $g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw}$

= $0.1 \cdot 39772 \cdot 1 + 162$

= 4138.9 Kgf

Saddle Reaction Force due to Transverse Load [Fwt]:

= $F_t / \text{Num of Saddles} \cdot B / E$

= $4771/2 \cdot 1535/1613$

= 2270.7 Kgf

Saddle Reaction Force due to Longitudinal Load [Fwl]:

= $F_l \cdot B / L_s$

= $4139 \cdot 1535/3600$

= 1764.8 Kgf

Saddle Reaction Force due to Vertical Transportation Acceleration [Frv]:

= Saddle Load $\cdot g_y$

= $128 \cdot 0.05$

= 6.412 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)

= $128 + 6.41 + \max(1765, 2271)$

= 2405.3 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 3346.41 Kgf

Transverse Shear Load 2385.37 Kgf

Longitudinal Shear Load 4138.93 Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

= $\min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$

= $\min(2(1750/2 + 42 + 14) \sin(120/2), 2 \cdot 896)$

= 1612.539 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434

K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

= $-Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$

= $-2405 \cdot 1200 [1 - (1 - 1200/6298 + (896^2 - 0^2)/$

$(2 \cdot 1200 \cdot 6298)) / (1 + (4 \cdot 0)/(3 \cdot 6298))]$

= -396.7 Kg-m.

Moment per Equation 4.15.2 [M2]:

= $Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$

= $2405 \cdot 6298/4 ((1 + 2(896^2 - 0^2)/(6298^2)) / (1 + (4 \cdot 0)/$

$(3 \cdot 6298)) - 4 \cdot 1200/6298)$

= 1054.1 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= $P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$

= $0 \cdot 896 / (2 \cdot 42) - 1054 / (\pi \cdot 896^2 \cdot 42)$

= -0.10 N./mm²



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Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 896 / (2 \cdot 42) + 1054 / (\pi \cdot 896^2 \cdot 42)$$

$$= 0.10 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 896 / (2 \cdot 42) - 397 / (0.11 \cdot \pi \cdot 896^2 \cdot 42)$$

$$= 0.34 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 896 / (2 \cdot 42) + 397 / (0.19 \cdot \pi \cdot 896^2 \cdot 42)$$

$$= -0.19 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 2405 (6298 - 2 \cdot 1200) / (6298 + (4 \cdot 0 / 3))$$

$$= 1488.7 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 1489 / (896 \cdot 42)$$

$$= 0.45 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{896 \cdot 42}$$

$$= 151.312 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(400 + 1.56 \cdot \sqrt{896 \cdot 42}, 2 \cdot 1200)$$

$$= 702.62 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot r))$$

$$= -0.76 \cdot 2405 \cdot 0.1 / (703(42 + 1 \cdot 14))$$

$$= -0.05 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r_{*}]:

$$= -Q / (4(t + \eta \cdot r) b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot r)^2)$$

$$= -2405 / (4(42 + 1 \cdot 14) 703) -$$

$$12 \cdot 0.053 \cdot 2405 \cdot 896 / (6298(42 + 1 \cdot 14)^2)$$

$$= -0.83 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2500.0000	mm
Baseplate Thickness	Bpthk	36.0000	mm
Baseplate Width	Bpwid	425.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	240.0	N./mm ²
Height of Web at Center	Hw, c	600.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)



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Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(875 \cdot 42)} + 780)42^3)/12$$

$$= 666.209 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 666 + 453(176 - 21)^2$$

$$= 109945.133 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	1079.1	42.0	21.0	453.2	951727.8	0.110E+06
Wearplate	780.0	14.0	49.0	109.2	535080.0	0.177E+05
Web	14.0	568.0	340.0	79.5	2703680.5	0.427E+05
BasePlate	425.0	36.0	642.0	153.0	9822602.0	0.332E+06
Totals	...	...	...	794.9	14013090.0	0.502E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 14013090/795$$

$$= 176.282 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120)\cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 2405$$

$$= 489.5348 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.1405 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 143.9974 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1535 - 875 \cdot \sin(120/2) / 1.05$$

$$= 811.381 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 397.2070 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 0.1367 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 159.9971 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt})\text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(2405 + 941)425 / (4 \cdot 2500 \cdot 160)}$$

$$= 5.114 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (875 + 42 + 14)$$

$$= 1612.539 \text{ mm}$$

Distance between Ribs [e]:



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$$\begin{aligned}
 &= \text{Web Length} / (\text{Nribs} - 1) \\
 &= 1613 / (4 - 1) \\
 &= 537.513 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned}
 &= e \cdot \text{Bpwid} / 2 \\
 &= 538 \cdot 425 / 2 \\
 &= 1142.215 \text{ cm}^2
 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned}
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 3346 / (2500 \cdot 425) \\
 &= 0.315 \text{ Kg/cm}^2
 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned}
 &= \text{Ap} \cdot \text{Bp} \\
 &= 1142 \cdot 0.31 \\
 &= 359.748 \text{ Kg}
 \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 54 + 37.6 \\
 &= 91.666 \text{ cm}^2
 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned}
 &= P / \text{Ar} \\
 &= 360 / 91.7 \\
 &= 0.385 \text{ N./mm}^2
 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	400.0	...	56.0	...	0.747E+04

Rib dimension [D]:

$$\begin{aligned}
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 400 - 14 \\
 &= 386.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= \text{AY} / \text{A} \\
 &= 0 / 91.7 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \text{Saddle Width} / 2 \\
 &= 400 / 2 \\
 &= 200.000 \text{ mm}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{ \text{Total Inertia} / \text{Total Area} } \\
 &= \sqrt{ 7467 / 91.7 } \\
 &= 90.253 \text{ mm}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\text{theta} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 1535 - \cos(120 / 2) (875 + 42 + 14) - 36 \\
 &= 1033.500 \text{ mm}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{ 2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress} } \\
 &= \sqrt{ 2 \cdot \pi^2 \cdot 199943008 / 240 }
 \end{aligned}$$



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Saddle Calcs: Road Transport:

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$$= 128.242$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1034/90.3$$

$$= 11.451$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 4139 / (2 \cdot 2500) \cdot 538 \cdot 1034 / 2$$

$$= 229.930 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($11.5 < 128$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.5)^2 / (2 \cdot 128^2)) 240 /$$

$$(5/3 + 3 \cdot (11.5) / (8 \cdot 128) - (11.5^3) / (8 \cdot 128^3))$$

$$= 140.6 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 0.38 / 141 + (230 \cdot 200 / 74666672) / 160$$

$$= 0.040$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	386.0	0.0	54.0	0.0	0.747E+04
Web	537.5	14.0	0.0	75.3	0.0	12.3
Totals	...	...	...	129.3	...	0.748E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/129$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400/2$$

$$= 200.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + W_{pdthk})^2 - (Pitch/2)^2 } - B_{pthk}$$

$$= 1535 - \sqrt{ (931 + 14)^2 - (538/2)^2 } - 36$$

$$= 607.635 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 7479/129 }$$

$$= 76.055 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 608/76.1$$

$$= 7.989$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 4139 / (2 \cdot 2500)$$

$$= 0.828 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.83 \cdot 538 \cdot 608$$



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Saddle Calcs: Road Transport:

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= 270.370 Kg-m.

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

= Bending Moment / Section Modulus

= 270/373932

= 7.091 N./mm²

Compressive Allowable, $KL/r < C_c$ (7.99 < 128) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$

= $(1 - (7.99)^2 / (2 \cdot 128^2)) 240 /$

$(5/3 + 3 \cdot (7.99) / (8 \cdot 128) - (7.99^3) / (8 \cdot 128^3))$

= 141.7 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$

= $0.55/142 + (270 \cdot 200/7479) / 160$

= 0.048

Review notes about nozzle loadings and supports in the Warnings report.

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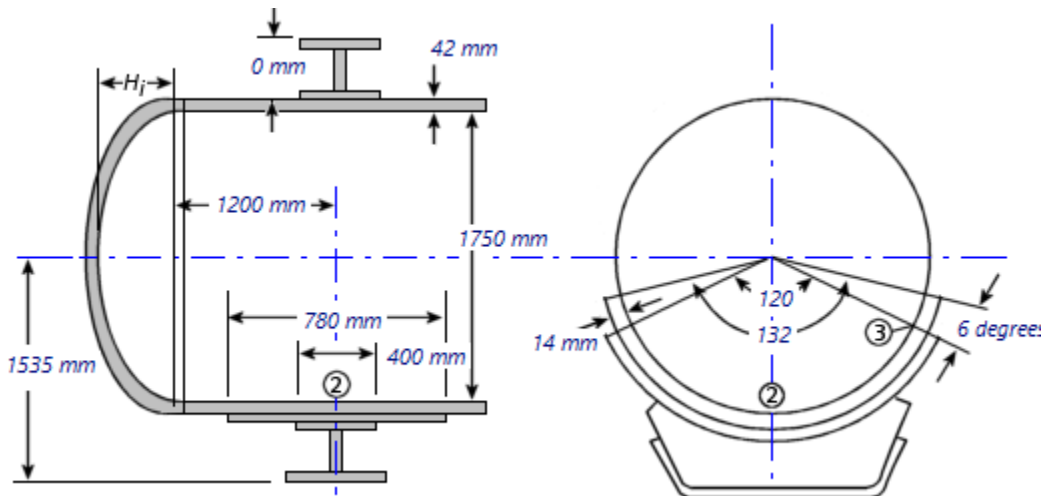
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	897.50	mm
Shell Thickness used in this Case	t	42.000	mm
Stiffened Vessel Length per 4.15.6	L	6298.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1200.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	780.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	118.00	N./mm ²
Shell Allowable Stress used in Calculation		118.00	N./mm ²
Head Allowable Stress used in Calculation		148.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1535.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	61250.86	Kgf
Pressure used in Analysis	P	4958.577 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	54.39	118.00
Long. Stress at Bottom of Midspan	59.73	118.00
Long. Stress at Top of Saddles	66.46	118.00
Long. Stress at Bottom of Saddles	51.85	118.00
Tangential Shear in Shell	12.43	94.40
Circ. Stress at Horn of Saddle	23.42	147.50
Circ. Compressive Stress in Shell	1.25	118.00



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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr}(\text{Ft}/\text{Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$
 $= 3 (775/2 + 0) \cdot 1535/1613$
 $= 1106.1 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$
 $= \max(402, 0) \cdot 1535/3600$
 $= 171.4 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot \text{B} / \text{Ls}$
 $= \max(3778, 0) \cdot 1535/3600$
 $= 1610.8 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= \text{Ftr}(\text{Ft}/\text{Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$
 $= 3 (3778/2 + 0) \cdot 1535/1613$
 $= 5394.2 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 55857 + \max(171, 1106, 1611, 5394)$
 $= 61250.9 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766 (0 + \pi/4(2.03 \cdot 1.2)^2)$
 $= 3942.978 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	62191.95	Kgf
Transverse Shear Load	1888.89	Kgf
Longitudinal Shear Load	3777.78	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot \text{Rm})$
 $= \min(2(1750/2 + 42 + 14) \sin(120/2), 2 \cdot 897)$
 $= 1612.539 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(2a \cdot L)) / (1 + (4\text{hm})/3L)]$
 $= -61251 \cdot 1200 [1 - (1 - 1200/6298 + (897^2 - 0^2) / (2 \cdot 1200 \cdot 6298)) / (1 + (4 \cdot 0) / (3 \cdot 6298))]$
 $= -10087.9 \text{ Kg-m}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L / 4 ((1 + 2(\text{Rm}^2 - \text{hm}^2)/(L^2)) / (1 + (4\text{hm}^2)/(3L)) - 4a/L)$
 $= 61251 \cdot 6298 / 4 ((1 + 2(897^2 - 0^2)/(6298^2)) / (1 + (4 \cdot 0) /$



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$$(3 \cdot 6298) - 4 \cdot 1200 / 6298 \\ = 26855.9 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M2 / (\pi \cdot R_m^2 t) \\ = 4959 \cdot 897 / (2 \cdot 39) - 26856 / (\pi \cdot 897^2 \cdot 39) \\ = 54.39 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M2 / (\pi \cdot R_m^2 \cdot t) \\ = 4959 \cdot 897 / (2 \cdot 39) + 26856 / (\pi \cdot 897^2 \cdot 39) \\ = 59.73 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t) \\ = 4959 \cdot 897 / (2 \cdot 39) - 10088 / (0.11 \cdot \pi \cdot 897^2 \cdot 39) \\ = 66.46 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t) \\ = 4959 \cdot 897 / (2 \cdot 39) + 10088 / (0.19 \cdot \pi \cdot 897^2 \cdot 39) \\ = 51.85 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3)) \\ = 61251 (6298 - 2 \cdot 1200) / (6298 + (4 \cdot 0 / 3)) \\ = 37909.8 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t) \\ = 1.17 \cdot 37910 / (897 \cdot 39) \\ = 12.43 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t} \\ = 0.78 \cdot \sqrt{897 \cdot 39} \\ = 145.930 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a) \\ = \min(400 + 1.56 \cdot \sqrt{897 \cdot 39}, 2 \cdot 1200) \\ = 691.86 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1) \\ = \min(118 / 118, 1) \\ = 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr)) \\ = -0.76 \cdot 61251 \cdot 0.1 / (692(39 + 1 \cdot 14)) \\ = -1.25 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r_{*}]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2) \\ = -61251 / (4(39 + 1 \cdot 14)692) - 12 \cdot 0.053 \cdot 61251 \cdot 897 / (6298(39 + 1 \cdot 14)^2) \\ = -23.42 \text{ N./mm}^2$$

Distance between Saddle Supports [L_s]:

$$= 3600.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature}) \\ = 0.13139\text{E-04} \cdot 3600 (275 - 21.1)$$



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= 12.009 mm

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2500.0000	mm
Baseplate Thickness	Bpthk	36.0000	mm
Baseplate Width	Bpwid	425.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	194.0	N./mm ²
Height of Web at Center	Hw,c	600.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$
 $= ((1.56 \cdot \sqrt{(878 \cdot 39)} + 780) 39^3) / 12$
 $= 528.271 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$
 $= 528 + 417 (182 - 19.5)^2$
 $= 109993.156 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	1068.7	39.0	19.5	416.8	812724.8	0.110E+06
Wearplate	780.0	14.0	46.0	109.2	502320.0	0.201E+05
Web	14.0	568.0	337.0	79.5	2679823.5	0.406E+05
BasePlate	425.0	36.0	639.0	153.0	9776700.0	0.320E+06
Totals	...	...	...	758.5	13771569.0	0.491E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 13771569/759$
 $= 181.563 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle}/2$
 $= 180 - 120/2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.2 \cdot 61251$
 $= 12465.8779 \text{ Kg}$

Tension Stress, St = (Fh/As) = 3.5775 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 116.3996 N./mm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$
 $= 1535 - 878 \cdot \sin(120/2) / 1.05$
 $= 808.900 \text{ mm}$

Bending Moment, M = Fh * d = 10083.8457 Kg-m.



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Bending Stress, $S_b = (M \cdot C_1 / I) = 3.6568 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 129.3329 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt{(3(61251 + 941) \cdot 425) / (4 \cdot 2500 \cdot 129)}$
 $= 24.521 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120/2) (875 + 42 + 14)$
 $= 1612.539 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 1613 / (4 - 1)$
 $= 537.513 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 538 \cdot 425 / 2$
 $= 1142.215 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 62192 / (2500 \cdot 425)$
 $= 5.853 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 1142 \cdot 5.85$
 $= 6685.797 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 54 + 37.6$
 $= 91.666 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 6686 / 91.7$
 $= 7.153 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	400.0	...	56.0	...	0.747E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 400 - 14$
 $= 386.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0 / 91.7$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:



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= Saddle Width / 2
= 400/2
= 200.000 mm

Radius of Gyration [r]:
= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{7467/91.7}$
= 90.253 mm

Length of Outer Rib [L]:
= Saddle Height - $\cos(\theta/2) \cdot (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
= 1535 - $\cos(120/2) \cdot (875 + 42 + 14) - 36$
= 1033.500 mm

Intermediate Term [Cc]:
= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
= $\sqrt{2 \cdot \pi^2 \cdot 199943008 / 194}$
= 142.636

Slenderness ratio [KL/r]:
= KL/r
= 1 · 1034/90.3
= 11.451

Bending Moment [Rm]:
= $FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$
= 3778 / $(2 \cdot 2500) \cdot 538 \cdot 1034 / 2$
= 209.867 Kg-m.

Compressive Allowable, $KL/r < Cc$ (11.5 < 143) per AISC E2-1 [Sca]:
= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
= $(1 - (11.5)^2 / (2 \cdot 143^2)) 194 / (5/3 + 3 \cdot (11.5) / (8 \cdot 143) - (11.5)^3 / (8 \cdot 143^3))$
= 114 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
= $Sc/Sca + (Rm \cdot C1 / I) / Sba$
= 7.15/114 + $(210 \cdot 200 / 74666672) / 129$
= 0.105

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	386.0	0.0	54.0	0.0	0.747E+04
Web	537.5	14.0	0.0	75.3	0.0	12.3
Totals	...	...	...	129.3	...	0.748E+04

Distance to Centroid from Datum [ytot]:
= AY / A
= 0/129
= 0.000 mm

Distance to Centroid [C1]:
= Saddle Width / 2
= 400/2
= 200.000 mm

Length of Inner Rib [L]:
= Saddle Height - $\sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$
= 1535 - $\sqrt{(931 + 14)^2 - (538/2)^2} - 36$
= 607.635 mm

Radius of Gyration [r]:



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$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{(7479/129)}$$

$$= 76.055 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 608/76.1$$

$$= 7.989$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 3778 / (2 \cdot 2500)$$

$$= 0.756 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.76 \cdot 538 \cdot 608$$

$$= 246.779 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 247/373932$$

$$= 6.472 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($7.99 < 143$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.99)^2 / (2 \cdot 143^2)) 194 / (5/3 + 3 \cdot (7.99) / (8 \cdot 143) - (7.99)^3 / (8 \cdot 143^3))$$

$$= 114.8 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 10.1/115 + (247 \cdot 200/7479) / 129$$

$$= 0.138$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	6
Total Number of Bolts in Tension/Baseplate	Nbt	3
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	118.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8674 cm ²
Saddle Load Q0 (Weight)	Q0	56797.8 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	1610.8 Kg
Maximum Transverse Force	Ft	1888.9 Kg
Maximum Longitudinal Force	F1	3777.8 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 3778 / (4.87 \cdot 6)$$

$$= 12.7 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1889 / (4.87 \cdot 6)$$

$$= 6.3 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss



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Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F_l / (BoltShearAllowable \cdot Nbolts)$
 = $3778 / (82.8 \cdot 6)$
 = 0.7457 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot F_t + \text{Sum of X Moments}$
 = $1535 \cdot 1889 + 0$
 = 2899.51 Kg-m.

Eccentricity (e):
 = $Rmom / QO$
 = 2900/56798
 = 51.05 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0, 0.75, 0]
 = 0.7457 cm²

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: A

Govrn. thk, tg = 14, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.99$, Temp. Reduction = 1 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve A	2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	1 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	897.50	mm
Shell Thickness used in this Case	t	42.000	mm
Stiffened Vessel Length per 4.15.6	L	6298.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1200.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	780.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	118.00	N./mm ²
Shell Allowable Stress used in Calculation		118.00	N./mm ²
Head Allowable Stress used in Calculation		0.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1535.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:



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Saddle Force Q, Operating Case 9959.63 Kg
 Pressure used in Analysis P 4958.577 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	56.62	118.00
Long. Stress at Bottom of Midspan	57.49	118.00
Long. Stress at Top of Saddles	58.59	118.00
Long. Stress at Bottom of Saddles	56.21	118.00
Tangential Shear in Shell	2.02	94.40
Circ. Stress at Horn of Saddle	3.81	147.50
Circ. Compressive Stress in Shell	0.20	118.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (775/2 + 0) \cdot 1535/1613$
 $= 1106.1\ Kg$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(402, 0) \cdot 1535/3600$
 $= 171.4\ Kg$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(3778, 0) \cdot 1535/3600$
 $= 1610.8\ Kg$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (3778/2 + 0) \cdot 1535/1613$
 $= 5394.2\ Kg$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 4565 + \max(171, 1106, 1611, 5394)$
 $= 9959.6\ Kg$

Longitudinal Wind Force [FI]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 766 (0 + \pi/4(2.03 \cdot 1.2)^2)$
 $= 3942.978\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	10900.72	Kg
Transverse Shear Load	1888.89	Kgf
Longitudinal Shear Load	3777.78	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \min(2(ShellID/2 + t + WearPadThickness) \sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(1750/2 + 42 + 14) \sin(120/2), 2 \cdot 897)$
 $= 1612.539\ mm$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			



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The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(R_m^2 - h m^2)}{(2a \cdot L)} \right) / (1 + (4hm)/3L) \right]$$

$$= -9960 \cdot 1200 \left[1 - \left(1 - \frac{1200}{6298} + \frac{(897^2 - 0^2)}{(2 \cdot 1200 \cdot 6298)} \right) / (1 + (4 \cdot 0)/(3 \cdot 6298)) \right]$$

$$= -1640.3 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(\frac{(1 + 2(R_m^2 - h m^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L}{(3 \cdot 6298)} \right) - 4 \cdot 1200 / 6298$$

$$= 9960 \cdot 6298 / 4 \left(\frac{(1 + 2(897^2 - 0^2)/(6298^2)) / (1 + (4 \cdot 0)/(3 \cdot 6298)) - 4 \cdot 1200 / 6298}{(3 \cdot 6298)} \right) - 4 \cdot 1200 / 6298$$

$$= 4366.9 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= 4959 \cdot 897 / (2 \cdot 39) - 4367 / (\pi \cdot 897^2 \cdot 39)$$

$$= 56.62 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 4959 \cdot 897 / (2 \cdot 39) + 4367 / (\pi \cdot 897^2 \cdot 39)$$

$$= 57.49 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 4959 \cdot 897 / (2 \cdot 39) - 1640 / (0.11 \cdot \pi \cdot 897^2 \cdot 39)$$

$$= 58.59 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 4959 \cdot 897 / (2 \cdot 39) + 1640 / (0.19 \cdot \pi \cdot 897^2 \cdot 39)$$

$$= 56.21 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h m / 3))$$

$$= 9960 (6298 - 2 \cdot 1200) / (6298 + (4 \cdot 0 / 3))$$

$$= 6164.3 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 6164 / (897 \cdot 39)$$

$$= 2.02 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(897 \cdot 39)}$$

$$= 145.930 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(400 + 1.56 \cdot \sqrt{(897 \cdot 39)}, 2 \cdot 1200)$$

$$= 691.86 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(118 / 118, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot t_r))$$

$$= -0.76 \cdot 9960 \cdot 0.1 / (692(39 + 1 \cdot 14))$$



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= -0.20 N./mm²

Circ. Comp. Stress at Horn of Saddle, L<8Rm (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2)$$

$$= -9960 / (4(39 + 1 \cdot 14)692) -$$

$$12 \cdot 0.053 \cdot 9960 \cdot 897 / (6298(39 + 1 \cdot 14)^2)$$

$$= -3.81 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2500.0000	mm
Baseplate Thickness	Bpthk	36.0000	mm
Baseplate Width	Bpwid	425.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	194.0	N./mm ²
Height of Web at Center	Hw,c	600.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(878 \cdot 39)} + 780)39^3) / 12$$

$$= 528.271 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 528 + 417(182 - 19.5)^2$$

$$= 109993.156 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	1068.7	39.0	19.5	416.8	812724.8	0.110E+06
Wearplate	780.0	14.0	46.0	109.2	502320.0	0.201E+05
Web	14.0	568.0	337.0	79.5	2679823.5	0.406E+05
BasePlate	425.0	36.0	639.0	153.0	9776700.0	0.320E+06
Totals	...	...	...	758.5	13771569.0	0.491E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 13771569 / 759$$

$$= 181.563 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120)\cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K_1 \cdot Q$$

$$= 0.2 \cdot 9960$$

$$= 2026.9999 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.5817 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 116.3996 \text{ N./mm}^2$$



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Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$
 $= 1535 - 878 \cdot \sin(120/2) / 1.05$
 $= 808.900 \text{ mm}$

Bending Moment, $M = F_h \cdot d = 1639.6722 \text{ Kg-m.}$

Bending Stress, $S_b = (M \cdot C_1 / I) = 0.5946 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 129.3329 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$
 $= \sqrt{3(9960 + 941) \cdot 425 / (4 \cdot 2500 \cdot 129)}$
 $= 10.266 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120/2) (875 + 42 + 14)$
 $= 1612.539 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 1613 / (4 - 1)$
 $= 537.513 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 538 \cdot 425 / 2$
 $= 1142.215 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 10901 / (2500 \cdot 425)$
 $= 1.026 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 1142 \cdot 1.03$
 $= 1171.855 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 54 + 37.6$
 $= 91.666 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 1172 / 91.7$
 $= 1.254 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	400.0	...	56.0	...	0.747E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 400 - 14$
 $= 386.000 \text{ mm}$



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Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/91.7$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400/2$$

$$= 200.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{7467/91.7}$$

$$= 90.253 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1535 - \cos(120/2) (875 + 42 + 14) - 36$$

$$= 1033.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008/194)}$$

$$= 142.636$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1034/90.3$$

$$= 11.451$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 3778 / (2 \cdot 2500) \cdot 538 \cdot 1034/2$$

$$= 209.867 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($11.5 < 143$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (11.5)^2 / (2 \cdot 143^2)) 194 / (5/3 + 3 \cdot (11.5) / (8 \cdot 143) - (11.5)^3 / (8 \cdot 143^3))$$

$$= 114 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 1.25/114 + (210 \cdot 200/74666672) / 129$$

$$= 0.054$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	386.0	0.0	54.0	0.0	0.747E+04
Web	537.5	14.0	0.0	75.3	0.0	12.3
Totals	...	...	...	129.3	...	0.748E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/129$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400/2$$

$$= 200.000 \text{ mm}$$



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Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1535 - \sqrt{(931 + 14)^2 - (538/2)^2} - 36$$

$$= 607.635 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{7479/129}$$

$$= 76.055 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= \text{KL}/r$$

$$= 1 \cdot 608/76.1$$

$$= 7.989$$

Unit Force [Force,u]:

$$= \text{FI} / (2 \cdot \text{Baseplate Length})$$

$$= 3778 / (2 \cdot 2500)$$

$$= 0.756 \text{ Kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.76 \cdot 538 \cdot 608$$

$$= 246.779 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 247/373932$$

$$= 6.472 \text{ N./mm}^2$$

Compressive Allowable, $\text{KL}/r < C_c$ ($7.99 < 143$) per AISC E2-1 [Sca]:

$$= (1 - (\text{KL}/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (\text{KL}/r) / (8 \cdot C_c) - (\text{KL}/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.99)^2 / (2 \cdot 143^2)) 194 / (5/3 + 3 \cdot (7.99) / (8 \cdot 143) - (7.99^3) / (8 \cdot 143^3))$$

$$= 114.8 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= \text{Sc}/\text{Sca} + (\text{Mbase,c} \cdot C1/I) / \text{Sba}$$

$$= 1.78/115 + (247 \cdot 200/7479) / 129$$

$$= 0.066$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	6
Total Number of Bolts in Tension/Baseplate	Nbt	3
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	118.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8674 cm ²
Saddle Load Q0 (Weight)	Q0	5506.5 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	1610.8 Kgf
Maximum Transverse Force	Ft	1888.9 Kgf
Maximum Longitudinal Force	F1	3777.8 Kgf
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

$$= 17039.3 \text{ Kgf}$$

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= \max(F_f, F_1) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$



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$$= 17039 / (4.87 \cdot 6)$$

$$= 57.2 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1889 / (4.87 \cdot 6)$$

$$= 6.3 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$$

$$= 3778 / (82.8 \cdot 6)$$

$$= 0.7457 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 1535 \cdot 1889 + 0$$

$$= 2899.51 \text{ Kg-m.}$$

Eccentricity (e):

$$= R_{mom} / QO$$

$$= 2900 / 5507$$

$$= 526.55 \text{ mm} > B_{plen} / 6 \text{ --> Uplift in Transverse direction}$$

f = B_{plen} / 2 - Edgedis

$$= 2500 / 2 - 50.8$$

$$= 1199.20 \text{ mm}$$

Modular Ratio Of Steel/Concrete (n₁):

$$= E_S / E_C$$

$$= 203402 / 21526$$

$$= 9.45$$

K₁ = 3 (e - 0.5 • B_{plen})

$$= 3 (527 - 0.5 \cdot 2500)$$

$$= -2170.36 \text{ mm}$$

K₂ = 6 • n₁ • A_t / B_{pwid} • (f + e)

$$= 6 \cdot 9.45 \cdot 14.6 / 425 \cdot (1199 + 527)$$

$$= 336159.57 \text{ mm}^2$$

K₃ = -K₂ • (0.5 • B_{plen} + f)

$$= -336160 \cdot (0.5 \cdot 2500 + 1199)$$

$$= -823322022.41 \text{ mm}^3$$

Iteratively Solving for the Effective Bearing Length:

$$Y^3 + K_1 \cdot Y^2 + K_2 \cdot Y + K_3 = 0$$

$$Y^3 + -2170 \cdot Y^2 + 336160 \cdot Y + -823321984 = 0$$

$$Y = 2188.64 \text{ mm}$$

Num = (B_{plen} / 2 - Y / 3 - e)

$$= (2500 / 2 - 2189 / 3 - 527)$$

$$= -6.10$$

Denom = (B_{plen} / 2 - Y / 3 + f)

$$= (2500 / 2 - 2189 / 3 + 1199)$$

$$= 1719.65$$

Total Bolt Tension Force [Tforce]:



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= - QO • Num / Denom
 = - 5507 • -6.1/1720
 = 19.52 Kgf

Bolt Area Required due to Transverse Load [Bltareart]:
 = Tforce / (Stba • Nbt)
 = 19.5 / (138 • 3)
 = 0.0046 cm²

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0, 0.75, 0.0046]
 = 0.7457 cm²

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)
 = 2(QO + Tforce) / (Y • Bpwid)
 = 2(5507 + 19.5) / (2189 • 425)
 = 116.52 KPa.

Distance from Baseplate Edge to the Web [ADIST]:
 = (Bplen - WebIngh) / 2
 = (2500 - 2449) / 2
 = 25.4000 mm

Overturning Moment due To Bolt Tension [Mt]:
 = Tforce • Adist
 = 19.5 • 25.4
 = 0.50 Kg-m.

Equivalent Bearing Pressure (f1):
 = fc • (Y - Adist) / Y
 = 117 • (2189 - 25.4) / 2189
 = 115.16 KPa.

Overturning Moment due to Bearing Pressure [Mc]:
 = (Adist² • Bpwid / 6) • (f1 + 2 • fc)
 = (25.4² • 425 / 6) • (115 + 2 • 117)
 = 1.62 Kg-m.

Baseplate Required Thickness [Treq]:
 = $\sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))}$
 = $\sqrt{(6 \cdot \max(0.5, 1.62) / (425 \cdot 177))}$
 = 1.1266 mm

MDMT Calculations for: New Saddle, Saddle to Wearplate Junction, Curve: A

Govrn. thk, tg = 14, c = 3 mm, E* = 1
 Thickness Ratio = tr • E* / (tg - c) = 0.99, Temp. Reduction = 1 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve A	2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	1 °C

Review notes about nozzle loadings and supports in the Warnings report.

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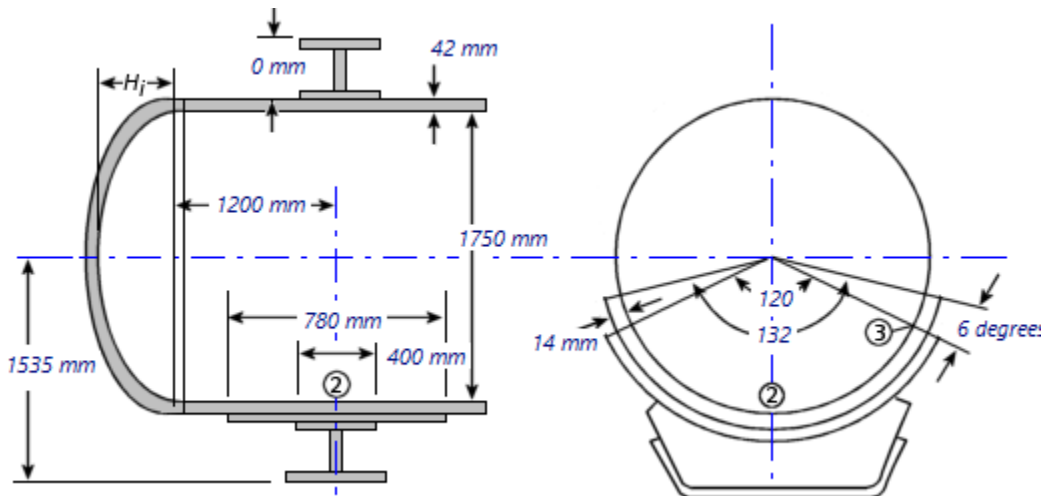
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	897.50	mm
Shell Thickness used in this Case	t	42.000	mm
Stiffened Vessel Length per 4.15.6	L	6298.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1200.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	780.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	118.00	N./mm ²
Shell Allowable Stress used in Calculation		216.00	N./mm ²
Head Allowable Stress used in Calculation		216.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1535.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		57377.88	Kgf
Pressure used in Analysis	P	6443.797	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	71.65	216.00
Long. Stress at Bottom of Midspan	76.65	216.00
Long. Stress at Top of Saddles	82.96	216.00
Long. Stress at Bottom of Saddles	69.27	216.00
Tangential Shear in Shell	11.65	127.92
Circ. Stress at Horn of Saddle	21.94	323.99
Circ. Compressive Stress in Shell	1.17	216.00



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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (387/2 + 0) \cdot 1535/1613$$

$$= 553.1 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(201, 0) \cdot 1535/3600$$

$$= 85.7 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 56825 + \max(85.7, 553, 0, 0)$$

$$= 57377.9 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (2.03 \cdot 1.2)^2)$$

$$= 3942.978 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 58318.97 Kgf

Transverse Shear Load 193.66 Kgf

Longitudinal Shear Load 201.04 Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1750/2 + 42 + 14)\sin(120/2), 2 \cdot 897)$$

$$= 1612.539 \text{ mm}$$

The Computed K values from Table 4.15.1:

$$K1 = 0.1066 \quad K2 = 1.1707 \quad K3 = 0.8799 \quad K4 = 0.4011$$

$$K5 = 0.7603 \quad K6 = 0.0529 \quad K7 = 0.0529 \quad K8 = 0.3405$$

$$K9 = 0.2711 \quad K10 = 0.0581 \quad K1^* = 0.1923 \quad K6p = 0.0434$$

$$K7p = 0.0434$$

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -57378 \cdot 1200 [1 - (1 - 1200/6298 + (897^2 - 0^2)/(2 \cdot 1200 \cdot 6298))/(1 + (4 \cdot 0)/(3 \cdot 6298))]$$

$$= -9450.0 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 57378 \cdot 6298/4 ((1 + 2(897^2 - 0^2)/(6298^2))/(1 + (4 \cdot 0)/(3 \cdot 6298)) - 4 \cdot 1200/6298)$$

$$= 25157.8 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 6444 \cdot 897/(2 \cdot 39) - 25158/(\pi \cdot 897^2 \cdot 39)$$

$$= 71.65 \text{ N/mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$



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$$= 6444 \cdot 897 / (2 \cdot 39) + 25158 / (\pi \cdot 897^2 \cdot 39)$$

$$= 76.65 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 6444 \cdot 897 / (2 \cdot 39) - 9450 / (0.11 \cdot \pi \cdot 897^2 \cdot 39)$$

$$= 82.96 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 6444 \cdot 897 / (2 \cdot 39) + 9450 / (0.19 \cdot \pi \cdot 897^2 \cdot 39)$$

$$= 69.27 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 57378 (6298 - 2 \cdot 1200) / (6298 + (4 \cdot 0 / 3))$$

$$= 35512.7 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [$\tau2$]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 35513 / (897 \cdot 39)$$

$$= 11.65 \text{ N./mm}^2$$

Decay Length (4.15.20) [$x1, x2$]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(897 \cdot 39)}$$

$$= 145.930 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(400 + 1.56 \cdot \sqrt{(897 \cdot 39)}, 2 \cdot 1200)$$

$$= 691.86 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [$\sigma6, r$]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 57378 \cdot 0.1 / (692(39 + 1 \cdot 14))$$

$$= -1.17 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [$\sigma7, r^*$]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -57378 / (4(39 + 1 \cdot 14)692) - 12 \cdot 0.053 \cdot 57378 \cdot 897 / (6298(39 + 1 \cdot 14)^2)$$

$$= -21.94 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2500.0000	mm
Baseplate Thickness	Bpthk	36.0000	mm
Baseplate Width	Bpwid	425.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	240.0	N./mm ²
Height of Web at Center	Hw,c	600.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(878 \cdot 39)} + 780)^3) / 12$$

$$= 528.271 \text{ cm}^4$$



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Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 528 + 417 (182 - 19.5)^2$$

$$= 109993.156 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	1068.7	39.0	19.5	416.8	812724.8	0.110E+06
Wearplate	780.0	14.0	46.0	109.2	502320.0	0.201E+05
Web	14.0	568.0	337.0	79.5	2679823.5	0.406E+05
BasePlate	425.0	36.0	639.0	153.0	9776700.0	0.320E+06
Totals	...	...	...	758.5	13771569.0	0.491E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 13771569/759$$

$$= 181.563 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 57378$$

$$= 11677.6416 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 3.3513 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 143.9974 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1535 - 878 \cdot \sin(120/2) / 1.05$$

$$= 808.900 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 9446.2295 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 3.4256 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 159.9971 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt[3]{(3(57378 + 941) \cdot 425) / (4 \cdot 2500 \cdot 160)}$$

$$= 21.349 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (875 + 42 + 14)$$

$$= 1612.539 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 1613 / (4 - 1)$$

$$= 537.513 \text{ mm}$$



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Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 538 \cdot 425 / 2$$

$$= 1142.215 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 58319 / (2500 \cdot 425)$$

$$= 5.489 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1142 \cdot 5.49$$

$$= 6269.441 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 54 + 37.6$$

$$= 91.666 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 6269 / 91.7$$

$$= 6.707 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	400.0	...	56.0	...	0.747E+04

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 400 - 14$$

$$= 386.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 91.7$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 400 / 2$$

$$= 200.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (7467 / 91.7) }$$

$$= 90.253 \text{ mm}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 1535 - \cos(120 / 2) (875 + 42 + 14) - 36$$

$$= 1033.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 199943008 / 240) }$$

$$= 128.242$$

Slenderness ratio [KL/r]:

$$= KL / r$$



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$$= 1 \cdot 1034/90.3$$

$$= 11.451$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 201 / (2 \cdot 2500) \cdot 538 \cdot 1034 / 2$$

$$= 11.168 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($11.5 < 128$) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr)^3 / (8 \cdot Cc^3))$$

$$= (1 - (11.5)^2 / (2 \cdot 128^2)) 240 /$$

$$(5/3 + 3 \cdot (11.5) / (8 \cdot 128) - (11.5^3) / (8 \cdot 128^3))$$

$$= 140.6 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 6.71/141 + (11.2 \cdot 200/74666672) / 160$$

$$= 0.050$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	386.0	0.0	54.0	0.0	0.747E+04
Web	537.5	14.0	0.0	75.3	0.0	12.3
Totals	...	...	...	129.3	...	0.748E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/129$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400/2$$

$$= 200.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + Wpdthk)^2 - (Pitch/2)^2 } - Bpthk$$

$$= 1535 - \sqrt{ (931 + 14)^2 - (538/2)^2 } - 36$$

$$= 607.635 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (7479/129) }$$

$$= 76.055 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 608/76.1$$

$$= 7.989$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 201 / (2 \cdot 2500)$$

$$= 0.040 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.04 \cdot 538 \cdot 608$$

$$= 13.133 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma B, \text{base}, c$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$



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= 13.1/373932
 = 0.344 N./mm²

Compressive Allowable, $KL/r < C_c$ ($7.99 < 128$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.99)^2 / (2 \cdot 128^2)) 240 / (5/3 + 3 \cdot (7.99) / (8 \cdot 128) - (7.99^3) / (8 \cdot 128^3))$$

 = 141.7 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$
 = $9.51 / 142 + (13.1 \cdot 200 / 7479) / 160$
 = 0.069

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	6
Total Number of Bolts in Tension/Baseplate	Nbt	3
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	118.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8674 cm ²
Saddle Load QO (Weight)	QO	57765.9 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	85.7 Kg
Maximum Transverse Force	Ft	193.7 Kg
Maximum Longitudinal Force	F1	201.0 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
 = $F_1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $201 / (4.87 \cdot 6)$
 = 0.7 N./mm². Must be less than 29.0 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
 = $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $194 / (4.87 \cdot 6)$
 = 0.7 N./mm². Must be less than 29.0 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F_1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 = $201 / (29 \cdot 6)$
 = 0.1134 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot F_t + \text{Sum of X Moments}$
 = $1535 \cdot 194 + 0$
 = 297.28 Kg-m.

Eccentricity (e):
 = R_{mom} / QO
 = $297 / 57766$
 = 5.15 mm < $B_{plen} / 6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)



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Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0, 0.11, 0]
 = 0.1134 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	897.50	mm
Shell Thickness used in this Case	t	42.000	mm
Stiffened Vessel Length per 4.15.6	L	6298.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1200.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	780.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	14.0	mm
Wear Plate Allowable Stress	fw or Sr	118.00	N./mm ²
Shell Allowable Stress used in Calculation		216.00	N./mm ²
Head Allowable Stress used in Calculation		216.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1535.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		4094.02	Kgf
Pressure used in Analysis	P	6443.797	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	73.97	216.00
Long. Stress at Bottom of Midspan	74.33	216.00
Long. Stress at Top of Saddles	74.78	216.00
Long. Stress at Bottom of Saddles	73.80	216.00
Tangential Shear in Shell	0.83	127.92
Circ. Stress at Horn of Saddle	1.57	323.99
Circ. Compressive Stress in Shell	0.08	216.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 = Ftr(Ft/Num of Saddles + Z Force Load) • B / E
 = 3 (387/2 + 0) • 1535/1613
 = 553.1 Kgf

Longitudinal Saddle Reaction Force [Fwl]:
 = max(Fl,wind; Sum of X Forces) • B / Ls
 = max(201, 0) • 1535/3600
 = 85.7 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:
 = Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 3541 + max(85.7, 553, 0, 0)



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= 4094.0 Kgf

Longitudinal Wind Force [FI]:

= WindScalar • WindPress (Platform Area + ($\pi/4$ (OD • WindDiaMult)²))
 = 1.1 • 766 (0 + $\pi/4$ (2.03 • 1.2)²)
 = 3942.978 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	5035.11	Kgf
Transverse Shear Load	193.66	Kgf
Longitudinal Shear Load	201.04	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)
 = min(2(1750/2 + 42 + 14)sin(120/2), 2•897)
 = 1612.539 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]
 = -4094•1200[1-(1-1200/6298+(897²-0²)/
 (2•1200•6298))/(1+(4•0)/(3•6298))]
 = -674.3 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)
 = 4094•6298/4((1+2(897²-0²)/(6298²))/(1+(4•0)/
 (3•6298))-4•1200/6298)
 = 1795.1 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= P • Rm/(2t) - M2/(π •Rm²t)
 = 6444 • 897/(2•39) - 1795/(π •897²•39)
 = 73.97 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= P • Rm/(2t) + M2/(π • Rm² • t)
 = 6444 • 897/(2 • 39) + 1795/(π • 897² • 39)
 = 74.33 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

= P • Rm/(2t) - M1/(K1• π •Rm²t)
 = 6444•897/(2•39)--674/(0.11• π •897²•39)
 = 74.78 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)
 = 6444•897/(2•39)+-674/(0.19• π •897²•39)
 = 73.80 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))
 = 4094 (6298 - 2 • 1200)/(6298 + (4 • 0/3))
 = 2533.9 Kgf



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Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 2534 / (897 \cdot 39)$$

$$= 0.83 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(897 \cdot 39)}$$

$$= 145.930 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B_1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(400 + 1.56 \cdot \sqrt{(897 \cdot 39)}, 2 \cdot 1200)$$

$$= 691.86 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 4094 \cdot 0.1 / (692(39 + 1 \cdot 14))$$

$$= -0.08 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -4094 / (4(39 + 1 \cdot 14)692) -$$

$$12 \cdot 0.053 \cdot 4094 \cdot 897 / (6298(39 + 1 \cdot 14)^2)$$

$$= -1.57 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2500.0000	mm
Baseplate Thickness	Bpthk	36.0000	mm
Baseplate Width	Bpwid	425.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	240.0	N./mm ²
Height of Web at Center	Hw,c	600.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(878 \cdot 39)} + 780)39^3) / 12$$

$$= 528.271 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 528 + 417(182 - 19.5)^2$$

$$= 109993.156 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	1068.7	39.0	19.5	416.8	812724.8	0.110E+06
Wearplate	780.0	14.0	46.0	109.2	502320.0	0.201E+05
Web	14.0	568.0	337.0	79.5	2679823.5	0.406E+05
BasePlate	425.0	36.0	639.0	153.0	9776700.0	0.320E+06
Totals	...	...	...	758.5	13771569.0	0.491E+06

Distance to Centroid [C1]:



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$$\begin{aligned}
 &= AY / A \\
 &= 13771569/759 \\
 &= 181.563 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Angle } [\beta]: \\
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 120/2 \\
 &= 120.0
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Coefficient [K1]:} \\
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\
 &= 0.2035
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Force [Fh]:} \\
 &= K1 \cdot Q \\
 &= 0.2 \cdot 4094 \\
 &= 833.2211 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh / As) = 0.2391 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 143.9974 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Dimension [d]:} \\
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 1535 - 878 \cdot \sin(120/2) / 1.05 \\
 &= 808.900 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 674.0057 \text{ Kg-m.}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 0.2444 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 159.9971 \text{ N./mm}^2
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\
 &= \sqrt{3(4094 + 941) \cdot 425 / (4 \cdot 2500 \cdot 160)} \\
 &= 6.273 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

$$\begin{aligned}
 \text{Web Length Dimension [Web Length]:} \\
 &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120/2) (875 + 42 + 14) \\
 &= 1612.539 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Distance between Ribs [e]:} \\
 &= \text{Web Length} / (N_{\text{ribs}} - 1) \\
 &= 1613 / (4 - 1) \\
 &= 537.513 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Baseplate Pressure Area [Ap]:} \\
 &= e \cdot B_{\text{pwid}} / 2 \\
 &= 538 \cdot 425 / 2 \\
 &= 1142.215 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Bearing Pressure [Bp]:} \\
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 5035 / (2500 \cdot 425) \\
 &= 0.474 \text{ Kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Axial Load [P]:} \\
 &= Ap \cdot Bp \\
 &= 1142 \cdot 0.47 \\
 &= 541.287 \text{ Kgf}
 \end{aligned}$$

$$\text{Area of the Rib and Web [Ar]:}$$



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= Rib Area + Web Area

= 54 + 37.6

= 91.666 cm²

Compressive Stress [Sc]:

= P/Ar

= 541/91.7

= 0.579 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	400.0	...	56.0	...	0.747E+04

Rib dimension [D]:

= Saddle Width - Web Thickness

= 400 - 14

= 386.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/91.7

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 400/2

= 200.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{7467/91.7}$

= 90.253 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk

= 1535 - cos(120/2)(875 + 42 + 14) - 36

= 1033.500 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 199943008/240}$

= 128.242

Slenderness ratio [KL/r]:

= KL/r

= 1 • 1034/90.3

= 11.451

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 201 / (2 • 2500) • 538 • 1034/2

= 11.168 Kg-m.

Compressive Allowable, KL/r < Cc (11.5 < 128) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (11.5)^2 / (2 \cdot 128^2)) 240 /$

$(5/3 + 3 \cdot (11.5) / (8 \cdot 128) - (11.5^3) / (8 \cdot 128^3))$

= 140.6 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba



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$$= 0.58/141 + (11.2 \cdot 200/74666672)/160$$

$$= 0.006$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	386.0	0.0	54.0	0.0	0.747E+04
Web	537.5	14.0	0.0	75.3	0.0	12.3
Totals	...	...	...	129.3	...	0.748E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/129$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400/2$$

$$= 200.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1535 - \sqrt{(931 + 14)^2 - (538/2)^2} - 36$$

$$= 607.635 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(7479/129)}$$

$$= 76.055 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 608/76.1$$

$$= 7.989$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 201/(2 \cdot 2500)$$

$$= 0.040 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.04 \cdot 538 \cdot 608$$

$$= 13.133 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 13.1/373932$$

$$= 0.344 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($7.99 < 128$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.99)^2 / (2 \cdot 128^2)) 240 / (5/3 + 3 \cdot (7.99) / (8 \cdot 128) - (7.99^3) / (8 \cdot 128^3))$$

$$= 141.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 0.82/142 + (13.1 \cdot 200/7479)/160$$

$$= 0.008$$

Input Data for Base Plate Bolting Calculations:



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Total Number of Bolts per BasePlate	Nbolts	6
Total Number of Bolts in Tension/Baseplate	Nbt	3
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	118.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8674 cm ²
Saddle Load Q0 (Weight)	Q0	4482.1 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	85.7 Kg
Maximum Transverse Force	Ft	193.7 Kg
Maximum Longitudinal Force	F1	201.0 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 201 / (4.87 \cdot 6)$$

$$= 0.7 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 194 / (4.87 \cdot 6)$$

$$= 0.7 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (Q0 > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 201 / (29 \cdot 6)$$

$$= 0.1134 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1535 \cdot 194 + 0$$

$$= 297.28 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 297 / 4482$$

$$= 66.32 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.11, 0]$$

$$= 0.1134 \text{ cm}^2$$

Review notes about nozzle loadings and supports in the Warnings report.

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 Nozzle Calcs.: N3-Tube Inlet/L Nozl: 17 9:34am Sep 9,2026

Input, Nozzle Desc: N3-Tube Inlet/L

From: 30

Pressure for Reinforcement Calculations	P	1267.154	KPa.
Temperature for Internal Pressure	Temp	250	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	118.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	118.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1750.00	mm
Design Length of Section	L	1200.0001	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1013.18	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	148.00	N./mm ²
Allowable Stress At Ambient	Sna	148.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		12.7500	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	17.4498	mm
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	35.0000	mm
Height of Beveled Transition	L'	20.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
ASME Code Weld Type per UW-16		g	
Flange Class		300	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



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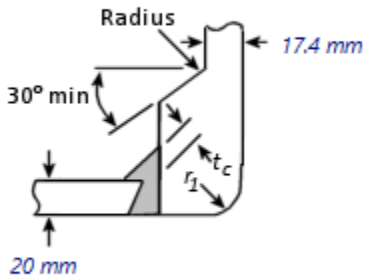
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Nozzle Calcs.: N3-Tube Inlet/L

Nozl: 17

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UW-16.1 [g]

Reinforcement CALCULATION, Description: N3-Tube Inlet/L

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	12.750 in.
Actual Thickness Used in Calculation	0.687 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 1267 \cdot 878 / (118 \cdot 1 - 0.6 \cdot 1267)$$

$$= 9.4904 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= 1267 \cdot 162 / (148 \cdot 1 + 0.4 \cdot 1267)$$

$$= 1.3818 \text{ mm}$$

Required Nozzle thickness under External Pressure per UG-28 : 1.0182 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 32.6 / 0.5773$$

$$= 56.3786 \text{ mm}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	589.9008	mm
--	----	----------	----

Parallel to Vessel Wall, opening length	d	294.9504	mm
---	---	----------	----

Normal to Vessel Wall (Thickness Limit), no pad	T1np	42.5000	mm
---	------	---------	----

Weld Strength Reduction Factor [fr1]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 148 / 118)$$

$$= 1.000$$



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Nozzle Calcs.: N3-Tube Inlet/L

Nozl: 17

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Weld Strength Reduction Factor [fr2]:

$$= \min(1, S_n/S_v)$$

$$= \min(1, 148/118)$$

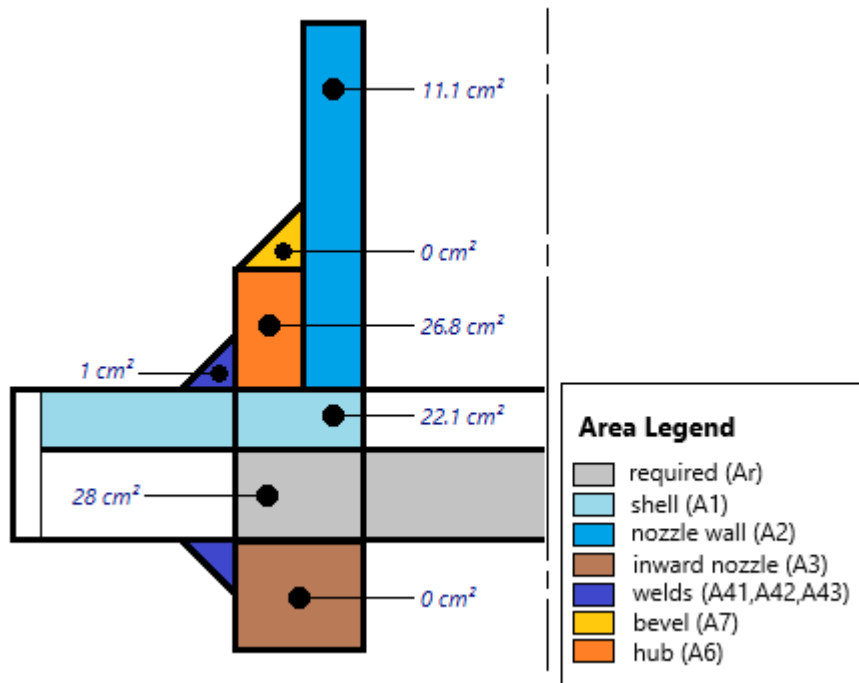
$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(1, 1)$$

$$= 1.000$$



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	27.992	7.211	NA
Area in Shell	A1	22.150	35.720	NA
Area in Nozzle Wall	A2	11.108	11.417	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub + Bevel	A6	26.752	26.752	NA
TOTAL AREA AVAILABLE	Atot	61.010	74.889	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (295 \cdot 9.49 \cdot 1 + 2 \cdot 47 \cdot 9.49 \cdot 1 \cdot (1 - 1))$$

$$= 27.992 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:



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$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 295(1 \cdot 17 - 1 \cdot 9.49) - 2 \cdot 47$$

$$(1 \cdot 17 - 1 \cdot 9.49) \cdot (1 - 1)$$

$$= 22.150 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 42.5)(14.4 - 1.38)1$$

$$= 11.108 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(Tnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$

$$= (2 \cdot \min(42.5, 300, 35)) \cdot (50 - 17.4) \cdot 1$$

$$= 26.752 \text{ cm}^2 \text{ Includes Bevel Area of } 3.967 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.3818 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 12.4904 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 12.4904 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.7700 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(12.5, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(4.38, 11.3)$$

$$= 11.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 17.4498 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	21.7,	Allowable	:	148.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	348.3 N./mm ²	Passed
Occasional	:	6.2,	Allowable	:	196.8 N./mm ²	Passed
Shear	:	12.3,	Allowable	:	103.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 17.4, tr = 1.38, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.096, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 20, tr = 9.49, c = 3 mm, E* = 1



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Thickness Ratio = $tr * E / (tg - c) = 0.56$, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 12 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -14 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -14 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1267.15 / 5170.01 = 0.245$

Weld Size Calculations, Description: N3-Tube Inlet/L

Intermediate Calc. for nozzle/shell Welds Tmin 17.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = $0.7 * Wo$ mm

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1450 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 15.1132 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 335.1133 mm

Input Echo, WRC297 Item 1, Description: N3-Tube Inlet/:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1750.000	mm
Shell Thickness	20.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 118.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	388.950	mm
Nozzle Thickness	50.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc 148.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 148.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure Dp 1267.15 KPa.



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Nozzle Calcs.: N3-Tube Inlet/L

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Radial Load	P	-2294.50	Kgf
Circumferential Shear	Vc	1529.68	Kgf
Longitudinal Shear	Vl	1529.68	Kgf
Circumferential Moment	Mc	1284.43	Kg-m.
Longitudinal Moment	Ml	1284.43	Kg-m.
Torsional Moment	Mt	1834.90	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (7.779) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 388.95 + 2 \cdot 1.65 \cdot \sqrt{(886.5(20.0 - 3.0))}$$

$$= 794.065 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1773.000 mm
Nozzle Outside Diameter	(d) =	388.950 mm
Vessel Thickness used	(T) =	17.000 mm
Nozzle Thickness used	(t) =	47.000 mm
T / t	=	0.362
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.240$$

Nr/P	=	0.066
Mr/P	=	0.132
M0/P	=	0.039
N0/P	=	0.084
MrD/Mc	=	0.281
NrDL/Mc	=	0.096
M0d/Mc	=	0.086
N0DL/Mc	=	0.078
MrD/Ml	=	0.105
NrDL/Ml	=	0.070
M0D/Ml	=	0.032
N0DL/Ml	=	0.107

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-11	-11	11	11
Outplane Bending (Ml)	-21	21	21	-21
Normal Pressure Stress	64	66	64	66
Outplane Stress Summary	56	63	123	44

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------



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Nozzle Calcs.: N3-Tube Inlet/L

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Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	61	-61	61	-61
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-70	70	70	-70
Inplane Pressure Stress	32	32	32	32
Inplane Stress Summary	20	38	177	-86

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	57	65	177	131

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-57	57	57	-57
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	32	32	32	32
Outplane Stress Summary	-9	69	123	-28

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	61	-61	61	-61
Inplane Membrane (Mc)	-10	-10	10	10
Inplane Bending (Mc)	-188	188	188	-188
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	64	66	64	66
Inplane Stress Summary	-68	187	330	-168

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1



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Nozzle Calcs.: N3-Tube Inlet/L

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Torsional Shear (Mt)	4	4	4	4
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Shear Stress Summary	3	3	5	5
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	68	187	331	168
------------------------	----	-----	-----	-----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-11	-11	11	11
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	5	4	5

Hoop Stress Summary	-1	0	22	24
---------------------	----	---	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-6	6	6	-6
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	0	0	17	-6
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	2	2	1	1
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	4	4	22	30
------------------------	---	---	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-8	-8	8	8



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Nozzle Calcs.: N3-Tube Inlet/L

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Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	5	4	5

Hoop Stress Summary	1	3	19	20
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	-3	-3	3	3
Axial Bending (Mc)	-20	20	20	-20
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-15	13	32	-20
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	1	1	2	2
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Two * Max Shear Stress	17	13	32	41
------------------------	----	----	----	----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	64	66	64	66	64	66	64	66
Circ. Pl (SUS)	-5	-5	17	17	-5	-5	15	15
Circ. Q (SUS)	-3	3	39	-39	-127	127	249	-249
Long. Pm (SUS)	32	32	32	32	32	32	32	32
Long. Pl (SUS)	-2	-2	12	12	-2	-2	14	14
Long. Q (SUS)	-9	9	131	-131	-39	39	75	-75
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	64.0	66.0	64.0	66.0	64.0	66.0	64.0	66.0
Pm+Pl (SUS)	59.0	61.0	81.0	83.0	59.0	61.0	79.0	81.0
Pm+Pl+Q (Total)	56.7	65.0	175.2	131.1	68.2	188.1	328.1	168.2

Vessel Stress Summation Comparison (N./mm²):



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Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	66.00	118.00	Passed
Pm+Pl (SUS)	83.03	177.00	Passed
Pm+Pl+Q (TOTAL)	328.12	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	4	5	4	5	4	5	4	5
Circ. Pl (SUS)	-5	-5	17	17	-2	-2	14	14
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-3	-3	3	3	-3	-3	3	3
Long. Q (SUS)	0	0	12	-12	-14	14	26	-26
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	4.0	5.0	4.0	5.0	4.0	5.0	4.0	5.0
Pm+Pl (SUS)	1.0	1.0	21.0	22.0	3.0	4.0	18.0	19.0
Pm+Pl+Q (Total)	2.0	2.2	21.2	29.1	17.1	13.1	31.1	40.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.00	148.00	Passed
Pm+Pl (SUS)	22.00	222.00	Passed
Pm+Pl+Q (TOTAL)	40.05	443.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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 Nozzle Calcs.: Noz N2 Fr30 Nozl: 18 9:34am Sep 9,2026

Input, Nozzle Desc: Noz N2 Fr30

From: 30

Pressure for Reinforcement Calculations	P	1250.000	KPa.
Temperature for Internal Pressure	Temp	250	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	118.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	118.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1750.00	mm
Design Length of Section	L	1200.0001	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1013.18	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	148.00	N./mm ²
Allowable Stress At Ambient	Sna	148.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		12.7500	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	17.4498	mm
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	35.0000	mm
Height of Beveled Transition	L'	20.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
ASME Code Weld Type per UW-16		g	
Flange Class		300	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



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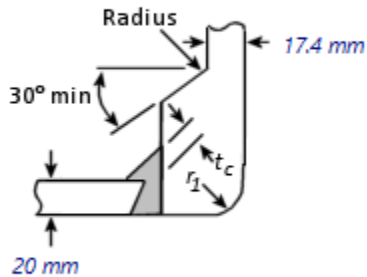
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Nozzle Calcs.: Noz N2 Fr30

Nozl: 18

9:34am

Sep 9, 2026



UW-16.1 [g]

Reinforcement CALCULATION, Description: Noz N2 Fr30

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	12.750 in.
Actual Thickness Used in Calculation	0.687 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 1250 \cdot 878 / (118 \cdot 1 - 0.6 \cdot 1250)$$

$$= 9.3611 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= 1250 \cdot 162 / (148 \cdot 1 + 0.4 \cdot 1250)$$

$$= 1.3631 \text{ mm}$$

Required Nozzle thickness under External Pressure per UG-28 : 1.0182 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 32.6 / 0.5773$$

$$= 56.3786 \text{ mm}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	589.9008	mm
--	----	----------	----

Parallel to Vessel Wall, opening length	d	294.9504	mm
---	---	----------	----

Normal to Vessel Wall (Thickness Limit), no pad	T1np	42.5000	mm
---	------	---------	----

Weld Strength Reduction Factor [fr1]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 148 / 118)$$

$$= 1.000$$



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Weld Strength Reduction Factor [fr2]:

$$= \min(1, S_n/S_v)$$

$$= \min(1, 148/118)$$

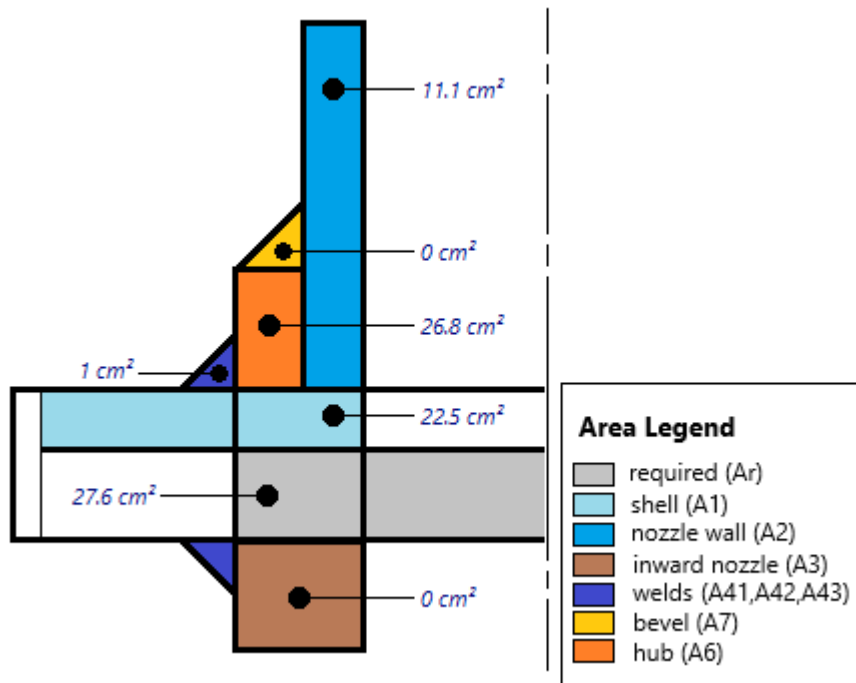
$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(1, 1)$$

$$= 1.000$$



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	27.610	7.211	NA
Area in Shell	A1	22.531	35.720	NA
Area in Nozzle Wall	A2	11.124	11.417	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub + Bevel	A6	26.752	26.752	NA
TOTAL AREA AVAILABLE	Atot	61.407	74.889	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (295 \cdot 9.36 \cdot 1 + 2 \cdot 47 \cdot 9.36 \cdot 1 \cdot (1 - 1))$$

$$= 27.610 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:



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$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 295(1 \cdot 17 - 1 \cdot 9.36) - 2 \cdot 47 \\
 &\quad (1 \cdot 17 - 1 \cdot 9.36) \cdot (1 - 1) \\
 &= 22.531 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot tnp)(tn - trn)fr2 \\
 &= (2 \cdot 42.5)(14.4 - 1.36)1 \\
 &= 11.124 \text{ cm}^2
 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned}
 &= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2 \\
 &= 10^2 \cdot 1 + (0)^2 \cdot 1 \\
 &= 1.000 \text{ cm}^2
 \end{aligned}$$

Area Available in the Hub Section [A6]:

$$\begin{aligned}
 &= (2 \cdot \min(Tnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2 \\
 &= (2 \cdot \min(42.5, 300, 35)) \cdot (50 - 17.4) \cdot 1 \\
 &= 26.752 \text{ cm}^2 \text{ Includes Bevel Area of } 3.967 \text{ cm}^2
 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.3631 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 12.3611 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 12.3611 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.7700 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned}
 &= \min[tb3, \max(tb1, tb2)] \\
 &= \min[11.3, \max(12.4, 4.5)] \\
 &= 11.3312 \text{ mm}
 \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned}
 &= \max(ta, tb) \\
 &= \max(4.36, 11.3) \\
 &= 11.3312 \text{ mm}
 \end{aligned}$$

Available Nozzle Neck Thickness = 17.4498 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	21.6,	Allowable	:	148.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	348.4 N./mm ²	Passed
Occasional	:	6.1,	Allowable	:	196.8 N./mm ²	Passed
Shear	:	12.3,	Allowable	:	103.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 17.4, tr = 1.36, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.094, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 20, tr = 9.36, c = 3 mm, E* = 1



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Thickness Ratio = $tr * E / (tg - c) = 0.55$, Temp. Reduction = 27 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 12 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -14 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -14 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1250.00 / 5170.01 = 0.242$

Weld Size Calculations, Description: Noz N2 Fr30

Intermediate Calc. for nozzle/shell Welds Tmin 17.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = $0.7 * Wo$ mm

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1450 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 15.1132 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 335.1133 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 294.95 mm

Input Echo, WRC297 Item 1, Description: Noz N2 Fr30 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1750.000	mm
Shell Thickness	20.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 118.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	388.950	mm
Nozzle Thickness	50.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc 148.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 148.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:



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*External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	1250.00	KPa.
Radial Load	P	-2294.51	Kgf
Circumferential Shear	Vc	1529.68	Kgf
Longitudinal Shear	Vl	1529.68	Kgf
Circumferential Moment	Mc	1284.43	Kg-m.
Longitudinal Moment	Ml	1284.43	Kg-m.
Torsional Moment	Mt	1834.90	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (7.779) is < 10
check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 388.95 + 2 \cdot 1.65 \cdot \sqrt{886.5(20.0 - 3.0)}$$

$$= 794.065 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1773.000 mm
Nozzle Outside Diameter	(d) =	388.950 mm
Vessel Thickness used	(T) =	17.000 mm
Nozzle Thickness used	(t) =	47.000 mm
T / t	=	0.362
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.240$$

Nr/P	=	0.066
Mr/P	=	0.132
M0/P	=	0.039
N0/P	=	0.084
MrD/Mc	=	0.281
NrDL/Mc	=	0.096
M0d/Mc	=	0.086
N0DL/Mc	=	0.078
MrD/Ml	=	0.105
NrDL/Ml	=	0.070
M0D/Ml	=	0.032
N0DL/Ml	=	0.107

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-11	-11	11	11
Outplane Bending (Ml)	-21	21	21	-21
Normal Pressure Stress	63	65	63	65
Outplane Stress Summary	55	62	122	43

Vessel Stresses (N./mm²):



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LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	61	-61	61	-61
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-70	70	70	-70
Inplane Pressure Stress	31	31	31	31
Inplane Stress Summary	20	38	176	-86

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	56	64	176	130

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-57	57	57	-57
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	31	31	31	31
Outplane Stress Summary	-9	69	123	-28

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	61	-61	61	-61
Inplane Membrane (Mc)	-10	-10	10	10
Inplane Bending (Mc)	-188	188	188	-188
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	63	65	63	65
Inplane Stress Summary	-68	186	330	-169

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to	Cu Left	Cl Left	Du Right	Dl Right
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circumferential plane)			Outside	Inside	Outside	Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-1	-1	1	1
Torsional Shear (Mt)			4	4	4	4
Shear Stress Summary			3	3	5	5

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Stress Intensities			Left	Left	Right	Right
			Outside	Inside	Outside	Inside
Two * Max Shear Stress			69	186	330	169

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	Bl
(Stresses in the			Top	Top	Bottom	Bottom
hoop direction)			Outside	Inside	Outside	Inside
Hoop Membrane (P)			6	6	6	6
Hoop Bending (P)			0	0	0	0
Hoop Membrane (Mc)			0	0	0	0
Hoop Bending (Mc)			0	0	0	0
Hoop Membrane (ML)			-11	-11	11	11
Hoop Bending (ML)			0	0	0	0
Hoop Pressure Stress			4	5	4	5
Hoop Stress Summary			-1	0	22	24

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	Bl
(Stresses Normal to			Top	Top	Bottom	Bottom
pipe cross-section)			Outside	Inside	Outside	Inside
Axial Membrane (P)			0	0	0	0
Axial Bending (P)			6	-6	6	-6
Axial Membrane (Mc)			0	0	0	0
Axial Bending (Mc)			0	0	0	0
Axial Membrane (ML)			-3	-3	3	3
Axial Bending (ML)			-6	6	6	-6
Axial Pressure Stress			2	2	2	2
Axial Stress Summary			0	0	17	-6

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	Bl
(Shear stress)			Outside	Inside	Outside	Inside
Shear due to (Vc)			0	0	0	0
Shear due to (Vl)			0	0	0	0
Shear due to Torsion			1	1	1	1
Shear Stress Summary			2	2	1	1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	Bl
(Stress Intensities			Outside	Inside	Outside	Inside
Two * Max Shear Stress			4	4	22	30

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Stresses in the			Left	Left	Right	Right
hoop direction)			Outside	Inside	Outside	Inside



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Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-8	-8	8	8
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	5	4	5

Hoop Stress Summary	1	3	19	20
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	-3	-3	3	3
Axial Bending (Mc)	-20	20	20	-20
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-15	13	32	-20
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	1	1	2	2
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	17	13	32	41

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	63	65	63	65	63	65	63	65
Circ. Pl (SUS)	-5	-5	17	17	-5	-5	15	15
Circ. Q (SUS)	-3	3	39	-39	-127	127	249	-249
Long. Pm (SUS)	31	31	31	31	31	31	31	31
Long. Pl (SUS)	-2	-2	12	12	-2	-2	14	14
Long. Q (SUS)	-9	9	131	-131	-39	39	75	-75
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	63.0	65.0	63.0	65.0	63.0	65.0	63.0	65.0
Pm+Pl (SUS)	58.0	60.0	80.0	82.0	58.0	60.0	78.0	80.0



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Pm+Pl+Q (Total)	55.7	64.0	174.2	131.1	69.2	187.1	327.1	169.2
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Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	65.00	118.00	Passed
Pm+Pl (SUS)	82.03	177.00	Passed
Pm+Pl+Q (TOTAL)	327.12	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	4	5	4	5	4	5	4	5
Circ. Pl (SUS)	-5	-5	17	17	-2	-2	14	14
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-3	-3	3	3	-3	-3	3	3
Long. Q (SUS)	0	0	12	-12	-14	14	26	-26
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	4.0	5.0	4.0	5.0	4.0	5.0	4.0	5.0
Pm+Pl (SUS)	1.0	1.0	21.0	22.0	3.0	4.0	18.0	19.0
Pm+Pl+Q (Total)	2.0	2.2	21.2	29.1	17.1	13.1	31.1	40.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.00	148.00	Passed
Pm+Pl (SUS)	22.00	222.00	Passed
Pm+Pl+Q (TOTAL)	40.05	443.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: 3" Nozzle

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Input, Nozzle Desc: 3" Nozzle

From: 30

Pressure for Reinforcement Calculations	P	1258.577	KPa.
Temperature for Internal Pressure	Temp	250	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	118.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	118.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1750.00	mm
Design Length of Section	L	1200.0001	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		628.44	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	148.00	N./mm ²
Allowable Stress At Ambient	Sna	148.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm
Flange Material		SA-336 F11	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0250	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
ASME Code Weld Type per UW-16		g	
Flange Class		300	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



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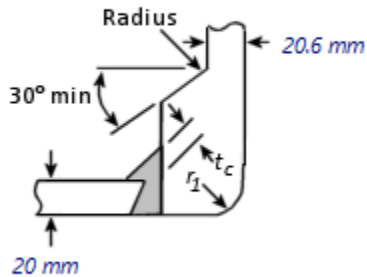
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Nozzle Calcs.: 3" Nozzle

Nozl: 19

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UW-16.1 [g]

Reinforcement CALCULATION, Description: 3" Nozzle

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.000 in.
Actual Thickness Used in Calculation	0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)

= $1259 \cdot 878 / (118 \cdot 1 - 0.6 \cdot 1259)$

= 9.4257 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)

= $1259 \cdot 41.1 / (148 \cdot 1 - 0.6 \cdot 1259)$

= 0.3513 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4764 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	164.4000 mm
Parallel to Vessel Wall, opening length	d	82.2000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	42.5000 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: 3" Nozzle.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.4764 mm
Wall Thickness per UG16(b),	tr16b	= 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 12.4257 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 12.4257 mm
Wall Thickness, shell/head, external pressure	trb2	= 3.7700 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000 mm
Wall Thickness per table UG-45	tb3	= 8.7300 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1, tb2)]

= min[8.73, max(12.4, 4.5)]

= 8.7300 mm



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 Nozzle Calcs.: 3" Nozzle Nozl: 19 9:34am Sep 9,2026

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(3.48, 8.73)
 = 8.7300 mm

Available Nozzle Neck Thickness = 20.5740 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 103156.6,	Allowable	: 148.0 N./mm ²	Failed
Expansion	: 0.0,	Allowable	: 222.0 N./mm ²	Passed
Occasional	: 1.2,	Allowable	: 196.8 N./mm ²	Passed
Shear	: 67321.5,	Allowable	: 103.6 N./mm ²	Failed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 20, tr = 9.43, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.55, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	12 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-14 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -14 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 1258.58/5170.01 = 0.243

Weld Size Calculations, Description: 3" Nozzle

Intermediate Calc. for nozzle/shell Welds Tmin 17.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1450 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.9695 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 221.9945 mm

Input Echo, WRC297 Item 1, Description: 3" Nozzle :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1750.000	mm
Shell Thickness	20.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc	118.00 N./mm ²



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 Nozzle Calcs.: 3" Nozzle Nozl: 19 9:34am Sep 9, 2026

Vessel Hot S.I. Allowable Smh 118.00 N./mm²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	76.200	mm
Nozzle Thickness	20.5740	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc	148.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	148.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTAINED loads.*

Design Internal Pressure	Dp	1258.58	KPa.
Radial Load	P	-5999.99	Kgf
Circumferential Shear	Vc	4000.00	Kgf
Longitudinal Shear	Vl	4000.00	Kgf
Circumferential Moment	Mc	895999.50	Kg-m.
Longitudinal Moment	Ml	895999.50	Kg-m.
Torsional Moment	Mt	0.128E+07	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (5.704) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 117.348 + 2 \cdot 1.65 \cdot \sqrt{886.5(20.0 - 3.0)}$
 $= 522.462 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1773.000 mm
Nozzle Outside Diameter	(d) =	117.348 mm
Vessel Thickness used	(T) =	17.000 mm
Nozzle Thickness used	(t) =	17.574 mm
T / t	=	0.967
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.676$$

Nr/P	=	0.153
Mr/P	=	0.241
M0/P	=	0.082
N0/P	=	0.148
MrD/Mc	=	0.289
NrDL/Mc	=	0.142
M0d/Mc	=	0.113
N0DL/Mc	=	-0.077
MrD/Ml	=	0.238
NrDL/Ml	=	0.128
M0D/Ml	=	0.092



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Nozzle Calcs.: 3" Nozzle

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NØDL/Ml = 0.062

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	30	30	30	30
Outplane Bending (P)	100	-100	100	-100
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-16087	-16087	16087	16087
Outplane Bending (ML)	-142650	142650	142650	-142650
Normal Pressure Stress	64	65	64	65
Outplane Stress Summary	-158543	126558	158933	-126567

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	31	31	31	31
Inplane Bending (P)	294	-294	294	-294
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-33198	-33198	33198	33198
Inplane Bending (ML)	-370483	370483	370483	-370483
Inplane Pressure Stress	32	32	32	32
Inplane Stress Summary	-403323	337053	404040	-337516

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	12	12	-12	-12
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	34135	34135	34135	34135
Shear Stress Summary	34148	34148	34123	34123

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	407998	342455	408702	342899

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	30	30	30	30
Outplane Bending (P)	100	-100	100	-100
Outplane Membrane (Mc)	20051	20051	-20051	-20051
Outplane Bending (Mc)	-176273	176273	176273	-176273
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	32	32	32	32
Outplane Stress Summary	-156059	196286	156384	-196362



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Nozzle Calcs.: 3" Nozzle

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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	31	31	31	31
Inplane Bending (P)	294	-294	294	-294
Inplane Membrane (Mc)	-36711	-36711	36711	36711
Inplane Bending (Mc)	-449618	449618	449618	-449618
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	64	65	64	65
Inplane Stress Summary	-485940	412708	486720	-413104

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-12	-12	12	12
Torsional Shear (Mt)	34135	34135	34135	34135
Shear Stress Summary	34123	34123	34148	34148

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	489433	417961	490214	418357

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	30	30	30	30
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-16087	-16087	16087	16087
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	4	2	4
Hoop Stress Summary	-16054	-16053	16120	16122

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	10	10	10	10
Axial Bending (P)	188	-188	188	-188
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-72949	-72949	72949	72949
Axial Bending (ML)	-253481	253481	253481	-253481
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-326230	180356	326631	-180708

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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Nozzle Calcs.: 3" Nozzle

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(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	12	12	-12	-12
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	33020	33020	33020	33020
Shear Stress Summary	33032	33032	33008	33008

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	329709	207222	330101	207606

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	30	30	30	30
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	20051	20051	-20051	-20051
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	4	2	4
Hoop Stress Summary	20084	20085	-20018	-20016

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	10	10	10	10
Axial Bending (P)	188	-188	188	-188
Axial Membrane (Mc)	-72949	-72949	72949	72949
Axial Bending (Mc)	-317668	317668	317668	-317668
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-390417	244543	390818	-244895

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-12	-12	12	12
Shear due to Torsion	33020	33020	33020	33020
Shear Stress Summary	33008	33008	33032	33032

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	415777	249296	416114	249647

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of	Stress Values at
---------	------------------



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 Nozzle Calcs.: 3" Nozzle Nozl: 19 9:34am Sep 9,2026

Stress Int. Location	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	64	65	64	65	64	65	64	65
Circ. Pl (SUS)	-16057	-16057	16117	16117	-36680	-36680	36742	36742
Circ. Q (SUS)	-142550	142550	142750	-142750	-449324	449324	449912	-449912
Long. Pm (SUS)	32	32	32	32	32	32	32	32
Long. Pl (SUS)	-33167	-33167	33229	33229	20081	20081	-20021	-20021
Long. Q (SUS)	-370189	370189	370777	-370777	-176173	176173	176373	-176373
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	12	12	-12	-12	-12	-12	12	12
Shear Q (SUS)	34135	34135	34135	34135	34135	34135	34135	34135
Pm (SUS)	64.0	65.0	64.0	65.0	64.0	65.0	64.0	65.0
Pm+Pl (SUS)	33135	33135	33261	33261	56729	56728	56795	56796
Pm+Pl+Q (Total)	407998	342454	408699	342898	489432	417961	490210	418357

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	65	117	Passed
Pm+Pl (SUS)	56796	176	Failed
Pm+Pl+Q (TOTAL)	490210	353	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	4	2	4	2	4	2	4
Circ. Pl (SUS)	-16057	-16057	16117	16117	20081	20081	-20021	-20021
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-72939	-72939	72959	72959	-72939	-72939	72959	72959
Long. Q (SUS)	-253293	253293	253669	-253669	-317480	317480	317856	-317856
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	12	12	-12	-12	-12	-12	12	12
Shear Q (SUS)	33020	33020	33020	33020	33020	33020	33020	33020
Pm (SUS)	2.0	4.0	2.0	4.0	2.0	4.0	2.0	4.0
Pm+Pl (SUS)	72938	72938	72960	72960	93021	93023	92979	92977
Pm+Pl+Q (Total)	329709	207221	330099	207605	415775	249295	416112	249647

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	4	147	Passed



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Nozzle Calcs.: 3" Nozzle

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<i>Pm+PL (SUS)</i>	<i>93023</i>	<i>221</i>	<i>Failed</i>
<i>Pm+PL+Q (TOTAL)</i>	<i>416112</i>	<i>443</i>	<i>Failed</i>

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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 Nozzle Calcs.: Noz N1 750 Nozl: 20 9:34am Sep 9,2026

Input, Nozzle Desc: Noz N1 750

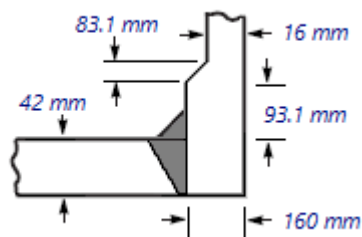
From: 50

Pressure for Reinforcement Calculations	P	4950.000	KPa.
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	118.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	118.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1750.00	mm
Shell Finished (Minimum) Thickness	t	42.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		3439.35	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	148.00	N./mm ²
Allowable Stress At Ambient	Sna	148.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		28.7400	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm
Hub Height of Integral Nozzle	h	93.1384	mm
Height of Beveled Transition	L'	83.1384	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		160.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	44.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: Noz N1 750



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 Nozzle Calcs.: Noz N1 750 Nozl: 20 9:34am Sep 9,2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 28.740 in.
 Actual Thickness Used in Calculation 0.630 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4950 \cdot 878 / (118 \cdot 1 - 0.6 \cdot 4950)$
 $= 37.7853 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4950 \cdot 368 / (148 \cdot 1 - 0.6 \cdot 4950)$
 $= 12.5611 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 144 / 0.5773$
 $= 249.4154 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1471.9921	mm
Parallel to Vessel Wall, opening length	d	735.9960	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	97.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 148 / 118)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 148 / 118)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



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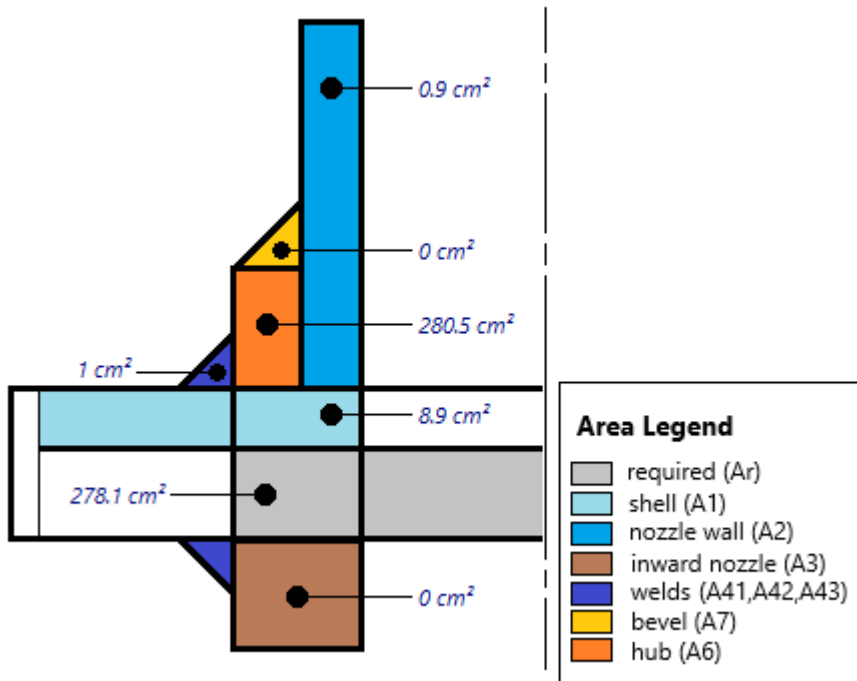
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Nozzle Calcs.: Noz N1 750

Nozl: 20

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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	278.098	NA	NA
Area in Shell	A1	8.940	NA	NA
Area in Nozzle Wall	A2	0.856	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	280.471	NA	NA
TOTAL AREA AVAILABLE	Atot	291.267	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (736 \cdot 37.8 \cdot 1 + 2 \cdot 157 \cdot 37.8 \cdot 1 \cdot (1 - 1))$$

$$= 278.098 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 736(1 \cdot 39 - 1 \cdot 37.8) - 2 \cdot 157$$

$$(1 \cdot 39 - 1 \cdot 37.8) \cdot (1 - 1)$$

$$= 8.940 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 97.5)(13 - 12.6)1$$

$$= 0.856 \text{ cm}^2$$



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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(97.5, 400, 93.1)) \cdot (160 - 16) \cdot 1$$

$$= 280.471 \text{ cm}^2 \text{ Includes Bevel Area of } 12.232 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 15.5611 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 40.7853 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 40.7853 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(40.8, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(15.6, 11.3)$$

$$= 15.5611 \text{ mm}$$

Available Nozzle Neck Thickness = 16.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 94.9,	Allowable	: 148.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 275.1 N./mm ²	Passed
Occasional	: 68.8,	Allowable	: 196.8 N./mm ²	Passed
Shear	: 16.3,	Allowable	: 103.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 42, tr = 37.8, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.97, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	32 °C
Min Metal Temp. at Required thickness (UCS 66.1)	31 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 31 °C

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 144 / 0.5773$$

$$= 249.4154 \text{ mm}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]



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Parallel to Vessel Wall (Diameter Limit) D1 1103.9940 mm
 Parallel to Vessel Wall, opening length d 551.9970 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 97.5000 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure] AR		185.408	185.408
Area Available in Shell A1		4.470	4.470
Area Available in Nozzle Wall A2		0.856	0.856
Area Available in Inward Nozzle A3		0.000	0.000
Area Available in Welds A4		1.000	1.000
Area Available in Pad A5		0.000	0.000
Area Available in Hub A6		280.471	280.471
TOTAL AREA AVAILABLE Atot		286.797	286.797

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: Noz N1 750

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (278 - 8.94 + 2 \cdot 157 \cdot 1 \cdot (1 \cdot 39 - 37.8)) \cdot 118)$
 $= 328445.66 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i\text{-Can}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (281 + 1 - 0 \cdot 1) \cdot 118$
 $= 339701.25 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (281 + 0 + 1 + (122)) \cdot 118$
 $= 487047.72 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (281 + 0 + 1 + 0 + (122)) \cdot 118$
 $= 487047.72 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 1050 \cdot 10 \cdot 0.49 \cdot 118$
 $= 97241. \text{ Kgf}$



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Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 446) \cdot (160 - 3) \cdot 0.7 \cdot 148$$

$$= 2326442. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 1050 \cdot (42 - 3) \cdot 0.74 \cdot 148$$

$$= 718338. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (97241 + 2326442) = 2423683 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (97241 + 0 + 718338 + 0) = 815579 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (97241 + 718338 + 0) = 815579 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 2423683 Kg, must exceed W = 328445 Kg or W1 = 339701 Kg
 Path 2-2 = 815579 Kg, must exceed W = 328445 Kg or W2 = 487047 Kg
 Path 3-3 = 815579 Kg, must exceed W = 328445 Kg or W3 = 487047 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5059 KPa.

The Drop for this Nozzle is : 174.9985 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 616.9984 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 689.57 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 736.00 mm

Warning:
Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: Noz N1 750 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1750.000	mm
Shell Thickness	42.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 118.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	729.996	mm
Nozzle Thickness	160.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	



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Nozzle Cold S.I. Allowable	SNmc	148.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	148.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	4950.00	KPa.
Radial Load	P	-6883.56	Kgf
Circumferential Shear	Vc	4971.46	Kgf
Longitudinal Shear	Vl	4971.46	Kgf
Circumferential Moment	Mc	11468.10	Kg-m.
Longitudinal Moment	Ml	11468.10	Kg-m.
Torsional Moment	Mt	13188.40	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Dv (.573) is > 0.5
 check the Limitations of WRC bulletin 297.

Warning - The ratio, Dn/Tn (6.562) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 1049.996 + 2 \cdot 1.65 \cdot \sqrt{897.5(42.0 - 3.0)}$
 $= 1667.391 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1795.000 mm
Nozzle Outside Diameter	(d) =	1049.996 mm
Vessel Thickness used	(T) =	39.000 mm
Nozzle Thickness used	(t) =	157.000 mm
T / t	=	0.248
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.968$$

Nr/P	=	0.051
Mr/P	=	0.082
M0/P	=	0.026
N0/P	=	0.058
MrD/Mc	=	0.256
NrDL/Mc	=	0.091
M0d/Mc	=	0.077
N0DL/Mc	=	0.085
MrD/Ml	=	0.070
NrDL/Ml	=	0.048
M0D/Ml	=	0.021
N0DL/Ml	=	0.062

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7



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Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-8	8	8	-8
Normal Pressure Stress	109	113	109	113

Outplane Stress Summary	105	114	131	104
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Vessel Stresses (N/mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	21	-21	21	-21
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-29	29	29	-29
Inplane Pressure Stress	54	54	54	54

Inplane Stress Summary	45	61	111	8
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Vessel Stresses (N/mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1

Shear Stress Summary	2	2	1	1
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Vessel Stresses (N/mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	105	114	131	104
------------------------	-----	-----	-----	-----

Vessel Stresses (N/mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	-5	-5	5	5
Outplane Bending (Mc)	-32	32	32	-32
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	54	54	54	54

Outplane Stress Summary	25	76	102	23
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Vessel Stresses (N/mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	21	-21	21	-21
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-108	108	108	-108



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Nozzle Calcs.: Noz N1 750

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Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	109	113	109	113

Inplane Stress Summary	18	196	247	-7
------------------------	----	-----	-----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	1	1	2	2
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	25	196	247	31
------------------------	----	-----	-----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-4	-4	4	4
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	17	11	17

Hoop Stress Summary	9	15	18	24
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	5	5	5	5

Axial Stress Summary	4	4	9	5
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
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Nozzle Stresses (N./mm²):



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LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	9	15	18	24

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-5	-5	5	5
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	17	11	17
Hoop Stress Summary	8	14	20	25

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-5	5	5	-5
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	0	9	13	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	8	14	20	25

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	109	113	109	113	109	113	109	113
Circ. Pl (SUS)	-2	-2	6	6	-4	-4	8	8
Circ. Q (SUS)	-1	1	15	-15	-87	87	129	-129
Long. Pm (SUS)	54	54	54	54	54	54	54	54
Long. Pl (SUS)	-1	-1	5	5	-3	-3	7	7



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Nozzle Calcs.: Noz N1 750

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Long. Q (SUS)	-8	8	50	-50	-25	25	39	-39
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	109.0	113.0	109.0	113.0	109.0	113.0	109.0	113.0
Pm+Pl (SUS)	107.0	111.0	115.0	119.0	105.0	109.0	117.0	121.0
Pm+Pl+Q (Total)	106.0	112.0	130.0	104.0	26.1	196.0	246.0	30.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	113.00	118.00	Passed
Pm+Pl (SUS)	121.00	177.00	Passed
Pm+Pl+Q (TOTAL)	246.01	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	11	17	11	17	11	17	11	17
Circ. Pl (SUS)	-2	-2	6	6	-3	-3	7	7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	-1	1	1	-1	-5	5	5	-5
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	11.0	17.0	11.0	17.0	11.0	17.0	11.0	17.0
Pm+Pl (SUS)	9.0	15.0	17.0	23.0	8.0	14.0	18.0	24.0
Pm+Pl+Q (Total)	9.0	15.0	17.0	23.0	9.0	14.0	18.0	24.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	17.00	148.00	Passed
Pm+Pl (SUS)	24.00	222.00	Passed
Pm+Pl+Q (TOTAL)	24.00	443.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: K7-Temperature

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Input, Nozzle Desc: K7-Temperature

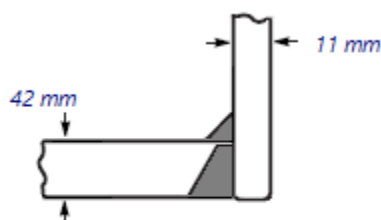
From: 50

Pressure for Reinforcement Calculations	P	4950.000	KPa.
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	118.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	118.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1750.00	mm
Shell Finished (Minimum) Thickness	t	42.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4439.35	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		1.5000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	11.0000	mm
Flange Material		SA-105	
Flange Type		RTJWn	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	44.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		2500	
Flange Grade		GR 1.1	
Flange Series		B	

The Pressure Design option was Design Pressure + static head.





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 Nozzle Calcs.: K7-Temperature Nozl: 21 9:34am Sep 9,2026

Reinforcement CALCULATION, Description: K7-Temperature

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.500 in.
 Actual Thickness Used in Calculation 0.433 in.

Note:
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4950 \cdot 878 / (118 \cdot 1 - 0.6 \cdot 4950)$
 $= 37.7853 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4950 \cdot 22 / (118 \cdot 1 - 0.6 \cdot 4950)$
 $= 0.9489 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	138.1000	mm
Parallel to Vessel Wall	Rn+tn+t	69.0500	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	20.0000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K7-Temperature.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.9489	mm
Wall Thickness per UG16(b),	tr16b = 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1 = 40.7853	mm
Wall Thickness	tb1 = max(trb1, tr16b) = 40.7853	mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000	mm
Wall Thickness per table UG-45	tb3 = 6.4200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(40.8, 4.5)]$
 $= 6.4200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.95, 6.42)$
 $= 6.4200 \text{ mm}$

Available Nozzle Neck Thickness = 11.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	89.0,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	206.0 N./mm ²	Passed
Occasional	:	5.8,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	66.4,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.



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 Nozzle Calcs.: K7-Temperature Nozl: 21 9:34am Sep 9,2026

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 11$, $t_r = 0.95$, $c = 3$ mm, $E* = 1$
 Thickness Ratio = $t_r * E*/(t_g - c) = 0.12$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:A)

Govrn. thk, $t_g = 11$, $t_r = 0.95$, $c = 3$ mm, $E* = 1$
 Thickness Ratio = $t_r * E*/(t_g - c) = 0.12$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $4950.00/42509.99 = 0.116$

Weld Size Calculations, Description: K7-Temperature

Intermediate Calc. for nozzle/shell Welds T_{min} 8.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6000 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

[Skipping the nozzle attachment weld strength calculations. Per UW-15\(b\)\(2\) the nozzles exempted by UG-36\(c\)\(3\)\(a\) \(small nozzles\) do not require a weld strength check.](#)

The Drop for this Nozzle is : 0.5162 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 242.5161 mm

Input Echo, WRC297 Item 1, Description: K7-Temperature:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1750.000	mm
Shell Thickness	42.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	S_{mc}	118.00 N./mm ²
Vessel Hot S.I. Allowable	S_{mh}	118.00 N./mm ²

[See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.](#)

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	38.100	mm
Nozzle Thickness	11.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	



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Nozzle Calcs.: K7-Temperature

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Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	4950.00	KPa.
Radial Load	P	-739.35	Kgf
Circumferential Shear	Vc	611.87	Kgf
Longitudinal Shear	Vl	611.87	Kgf
Circumferential Moment	Mc	96.84	Kg-m.
Longitudinal Moment	Ml	96.84	Kg-m.
Torsional Moment	Mt	127.42	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tv, (1.431) is < 5
check the Limitations of WRC bulletin 297.

Warning - The ratio, Dn/Tn (5.464) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 60.1 + 2 \cdot 1.65 \cdot \sqrt{897.5(42.0 - 3.0)}$$

$$= 677.495 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1795.000 mm
Nozzle Outside Diameter	(d) =	60.100 mm
Vessel Thickness used	(T) =	39.000 mm
Nozzle Thickness used	(t) =	8.000 mm
T / t	=	4.875
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.227$$

Nr/P	=	0.211
Mr/P	=	0.135
M0/P	=	0.308
N0/P	=	0.244
MrD/Mc	=	0.117
NrDL/Mc	=	0.159
M0d/Mc	=	0.298
N0DL/Mc	=	-0.130
MrD/Ml	=	0.117
NrDL/Ml	=	0.163
M0D/Ml	=	0.293
N0DL/Ml	=	-0.124

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	8	-8	8	-8



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Nozzle Calcs.: K7-Temperature

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Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	1	1	-1	-1
Outplane Bending (ML)	-18	18	18	-18
Normal Pressure Stress	109	113	109	113

Outplane Stress Summary	102	125	135	86
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-7	7	7	-7
Inplane Pressure Stress	54	54	54	54

Inplane Stress Summary	50	57	68	46
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5

Shear Stress Summary	7	7	4	4
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	103	126	136	87
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	1	1	-1	-1
Outplane Bending (Mc)	-18	18	18	-18
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	54	54	54	54

Outplane Stress Summary	47	66	81	26
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-7	7	7	-7



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Nozzle Calcs.: K7-Temperature

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Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	109	113	109	113

Inplane Stress Summary	104	116	122	105
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5

Shear Stress Summary	4	4	7	7
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	105	117	124	106
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	1	1	-1	-1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	19	13	19

Hoop Stress Summary	16	22	13	19
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	5	5	5	5
Axial Bending (P)	19	-19	19	-19
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-62	-62	62	62
Axial Bending (ML)	-52	52	52	-52
Axial Pressure Stress	6	6	6	6

Axial Stress Summary	-82	-17	147	3
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	7	7	-7	-7
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	27	27	27	27

Shear Stress Summary	35	35	19	19
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Nozzle Stresses (N./mm²):



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Nozzle Calcs.: K7-Temperature

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LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	121	81	150	42

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	1	1	-1	-1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	19	13	19
Hoop Stress Summary	16	22	13	19

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	5	5	5	5
Axial Bending (P)	19	-19	19	-19
Axial Membrane (Mc)	-62	-62	62	62
Axial Bending (Mc)	-55	55	55	-55
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	-86	-14	150	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-7	-7	7	7
Shear due to Torsion	27	27	27	27
Shear Stress Summary	19	19	35	35

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	110	53	159	73

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	109	113	109	113	109	113	109	113
Circ. Pl (SUS)	2	2	0	0	-1	-1	1	1
Circ. Q (SUS)	-10	10	26	-26	-4	4	10	-10
Long. Pm (SUS)	54	54	54	54	54	54	54	54
Long. Pl (SUS)	-1	-1	1	1	2	2	0	0



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 Nozzle Calcs.: K7-Temperature Nozl: 21 9:34am Sep 9,2026

Long. Q (SUS)	-4	4	10	-10	-10	10	26	-26
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	109.0	113.0	109.0	113.0	109.0	113.0	109.0	113.0
Pm+Pl (SUS)	111.0	115.0	109.0	113.0	108.0	112.0	110.0	114.0
Pm+Pl+Q (Total)	101.7	125.5	135.2	87.4	104.3	116.3	120.9	104.5

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	113.00	118.00	Passed
Pm+Pl (SUS)	115.02	177.00	Passed
Pm+Pl+Q (TOTAL)	135.23	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	13	19	13	19	13	19	13	19
Circ. Pl (SUS)	2	2	0	0	2	2	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	-57	-57	67	67	-57	-57	67	67
Long. Q (SUS)	-33	33	71	-71	-36	36	74	-74
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	7	7	-7	-7	-7	-7	7	7
Shear Q (SUS)	27	27	27	27	27	27	27	27
Pm (SUS)	13.0	19.0	13.0	19.0	13.0	19.0	13.0	19.0
Pm+Pl (SUS)	67.5	73.3	73.8	73.9	67.5	73.3	73.8	73.9
Pm+Pl+Q (Total)	120.1	78.4	147.0	43.5	109.6	53.8	155.1	70.9

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	19.00	118.00	Passed
Pm+Pl (SUS)	73.89	177.00	Passed
Pm+Pl+Q (TOTAL)	155.13	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



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Nozzle Calcs.: M1-Manway

Nozl: 22

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Input, Nozzle Desc: M1-Manway

From: 50

Pressure for Reinforcement Calculations	P	4958.577	KPa.
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	118.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	118.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1750.00	mm
Shell Finished (Minimum) Thickness	t	42.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		7039.35	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	148.00	N./mm ²
Allowable Stress At Ambient	Sna	148.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		22.8900	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.2700	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	110.0000	mm
Height of Beveled Transition	L'	72.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		150.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	450.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		44.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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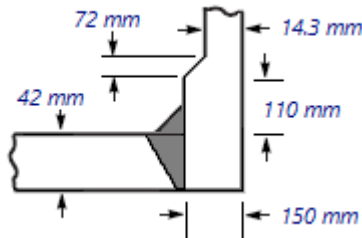
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Nozzle Calcs.: M1-Manway

Nozl: 22

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Reinforcement CALCULATION, Description: M1-Manway

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	22.890 in.
Actual Thickness Used in Calculation	0.562 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4959 \cdot 878 / (118 \cdot 1 - 0.6 \cdot 4959)$
 $= 37.8524 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 4959 \cdot 291 / (148 \cdot 1 + 0.4 \cdot 4959)$
 $= 9.6116 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 136 / 0.5773$
 $= 235.0913 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1117.7319	mm
Parallel to Vessel Wall, opening length	d	558.8660	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	97.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 148 / 118)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 148 / 118)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



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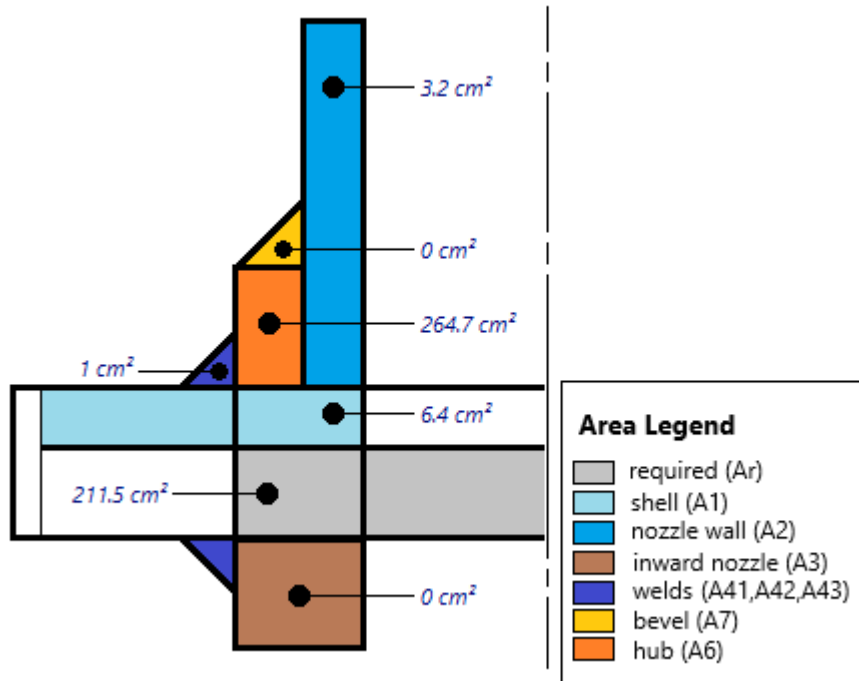
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Nozzle Calcs.: M1-Manway

Noz1: 22

9:34am

Sep 9, 2026



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	211.544	NA	NA
Area in Shell	A1	6.413	NA	NA
Area in Nozzle Wall	A2	3.234	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	264.673	NA	NA
TOTAL AREA AVAILABLE	Atot	275.321	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (559 \cdot 37.9 \cdot 1 + 2 \cdot 147 \cdot 37.9 \cdot 1 \cdot (1 - 1))$$

$$= 211.544 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot tr - F \cdot tr) - 2 \cdot tn(E1 \cdot tr - F \cdot tr) \cdot (1 - fr1)$$

$$= 559(1 \cdot 39 - 1 \cdot 37.9) - 2 \cdot 147$$

$$(1 \cdot 39 - 1 \cdot 37.9) \cdot (1 - 1)$$

$$= 6.413 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 97.5)(11.3 - 9.61)1$$

$$= 3.234 \text{ cm}^2$$



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 Nozzle Calcs.: M1-Manway Nozl: 22 9:34am Sep 9,2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i - c_n / 0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr2$$

$$= (2 \cdot \min(97.5, 450, 110)) \cdot (150 - 14.3) \cdot 1$$

$$= 264.673 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 12.6116 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 40.8524 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 40.8524 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(40.9, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(12.6, 11.3)$$

$$= 12.6116 \text{ mm}$$

Available Nozzle Neck Thickness = 14.2700 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 14.3, tr = 9.61, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.85, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-6 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 42, tr = 37.9, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.97, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	32 °C
Min Metal Temp. at Required thickness (UCS 66.1)	31 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 31 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 4958.58/10200.00 = 0.486

Weld Size Calculations, Description: M1-Manway

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------



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 Nozzle Calcs.: M1-Manway Nozl: 22 9:34am Sep 9, 2026

Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (212 - 6.41 + 2 \cdot 147 \cdot 1 \cdot (1 \cdot 39 - 37.9)) \cdot 118)$
 $= 250877.45 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (268 + 1 - 0 \cdot 1) \cdot 118$
 $= 323555.06 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (268 + 0 + 1 + (115)) \cdot 118$
 $= 461516.41 \text{ Kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (268 + 0 + 1 + 0 + (115)) \cdot 118$
 $= 461516.41 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 581 \cdot 10 \cdot 0.49 \cdot 118$
 $= 53844. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.14 \cdot 285) \cdot (150 - 3) \cdot 0.7 \cdot 148$
 $= 1390717. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.14/2.0) \cdot 581 \cdot (42 - 3) \cdot 0.74 \cdot 148$
 $= 397760. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (53844 + 1390717) = 1444561 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (53844 + 0 + 397760 + 0) = 451604 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (53844 + 397760 + 0) = 451604 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 1444561 Kgf, must exceed W = 250877 Kgf	or W1 = 323555 Kgf
Path 2-2 = 451604 Kgf, must exceed W = 250877 Kgf	or W2 = 461516 Kgf
Path 3-3 = 451604 Kgf, must exceed W = 250877 Kgf	or W3 = 461516 Kgf



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Nozzle Calcs.: M1-Manway

Nozl: 22

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Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5088 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 49.7020 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 541.7020 mm

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 Nozzle Calcs.: Noz N4 750 Nozl: 23 9:34am Sep 9,2026

Input, Nozzle Desc: Noz N4 750

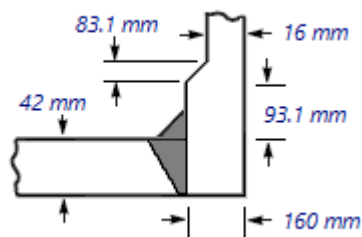
From: 50

Pressure for Reinforcement Calculations	P	4967.154	KPa.
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	118.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	118.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1750.00	mm
Shell Finished (Minimum) Thickness	t	42.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5739.35	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	148.00	N./mm ²
Allowable Stress At Ambient	Sna	148.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		28.7400	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.0000	mm
Hub Height of Integral Nozzle	h	93.1384	mm
Height of Beveled Transition	L'	83.1384	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		160.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	44.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: Noz N4 750



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 Nozzle Calcs.: Noz N4 750 Nozl: 23 9:34am Sep 9,2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 28.740 in.
 Actual Thickness Used in Calculation 0.630 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4967 \cdot 878 / (118 \cdot 1 - 0.6 \cdot 4967)$
 $= 37.9196 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4967 \cdot 368 / (148 \cdot 1 - 0.6 \cdot 4967)$
 $= 12.6055 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 144 / 0.5773$
 $= 249.4154 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1471.9921	mm
Parallel to Vessel Wall, opening length	d	735.9960	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	97.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 148 / 118)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 148 / 118)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



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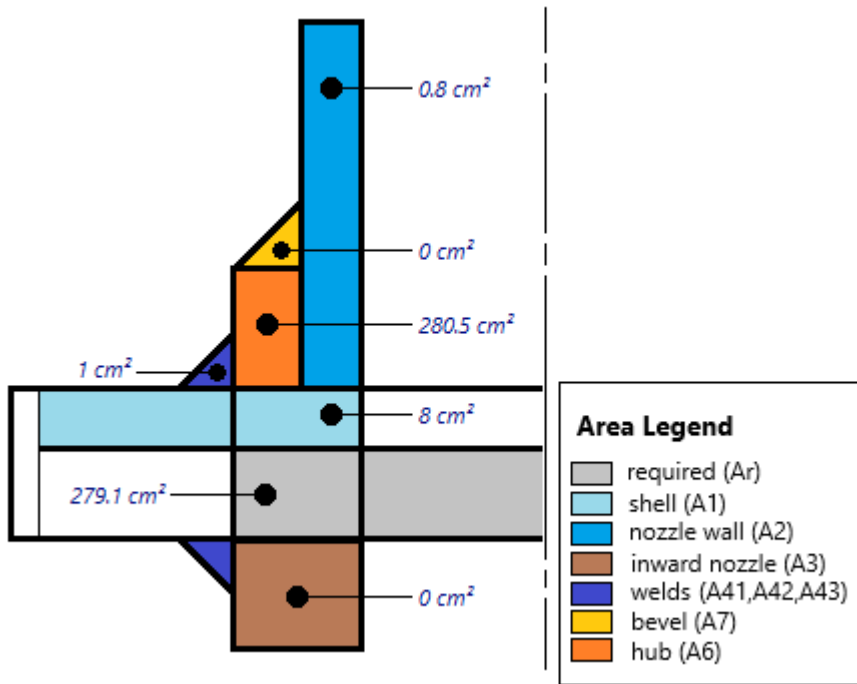
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Nozzle Calcs.: Noz N4 750

Nozl: 23

9:34am

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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	279.087	NA	NA
Area in Shell	A1	7.952	NA	NA
Area in Nozzle Wall	A2	0.769	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	280.471	NA	NA
TOTAL AREA AVAILABLE	Atot	290.192	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (736 \cdot 37.9 \cdot 1 + 2 \cdot 157 \cdot 37.9 \cdot 1 \cdot (1 - 1))$$

$$= 279.087 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 736(1 \cdot 39 - 1 \cdot 37.9) - 2 \cdot 157$$

$$(1 \cdot 39 - 1 \cdot 37.9) \cdot (1 - 1)$$

$$= 7.952 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 97.5)(13 - 12.6)1$$

$$= 0.769 \text{ cm}^2$$



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Nozzle Calcs.: Noz N4 750

Nozl: 23

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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(97.5, 400, 93.1)) \cdot (160 - 16) \cdot 1$$

$$= 280.471 \text{ cm}^2 \text{ Includes Bevel Area of } 12.232 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 15.6055 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 40.9196 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 40.9196 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(40.9, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(15.6, 11.3)$$

$$= 15.6055 \text{ mm}$$

Available Nozzle Neck Thickness = 16.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 95.1,	Allowable	: 148.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 274.9 N./mm ²	Passed
Occasional	: 69.1,	Allowable	: 196.8 N./mm ²	Passed
Shear	: 16.3,	Allowable	: 103.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 42, tr = 37.9, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.97, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	32 °C
Min Metal Temp. at Required thickness (UCS 66.1)	31 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 31 °C

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 144 / 0.5773$$

$$= 249.4154 \text{ mm}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]



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 Nozzle Calcs.: Noz N4 750 Nozl: 23 9:34am Sep 9,2026

Parallel to Vessel Wall (Diameter Limit) D1 1103.9940 mm
 Parallel to Vessel Wall, opening length d 551.9970 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 97.5000 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure] AR		186.067	186.067
Area Available in Shell A1		3.976	3.976
Area Available in Nozzle Wall A2		0.769	0.769
Area Available in Inward Nozzle A3		0.000	0.000
Area Available in Welds A4		1.000	1.000
Area Available in Pad A5		0.000	0.000
Area Available in Hub A6		280.471	280.471
TOTAL AREA AVAILABLE Atot		286.216	286.216

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: Noz N4 750

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (279 - 7.95 + 2 \cdot 157 \cdot 1 \cdot (1 \cdot 39 - 37.9)) \cdot 118)$
 $= 330317.31 \text{ Kg}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i\text{-Can}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (281 + 1 - 0 \cdot 1) \cdot 118$
 $= 339597.03 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (281 + 0 + 1 + (122)) \cdot 118$
 $= 486943.50 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (281 + 0 + 1 + 0 + (122)) \cdot 118$
 $= 486943.50 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 1050 \cdot 10 \cdot 0.49 \cdot 118$
 $= 97241. \text{ Kg}$



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 Nozzle Calcs.: Noz N4 750 Nozl: 23 9:34am Sep 9,2026

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 446) \cdot (160 - 3) \cdot 0.7 \cdot 148$$

$$= 2326442. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 1050 \cdot (42 - 3) \cdot 0.74 \cdot 148$$

$$= 718338. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (97241 + 2326442) = 2423683 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (97241 + 0 + 718338 + 0) = 815579 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (97241 + 718338 + 0) = 815579 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 2423683 Kgf, must exceed W = 330317 Kgf or W1 = 339597 Kgf
 Path 2-2 = 815579 Kgf, must exceed W = 330317 Kgf or W2 = 486943 Kgf
 Path 3-3 = 815579 Kgf, must exceed W = 330317 Kgf or W3 = 486943 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5042 KPa.

The Drop for this Nozzle is : 174.9985 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 616.9984 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: Noz N4 750 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1750.000	mm
Shell Thickness	42.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc	118.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	118.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	729.996	mm
Nozzle Thickness	160.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-336 F11	
Nozzle Cold S.I. Allowable	SNmc	148.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	148.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.



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Nozzle Calcs.: Noz N4 750

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Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	4967.15	KPa.
Radial Load	P	-6883.56	Kgf
Circumferential Shear	Vc	4971.46	Kgf
Longitudinal Shear	Vl	4971.46	Kgf
Circumferential Moment	Mc	11468.10	Kg-m.
Longitudinal Moment	Ml	11468.10	Kg-m.
Torsional Moment	Mt	13188.40	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Dv (.573) is > 0.5
check the Limitations of WRC bulletin 297.

Warning - The ratio, Dn/In (6.562) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 1049.996 + 2 \cdot 1.65 \cdot \sqrt{897.5(42.0 - 3.0)}$$

$$= 1667.391 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1795.000 mm
Nozzle Outside Diameter	(d) =	1049.996 mm
Vessel Thickness used	(T) =	39.000 mm
Nozzle Thickness used	(t) =	157.000 mm
T / t	=	0.248
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.968$$

Nr/P	=	0.051
Mr/P	=	0.082
M0/P	=	0.026
N0/P	=	0.058
MrD/Mc	=	0.256
NrDL/Mc	=	0.091
M0d/Mc	=	0.077
N0DL/Mc	=	0.085
MrD/Ml	=	0.070
NrDL/Ml	=	0.048
M0D/Ml	=	0.021
N0DL/Ml	=	0.062

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-4	-4	4	4
Outplane Bending (Ml)	-8	8	8	-8



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Nozzle Calcs.: Noz N4 750

Nozl: 23 9:34am

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Normal Pressure Stress	109	114	109	114
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Outplane Stress Summary	105	114	132	105
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	21	-21	21	-21
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-29	29	29	-29
Inplane Pressure Stress	54	54	54	54

Inplane Stress Summary	45	61	111	8
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1

Shear Stress Summary	2	2	1	1
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Two * Max Shear Stress	105	114	132	105
------------------------	-----	-----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	-5	-5	5	5
Outplane Bending (Mc)	-32	32	32	-32
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	54	54	54	54

Outplane Stress Summary	25	76	102	23
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	21	-21	21	-21
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-108	108	108	-108
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	109	114	109	114



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Nozzle Calcs.: Noz N4 750

Nozl: 23 9:34am

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Inplane Stress Summary	18	196	247	-6
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	1	1	2	2
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	26	196	247	31
------------------------	----	-----	-----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-4	-4	4	4
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	17	11	17

Hoop Stress Summary	9	15	18	24
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	5	5	5	5

Axial Stress Summary	4	4	9	5
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	9	15	18	24
------------------------	---	----	----	----



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Nozzle Calcs.: Noz N4 750

Nozl: 23

9:34am

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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-5	-5	5	5
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	17	11	17
Hoop Stress Summary	8	14	20	26

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-5	5	5	-5
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	0	9	13	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	8	14	20	26

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	109	114	109	114	109	114	109	114
Circ. Pl (SUS)	-2	-2	6	6	-4	-4	8	8
Circ. Q (SUS)	-1	1	15	-15	-87	87	129	-129
Long. Pm (SUS)	54	54	54	54	54	54	54	54
Long. Pl (SUS)	-1	-1	5	5	-3	-3	7	7
Long. Q (SUS)	-8	8	50	-50	-25	25	39	-39
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0



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 Nozzle Calcs.: Noz N4 750 Nozl: 23 9:34am Sep 9,2026

Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	109.0	114.0	109.0	114.0	109.0	114.0	109.0	114.0
Pm+Pl (SUS)	107.0	112.0	115.0	120.0	105.0	110.0	117.0	122.0
Pm+Pl+Q (Total)	106.0	113.0	130.0	105.0	26.1	197.0	246.0	29.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	114.00	118.00	Passed
Pm+Pl (SUS)	122.00	177.00	Passed
Pm+Pl+Q (TOTAL)	246.01	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	11	17	11	17	11	17	11	17
Circ. Pl (SUS)	-2	-2	6	6	-3	-3	7	7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	-1	1	1	-1	-5	5	5	-5
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	11.0	17.0	11.0	17.0	11.0	17.0	11.0	17.0
Pm+Pl (SUS)	9.0	15.0	17.0	23.0	8.0	14.0	18.0	24.0
Pm+Pl+Q (Total)	9.0	15.0	17.0	23.0	9.0	14.0	18.0	24.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	17.00	148.00	Passed
Pm+Pl (SUS)	24.00	222.00	Passed
Pm+Pl+Q (TOTAL)	24.00	443.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



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 Nozzle Calcs.: D4-Shell side d Nozl: 24 9:34am Page 196 of 216
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Input, Nozzle Desc: D4-Shell side d

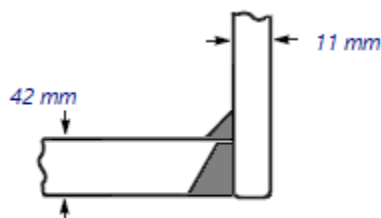
From: 50

Pressure for Reinforcement Calculations	P	4967.154	KPa.
Temperature for Internal Pressure	Temp	275	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	118.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	118.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1750.00	mm
Shell Finished (Minimum) Thickness	t	42.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		7970.49	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		1.5000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	11.0000	mm
Flange Material		SA-105	
Flange Type		RTJWn	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	44.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		2500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.





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 Nozzle Calcs.: D4-Shell side d Nozl: 24 9:34am Sep 9,2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.500 in.
 Actual Thickness Used in Calculation 0.433 in.

Note:
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4967 \cdot 878 / (118 \cdot 1 - 0.6 \cdot 4967)$
 $= 37.9196 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4967 \cdot 22 / (118 \cdot 1 - 0.6 \cdot 4967)$
 $= 0.9523 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	138.1000	mm
Parallel to Vessel Wall	Rn+tn+t	69.0500	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	20.0000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: D4-Shell side d.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.9523	mm
Wall Thickness per UG16(b),	tr16b = 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1 = 40.9196	mm
Wall Thickness	tb1 = max(trb1, tr16b) = 40.9196	mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000	mm
Wall Thickness per table UG-45	tb3 = 6.4200	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(40.9, 4.5)]$
 $= 6.4200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.95, 6.42)$
 $= 6.4200 \text{ mm}$

Available Nozzle Neck Thickness = 11.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	89.0,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	206.0 N./mm ²	Passed
Occasional	:	5.8,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	66.4,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.



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 Nozzle Calcs.: D4-Shell side d Nozl: 24 9:34am Sep 9,2026

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 11$, $t_r = 0.95$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.12$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:A)

Govrn. thk, $t_g = 11$, $t_r = 0.95$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.12$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $4967.15 / 42509.99 = 0.117$

Weld Size Calculations, Description: D4-Shell side d

Intermediate Calc. for nozzle/shell Welds T_{min} 8.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6000 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.5162 mm
 The Cut Length for this Nozzle is, Drop + H_o + H + T : 242.5161 mm

Input Echo, WRC297 Item 1, Description: D4-Shell side :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1750.000	mm
Shell Thickness	42.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	S_{mc} 118.00	N./mm ²
Vessel Hot S.I. Allowable	S_{mh} 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	38.100	mm
Nozzle Thickness	11.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	S_{Nmc} 118.00	N./mm ²



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Nozzle Hot S.I. Allowable SNmh 118.00 N./mm²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	4967.15	KPa.
Radial Load	P	-739.35	Kgf
Circumferential Shear	Vc	611.87	Kgf
Longitudinal Shear	Vl	611.87	Kgf
Circumferential Moment	Mc	96.84	Kg-m.
Longitudinal Moment	Ml	96.84	Kg-m.
Torsional Moment	Mt	127.42	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tv, (1.431) is < 5
 check the Limitations of WRC bulletin 297.
 Warning - The ratio, Dn/Tn (5.464) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 60.1 + 2 \cdot 1.65 \cdot \sqrt{897.5(42.0 - 3.0)}$
 $= 677.495 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1795.000 mm
Nozzle Outside Diameter	(d) =	60.100 mm
Vessel Thickness used	(T) =	39.000 mm
Nozzle Thickness used	(t) =	8.000 mm
T / t	=	4.875
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.227$$

Nr/P	=	0.211
Mr/P	=	0.135
M0/P	=	0.308
N0/P	=	0.244
MrD/Mc	=	0.117
NrDL/Mc	=	0.159
M0d/Mc	=	0.298
N0DL/Mc	=	-0.130
MrD/Ml	=	0.117
NrDL/Ml	=	0.163
M0D/Ml	=	0.293
N0DL/Ml	=	-0.124

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0



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Nozzle Calcs.: D4-Shell side d Nozl: 24 9:34am Sep 9,2026

Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	1	1	-1	-1
Outplane Bending (ML)	-18	18	18	-18
Normal Pressure Stress	109	114	109	114

Outplane Stress Summary	102	126	136	87
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-7	7	7	-7
Inplane Pressure Stress	54	54	54	54

Inplane Stress Summary	50	57	68	46
------------------------	----	----	----	----

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5

Shear Stress Summary	7	7	4	4
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	103	127	136	87
------------------------	-----	-----	-----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	1	1	-1	-1
Outplane Bending (Mc)	-18	18	18	-18
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	54	54	54	54

Outplane Stress Summary	47	66	81	27
-------------------------	----	----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-7	7	7	-7
Inplane Membrane (ML)	0	0	0	0



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Nozzle Calcs.: D4-Shell side d Nozl: 24 9:34am Sep 9,2026

Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	109	114	109	114
Inplane Stress Summary	105	117	123	105

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5
Shear Stress Summary	4	4	7	7

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	105	117	124	106

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	1	1	-1	-1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	19	13	19
Hoop Stress Summary	16	22	13	19

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	5	5	5	5
Axial Bending (P)	19	-19	19	-19
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-62	-62	62	62
Axial Bending (ML)	-52	52	52	-52
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	-82	-17	147	3

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	7	7	-7	-7
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	27	27	27	27
Shear Stress Summary	35	35	19	19

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
	Au	Al	Bu	Bl



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(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	121	81	150	42

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	1	1	-1	-1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	19	13	19
Hoop Stress Summary	16	22	13	19

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	5	5	5	5
Axial Bending (P)	19	-19	19	-19
Axial Membrane (Mc)	-62	-62	62	62
Axial Bending (Mc)	-55	55	55	-55
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	-86	-14	150	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-7	-7	7	7
Shear due to Torsion	27	27	27	27
Shear Stress Summary	19	19	35	35

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	110	53	159	73

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	109	114	109	114	109	114	109	114
Circ. Pl (SUS)	2	2	0	0	-1	-1	1	1
Circ. Q (SUS)	-10	10	26	-26	-4	4	10	-10
Long. Pm (SUS)	54	54	54	54	54	54	54	54
Long. Pl (SUS)	-1	-1	1	1	2	2	0	0
Long. Q (SUS)	-4	4	10	-10	-10	10	26	-26



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 Nozzle Calcs.: D4-Shell side d Nozl: 24 9:34am Sep 9,2026

Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	109.0	114.0	109.0	114.0	109.0	114.0	109.0	114.0
Pm+P1 (SUS)	111.0	116.0	109.0	114.0	108.0	113.0	110.0	115.0
Pm+P1+Q (Total)	101.7	126.5	135.2	88.4	104.3	117.3	120.9	105.5

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	114.00	118.00	Passed
Pm+P1 (SUS)	116.02	177.00	Passed
Pm+P1+Q (TOTAL)	135.23	353.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	13	19	13	19	13	19	13	19
Circ. P1 (SUS)	2	2	0	0	2	2	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. P1 (SUS)	-57	-57	67	67	-57	-57	67	67
Long. Q (SUS)	-33	33	71	-71	-36	36	74	-74
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	7	7	-7	-7	-7	-7	7	7
Shear Q (SUS)	27	27	27	27	27	27	27	27
Pm (SUS)	13.0	19.0	13.0	19.0	13.0	19.0	13.0	19.0
Pm+P1 (SUS)	67.5	73.3	73.8	73.9	67.5	73.3	73.8	73.9
Pm+P1+Q (Total)	120.1	78.4	147.0	43.5	109.6	53.8	155.1	70.9

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	19.00	118.00	Passed
Pm+P1 (SUS)	73.89	177.00	Passed
Pm+P1+Q (TOTAL)	155.13	353.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



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 ASME TS Calc:

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Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Main Shell

Shell Maximum Design Pressure	Psd,max	4950.00 KPa.
Shell Maximum Operating Pressure	Psox,max	4950.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	42.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1750.000 mm
Shell Temperature for Internal Pressure	Ts	275.00 °C
Shell Material		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K11789
Shell Allowable Stress at Temperature	Ss	118.00 N./mm ²
Shell Allowable Stress at Ambient		118.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	1250.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	1250.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	14.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1750.000 mm
Channel Design Temperature	TEMPC	250.00 °C
Channel Flange Matl		SA-387 11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K11789
Channel Allowable Stress at Temperature	Sc	118.00 N./mm ²
Channel Allowable Stress at Ambient		118.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	638
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	2500.00 mm
Straight Tube Length (bet. inner tubsht faces) L		2175.00 mm
Design Temperature of the Tubes		275.00 °C
Tube Material		SA-213 T11
Tube Material UNS Number		K11597
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		110.50 N./mm ²
Tube Allowable Stress At Ambient		118.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	166.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	849.916 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm



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ASME TS Calc:

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Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	5810.7 cm ²
Perimeter of Tube Layout	Cp	4024.3 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral with Channel, Conf. e

Tubesheet Design Metal Temperature	T	275.00 °C
Tubesheet Material		SA-336 F11

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K11572
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	325.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	2240.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		Yes
Is the Tubesheet Flat on Shellside?		No
Step 1 Diameter on the Shellside	ds1	0.00 mm
Step 1 Depth on the Shellside	hs1	0.00 mm
Step 2 Diameter on the Shellside	ds2	1747.00 mm
Step 2 Depth on the Shellside	hs2	7.00 mm
Calculated Tubesheet Diameter per Div 2, 4.18.4		2240.00 mm

Note: Tubesheet diameter is now: 2240.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes	AL	0.0 cm ²
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Junction Stress Reduction option	Analyze as Simply Supported
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Shell
Flange Outside Diameter	A	2240.000 mm
Flange Inside Diameter	B	1750.000 mm
Flange Face Outside Diameter	Fod	1883.000 mm
Flange Face Inside Diameter	Fid	1750.000 mm
Gasket Outside Diameter	Go	1872.000 mm
Gasket Inside Diameter	Gi	1782.000 mm
Small end Hub thk.	g0	44.0000 mm
Large end Hub thk.	g1	55.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch	Code	Sketch 1a
Gasket Thickness	tg	3.1750 mm
Full face Gasket Flange Option	Program	Selects
Length of Partition Gasket	lp	1800.000 mm
Width of Partition Gasket	wp	10.0000 mm
Partition Gasket Factor,	mPart	3.0000
Partition Gasket Design Seating Stress	yPart	68.95 N./mm ²
Partition Gasket Facing Sketch	Code	Sketch 1a
Partition Gasket Column for Gasket Seating	Code	Column II

Bolting Information:

Diameter of Bolt Circle	C	2070.000 mm
Nominal Bolt Diameter	dB	76.2000 mm



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ASME TS Calc:

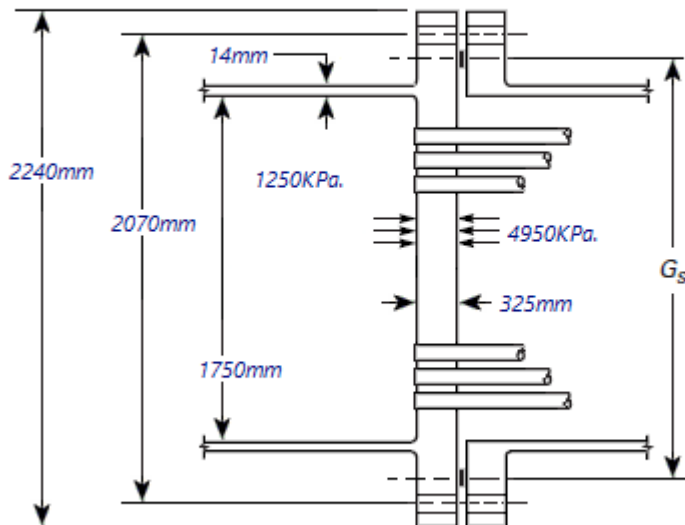
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Thread Series	TEMA Thread Series
Number of Bolts	n 28
Bolt Material	SA-193 B7
Bolt Allowable Stress At Temperature	Sb 159.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa 159.00 N./mm ²
Weld between Flange and Shell/Channel	0.0000 mm

Note: The full bolt load $Sa \cdot Ab$, was used for all load cases.

Tubesheet Integral with	Channel
Tubesheet Extended as Flange	Yes
Thickness of Extended Portion of Tubesheet	Tf 309.0000 mm
Is Bolt Load Transferred to the Tubesheet	Yes

Tubesheet Gasketed w/Shell and Integral w/Channel, Extended as a Flange



Configuration e:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Material Group C

Elastic Mod. at Design Temperature	275.0 °C	0.18798E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Channel - TM-1 Material Group C

Elastic Mod. at Design Temperature	250.0 °C	0.18966E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Tubes - TM-1 Material Group C

Elastic Mod. at Tubsht. Design Temp.	275.0 °C	0.18798E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

TubeSheet - TM-1 Material Group C

Elastic Mod. at Design Temperature	275.0 °C	0.18798E+09 KPa.
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ASME TS Calc:

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Elastic Mod. at Ambient Temperature 21.1 °C 0.20408E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1250 \cdot (25.4/2 - 0)) / (110 \cdot 1 + 0.4 \cdot 1250)$$

$$= 0.1430 + 0.0000 = 0.1430 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-2 at 275.00 °C
Elastic Modulus for Material 183830896.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	3024.92	12.04	50.0000	0.0075908	96.07

$$EMAWP = (4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 96.1) / (3 \cdot 12) = 10640 \text{ KPa.}$$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.1842	25.40	3024.92	21.45	50.0000	0.0023911	79.64

$$EMAWP = (4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 79.6) / (3 \cdot 21.4) = 4951 \text{ KPa.}$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.1842 mm
Allowable Internal Pressure at Corroded thickness	19664.33 KPa.
Required Internal Design Pressure	1250.00 KPa.
Allowable External Pressure at Corroded thickness	10640.22 KPa.
Required External Design Pressure	4950.00 KPa.
Required Thickness due to Shell Side pressure	1.1842 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Goc - Gic) / 2$	45.000 mm
Basic Gasket Width,	$b0 = N / 2$	22.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) \cdot 2.5$	11.949 mm
Gasket Reaction Diameter,	$G = Go - 2.0 \cdot b$	1848.103 mm

Results for ASME U-tube Tubesheet Calculations for Configuration e:

Per Edition 2025, Original Thickness :

Note:

The Tubesheet stress reduction option was Simply Supported. Performing the 1st pass by including the stiffness of integral shell/channel to ensure that the junction stress is < SPS.

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 850 + 25.4$$

$$= 1725.232 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$



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ASME TS Calc:

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$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= G_s / D_o \text{ (Configurations d, e, f)}$$

$$= 1848 / 1725$$

$$= 1.0712$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 1750 / 1725$$

$$= 1.0144$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 0 / 325$$

$$= 0.000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(d_t - 2t_t (E_t T / E) \cdot (S_t T / S) \cdot (p); d_t - 2t_t)$$

$$= \max(25.4 - 2 \cdot 2.11 (187982736 / 187982736) \cdot (0); 25.4 - 2 \cdot 2.11)$$

$$= 25.4000 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ 1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2) }$$

$$= 31.8 / \sqrt{ 1 - 4 \cdot \min(0; 4 \cdot 1725 \cdot 31.8) / (\pi \cdot 1725^2) }$$

$$= 31.7500 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.2000$$

$h/p = 10.236220$; $\mu^* = 0.200000$; $E^* = 29202954 \text{ KPa}$.

$E^*/E = 0.155349$; $\nu^* = 0.458572$

Note: As h/p (10.236) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:

β_s , k_s , λ_s , δ_s and w_s are all = 0

Channel coefficient [β_c]:

$$= ((12 (1 - \nu_c^2))^{1/4}) / \sqrt{ (D_c + t_c) t_c }$$

$$= ((12 (1 - 0.3^2))^{1/4}) / \sqrt{ (1750 + 14) 14 }$$

$$= 0.0116 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6 (1 - \nu_s^2))$$

$$= 0.012 \cdot 189658240 \cdot 14^3 / (6 (1 - 0.3^2))$$

$$= 1102568576.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$CqDelta = CqTheta = CmTheta = CmDelta = 1$

Channel coefficient [λ_c]:

$$= 6 (D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6 (1750 + 14 / \cos(0)) \cdot 1102568576 / 325^3 \cdot (1 +$$



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ASME TS Calc: Case: 1 9:34a Sep 9, 2026

$$325 \cdot 0.012 \cdot (1 + 1)/2 + (325 \cdot 0.012)^2/2 \cdot 1 \\ = 4020240.0 \text{ KPa.}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c))) \\ = 1750^2 / (4 \cdot 189658240 \cdot 14) \cdot (1 - 1750 \cdot 0.3 / (2(1750 + 14))) \\ = 0.24542E+00 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (C_m \Theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \\ \zeta_c (C_m \Theta + h \cdot C_q \Theta \cdot h \cdot \beta C / 2)] \\ = 1.01 \cdot 1102568576 [0.24542446 (1 + 0.012 \cdot 1 \cdot 325) - \\ 0.000 (1/325 + 1 \cdot 0.012/2)] \\ = 15112.6074 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o \\ = 2240/1725 \\ = 1.2984$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_c + E \cdot \ln(K)) \\ = (1 - 0.46) / 29202954 \cdot (4020240 + 187982736 \cdot \ln(1.3)) \\ = 0.9846$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2/16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t] \\ = 1725^2/16 \cdot [(1.07 - 1) \cdot (1.07^2 + 1) \cdot 4950 - \\ (1.01 - 1) \cdot (1.01^2 + 1) \cdot 1250] \\ = 134063800.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= M_{ts} + (\omega_c \cdot P_t) + W^* \cdot (C - G_s) / (2 \cdot \pi \cdot D_o) \\ = 134063800 + (18890760) + 1852157 \cdot (2070 - 1848) / (2 \cdot \pi \cdot 1725) \\ = 524751680.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M^* - D_o^2/32 \cdot F(P_s - P_t)) / (1 + F) \\ = (524751680 - 1725^2/32 \cdot 0.98 (4950 - 1250)) / (1 + 0.98) \\ = 93676656.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2/64 \cdot (3 + v^*) \cdot (P_s - P_t) \\ = 93676656 + 1725^2/64 \cdot (3 + 0.46) \cdot (4950 - 1250) \\ = 688808960.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|M_p|; |M_o|) \\ = \max(|93676656|; |688808960|) \\ = 688808960.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - h_g')^2) \\ = 6 \cdot 688808960 / (0.2 (325 - 8)^2)$$



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ASME TS Calc:

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$$= 205.6490 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 138$$

$$= 275.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 683722752 / (0.2 \cdot (281 - 8)^2)$$

$$= 275.9947 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + Cats + Catc$$

$$= 281 + 0 + 0$$

$$= 280.6233 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(HreqB; HreqS)$$

$$= \max(281 ; 24.2)$$

$$= 280.6233 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$| P_s - P_t |$]:

$$= | P_s - P_t |$$

$$= | 4950 - 1250 |$$

$$= 3700.000 \text{ KPa.}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / Do \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.2 \cdot 325 / 1725 \cdot \min(0.8 \cdot 138 ; 0.533 \cdot 222)$$

$$= 16.637 \text{ N./mm}^2$$

$| P_s - P_t |$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot | P_s - P_t | + Cats + Catc$$

$$= 1 / (4 \cdot 0.2 \cdot 110) \cdot (4 \cdot 5811 / 4024) \cdot | 4950 - 1250 | + 0$$

$$= 24.1976 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Channel Membrane Stress [$\sigma_{c,m}$]:

$$= Dc^2 / (4 \cdot t_c (Dc + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 1750^2 / (4 \cdot 14 (1750 + 14) \cdot \cos(0)) \cdot 1250$$

$$= 38.755 \text{ N./mm}^2$$

Axial Channel Bending Stress [$\sigma_{c,b}$]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \Theta) \cdot P_t -$$

$$6 \cdot ((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \Theta +$$

$$h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 247873 / 14^2 \cdot [(0.24542446 \cdot 1 \cdot 0.012 - 0.000 \cdot 1) \cdot 1250 -$$

$$6 \cdot ((1 - 0.46) / 29202954) \cdot 1725 / 325^3 \cdot (1 +$$

$$0.012 \cdot 325 / 2 \cdot 1) \cdot (93676656 + 1725^2 / 32 \cdot (4950 - 1250))]$$

$$= -118.136 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= | \sigma_{c,m} | + | \sigma_{c,b} | \text{ (Should be } \leq \text{ SPS}_{c,c} \text{)}$$

$$= 156.891 \text{ N./mm}^2 \text{ (Should be } \leq 394.02 \text{)}$$

Note: The Cylinder-to-Tubesheet Juncture stress is < SPS. So, the



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ASME TS Calc:

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Simply Supported option can be used.

Re-running the analysis by ignoring the integral channel/shell stiffness.
Only modified portions of the output will be re-printed.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate Shell Side parameters for the Simply Supported case:

β_s , k_s , λ_s , δ_s are all = 0

Intermediate Channel side parameters for the Simply Supported condition:

β_c , k_c , λ_c , δ_c are all = 0

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.46) / 29202954 \cdot (0 + 187982736 \cdot \ln(1.3))$$

$$= 0.9100$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (P_s , P_t) [Mts]:

$$= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$$

$$= 1725^2 / 16 \cdot [(1.07 - 1) \cdot (1.07^2 + 1) \cdot 4950 - (1.01 - 1) \cdot (1.01^2 + 1) \cdot 1250]$$

$$= 134063800.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= M_{ts} + (\omega_c \cdot P_t) + W^* \cdot (C - G_s) / (2 \cdot \pi \cdot D_o)$$

$$= 134063800 + (0) + 1852157 \cdot (2070 - 1848) / (2 \cdot \pi \cdot 1725)$$

$$= 505860896.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on center of tubesheet [M_o]:

$$= M_p + D_o^2 / 64 \cdot (3 + \nu^*) \cdot (P_s - P_t)$$

$$= 100871776 + 1725^2 / 64 \cdot (3 + 0.46) \cdot (4950 - 1250)$$

$$= 696004096.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= |M_o|$$

$$= |696004096|$$

$$= 696004096.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* \cdot (h - h_g')^2)$$

$$= 6 \cdot 696004096 / (0.2 \cdot (325 - 8)^2)$$

$$= 207.7972 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 138$$

$$= 275.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu^* \cdot (h - h_g')^2)$$

$$= 6 \cdot 696004096 / (0.2 \cdot (283 - 8)^2)$$

$$= 275.8515 \text{ N./mm}^2$$



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ASME TS Calc:

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Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned} &= h + \text{Cats} + \text{Catc} \\ &= 283 + 0 + 0 \\ &= 283.1322 \text{ mm} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned} &= \max(\text{HreqB}; \text{HreqS}) \\ &= \max(283; 72.3) \\ &= 283.1322 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

$$\begin{aligned} &= | \text{Ps} - \text{Pt} | \\ &= | 4950 - 1250 | \\ &= 3700.000 \text{ KPa.} \end{aligned}$$

Shear Stress check [τlimit]:

$$\begin{aligned} &= 4 \cdot \mu \cdot h / \text{Do} \cdot \min(0.8 \cdot S; 0.533 \cdot S_y) \\ &= 4 \cdot 0.2 \cdot 325 / 1725 \cdot \min(0.8 \cdot 138; 0.533 \cdot 222) \\ &= 16.637 \text{ N./mm}^2 \end{aligned}$$

| Ps - Pt | is ≤ the Shear Stress limit (τlimit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$\begin{aligned} &= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot \text{Ap} / \text{Cp}) \cdot | \text{Ps} - \text{Pt} | + \text{Cats} + \text{Catc} \\ &= 1 / (4 \cdot 0.2 \cdot 110) \cdot (4 \cdot 5811 / 4024) \cdot | 4950 - 1250 | + 0 \\ &= 24.1976 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Channel Membrane Stress [σc,m]:

$$\begin{aligned} &= \text{Dc}^2 / (4 \cdot \text{tc} \cdot (\text{Dc} + \text{tc}) \cdot \cos(\theta)) \cdot \text{Pt} \\ &= 1750^2 / (4 \cdot 14 \cdot (1750 + 14) \cdot \cos(0)) \cdot 1250 \\ &= 38.755 \text{ N./mm}^2 \end{aligned}$$

Axial Channel Bending Stress [σc,b]:

$$\begin{aligned} &= 6 \cdot \text{kc} / \text{tc}^2 \cdot [(\delta_c \cdot \text{Cm} \Delta \cdot \beta_c - \zeta_c \cdot \text{Cm} \Theta) \cdot \text{Pt} - \\ &\quad 6((1 - \nu^*) / E^*) \cdot \text{Do} / h^3 \cdot (\text{Cm} \Theta + \\ &\quad h \cdot \beta_c / 2 \cdot \text{Cm} \Delta) \cdot (\text{Mp} + \text{Do}^2 / 32 \cdot (\text{Ps} - \text{Pt}))] \\ &= 6 \cdot 0 / 14^2 \cdot [(0.000 \cdot 1 \cdot 0 - 0.000 \cdot 1) \cdot 1250 - \\ &\quad 6((1 - 0.46) / 29202954) \cdot 1725 / 325^3 \cdot (1 + \\ &\quad 0 \cdot 325 / 2 \cdot 1) \cdot (100871776 + 1725^2 / 32 \cdot (4950 - 1250))] \\ &= -0.000 \text{ N./mm}^2 \end{aligned}$$

Channel Membrane + Bending Stress [σc]:

$$\begin{aligned} &= | \sigma_{c,m} | + | \sigma_{c,b} | \text{ (Should be } \leq \text{SPS}_c) \\ &= 38.755 \text{ N./mm}^2 \text{ (Should be } \leq 394.02) \end{aligned}$$

Analysis Finished.

Tube Metal Area [At]:

$$\begin{aligned} &= \pi (d_o - t) t \\ &= \pi (25.4 - 2.11) \cdot 2.11 \\ &= 1.544 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$



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ASME TS Calc:

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$$= \pi \cdot 2.11 (25.4 - 2.11) 110$$

$$= 1739.535 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 110$$

$$= 1702.3129 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 110/110$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 110$$

$$= 1101.4966 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.11 \cdot (25.4 - 2.11) 1 \cdot 1)} - 0.75 \cdot 25.4 \right)$$

$$= 2.667 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.3334 mm

Minimum Required Groove Weld Leg [agr]: 1.3334 mm

Tube-Tubesheet Jt allowable, 1740 is $\geq$ tube strength 1740 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	207.8 $\leq$	276.0 N./mm ²	Ok
Tubesheet shear stress	8.2 $\leq$	110.4 N./mm ²	Ok
Stress in Channel at Tubesheet	38.8 $\leq$	394.0 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	283.132	325.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.333	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.333	2.110	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 6 Load Cases:

	--Reqd. Thk. + CA		----	Tubesheet	Stresses		Case	Pass/
Case#	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd	Type	Fail
D1uc	135.358	122.530	45	276	3	110	Ps+Pt D1	Ok
D2uc	311.214	...	252	276	11	110	Ps+Pt D2	Ok
D3uc	283.132	...	208	276	8	110	Ps+Pt D3	Ok
D1c	138.920	122.530	46	276	3	110	Ps+Pt-c D1	Ok
D2c	314.156	...	257	276	11	110	Ps+Pt-c D2	Ok
D3c	285.911	...	212	276	8	110	Ps+Pt-c D3	Ok



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ASME TS Calc:

Case: 1 9:34a Sep 9,2026

Max: 314.1555 122.530 mm 0.932 0.101 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	Integral Cyl. Req'd. Thk. Shell	Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	...	...	39	394	Ok
D2uc	...	...	...	...	...	394	Ok
D3uc	...	...	...	...	39	394	Ok
D1c	...	...	...	...	50	394	Ok
D2c	...	...	...	...	...	394	Ok
D3c	...	...	...	...	50	394	Ok
Max	...	...	...	...	0.126		

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	314.156	325.000	Ok
Tubesheet thickness flanged extension :	122.530	309.000	Ok
Tube thickness :	1.184	2.110	Ok
Tube-tubesheet fillet weld leg :	1.333	2.110	Ok
Tube-tubesheet groove weld leg :	1.333	2.110	Ok

Given Shell Thickness: 42.0000 mm

Given Channel Thickness: 14.0000 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	7554.25	1.000	5489.90	1.000
Tubesheet Shear Stress	48777.54	1.000	48777.55	1.000
Tubesheet-Channel Junction	2114.73	1.000	7629.79	1.000
Tube Pressure Stress	19664.32	1.000	10640.21	1.000
Tubesheet Extension Stress	No Calc	No Calc	5760.90	...
Minimum MAWP	2114.73		5489.90	



Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	7744.41	1.000	5595.30	1.000
Tubesheet Shear Stress	49695.00	1.000	49694.96	1.000
Tubesheet-Channel Junction	3211.56	1.000	9504.96	1.000
Tube Pressure Stress	20998.73	1.000	13593.03	1.000
Tubesheet Extension Stress	No Calc	No Calc	5760.90	...
Minimum MAPnc	3211.56		5595.30	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Channel Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):
= max(tubesheet thk/4, min(tubesheet thk, channel thickness))
= max(325/4, min(325, 14))
= 81.250 mm

Thickness Ratio = 0.88, Temperature Reduction per Fig. UCS 66.1 = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 45 °C
Min Metal Temp. at Required thickness (UCS 66.1) 39 °C

Shell Side MDMT calculation:

Governing thickness for the shell side:
= tubesheet thickness/4
= 325/4
= 81.250 mm

Thickness Ratio = 0.88, Temperature Reduction per Fig. UCS 66.1 = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 27 °C
Min Metal Temp. at Required thickness (UCS 66.1) 21 °C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:
= max(pt/Tubeside MAPnc, ps/Shellside MAPnc), must be ≤ 1
= max(1250/3212, 4950/5595)
= 0.885

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Problems/Failures Summary:

Step: 35

9:34am

Sep 9, 2026

Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: Insufficient BasePlate bolting area!**

**** Error: Baseplate Bolt Shear Stress Failure!**

**** Warning: A standard flange seems to be governing the Channelside MAWP!**

Please review all reports carefully!

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: